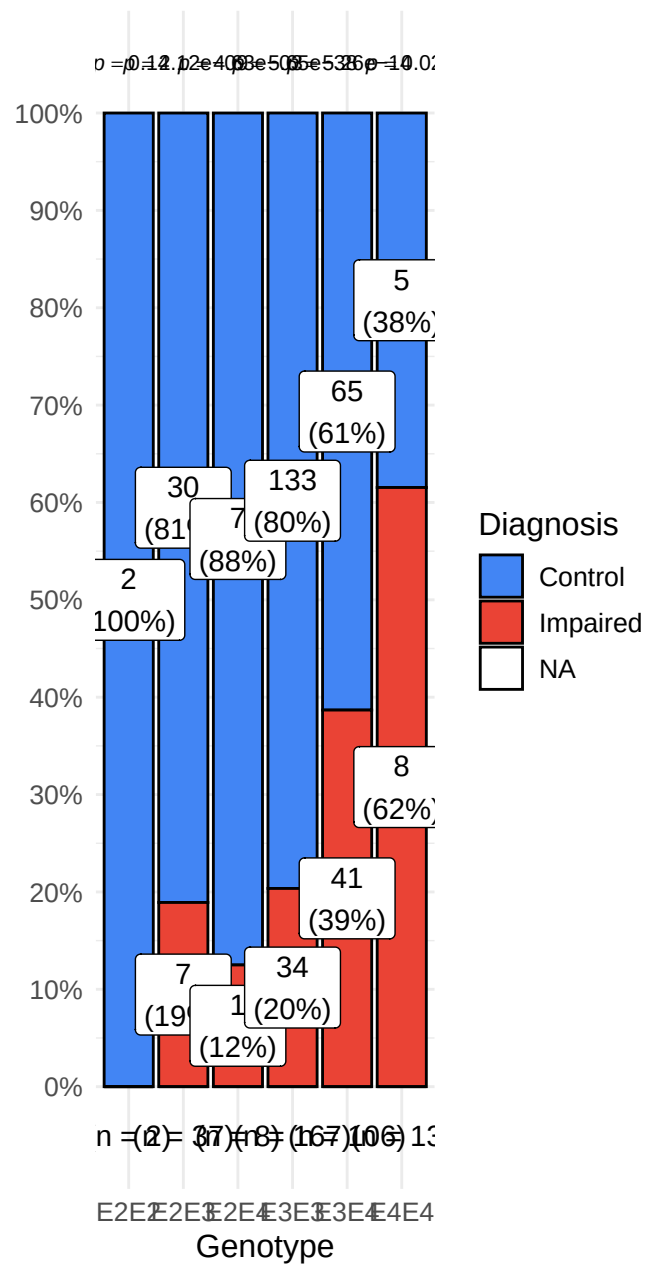
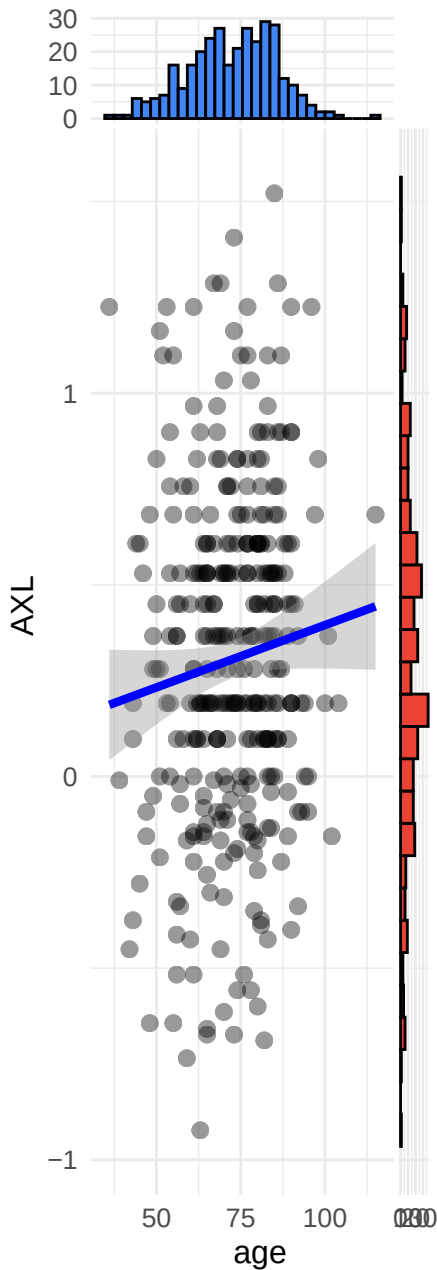
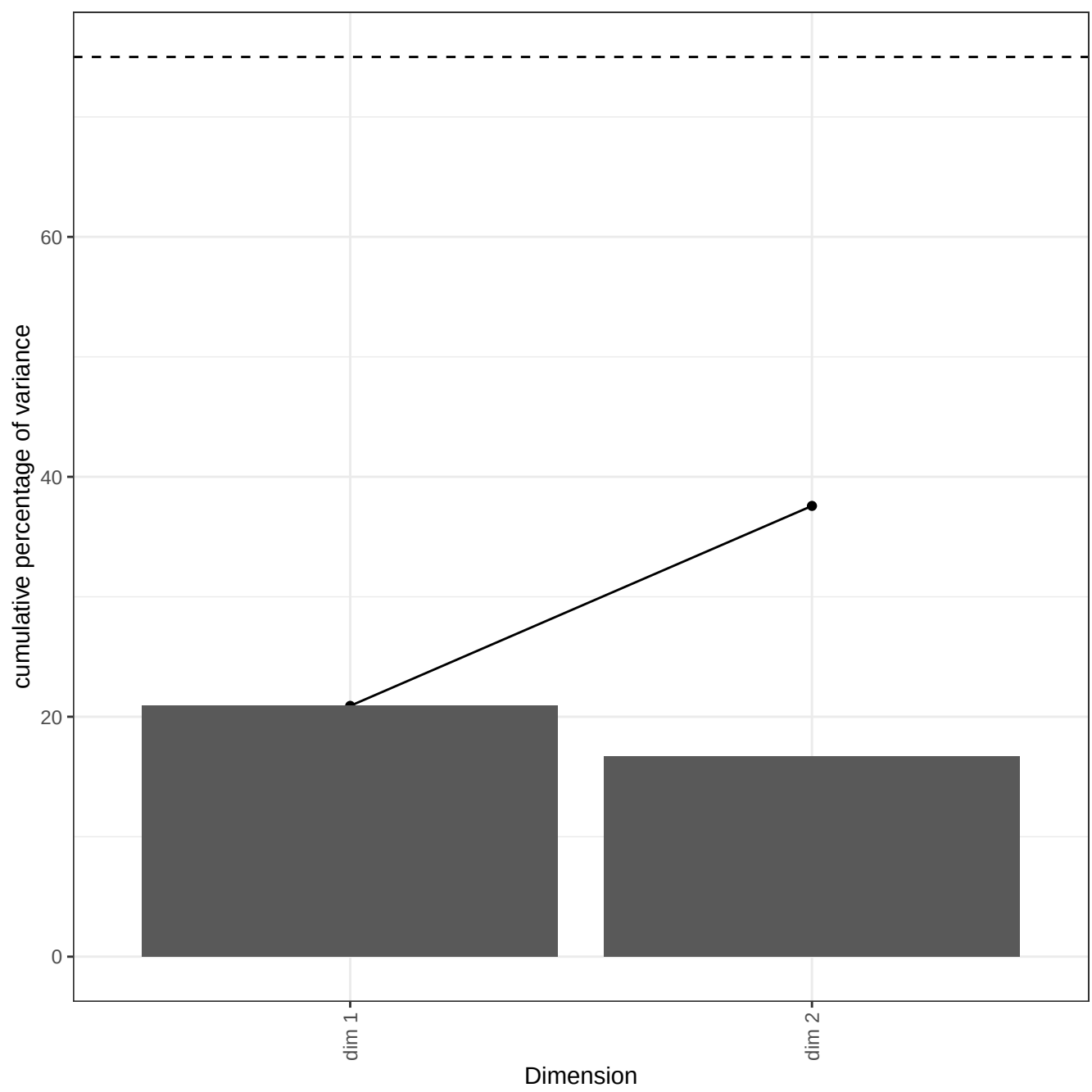
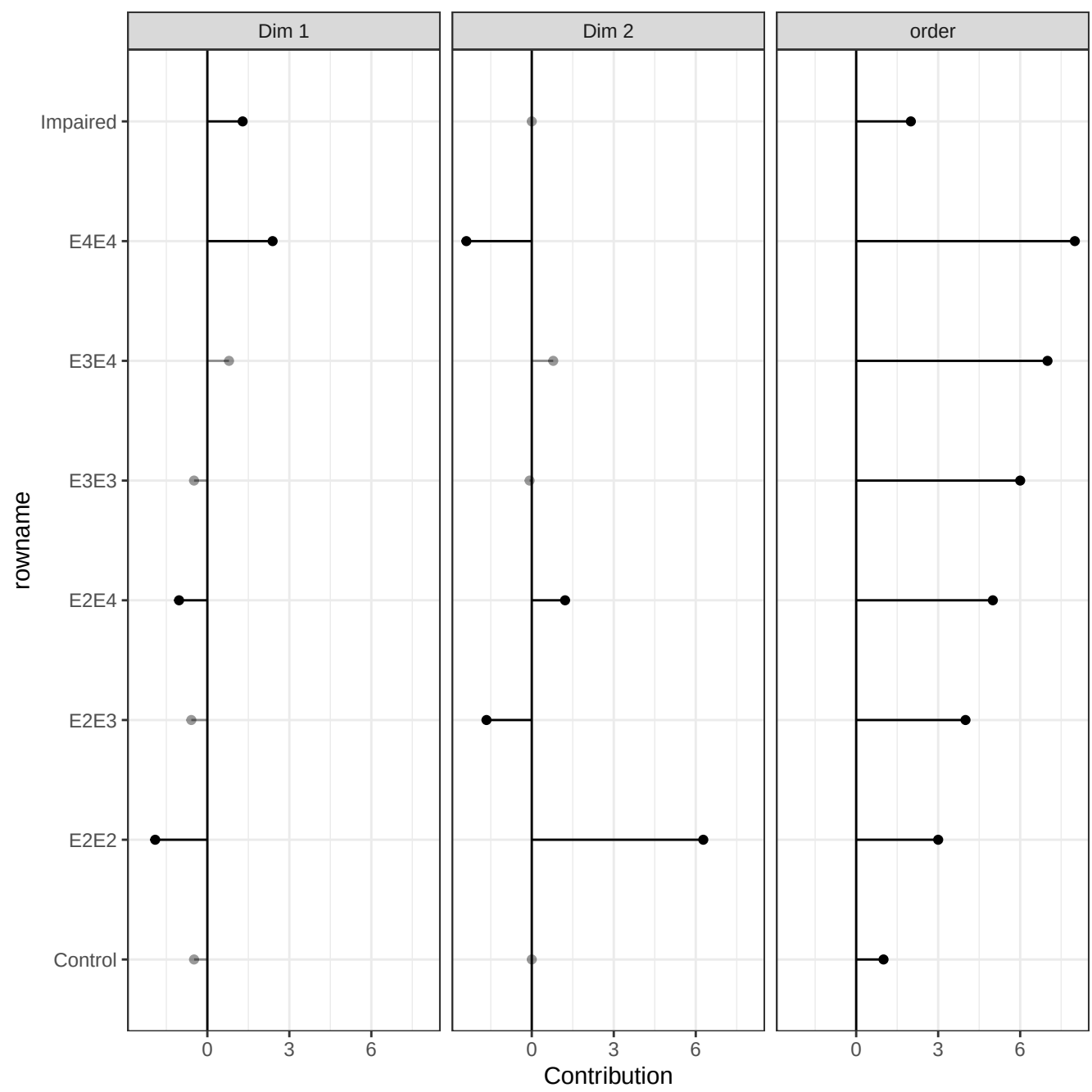
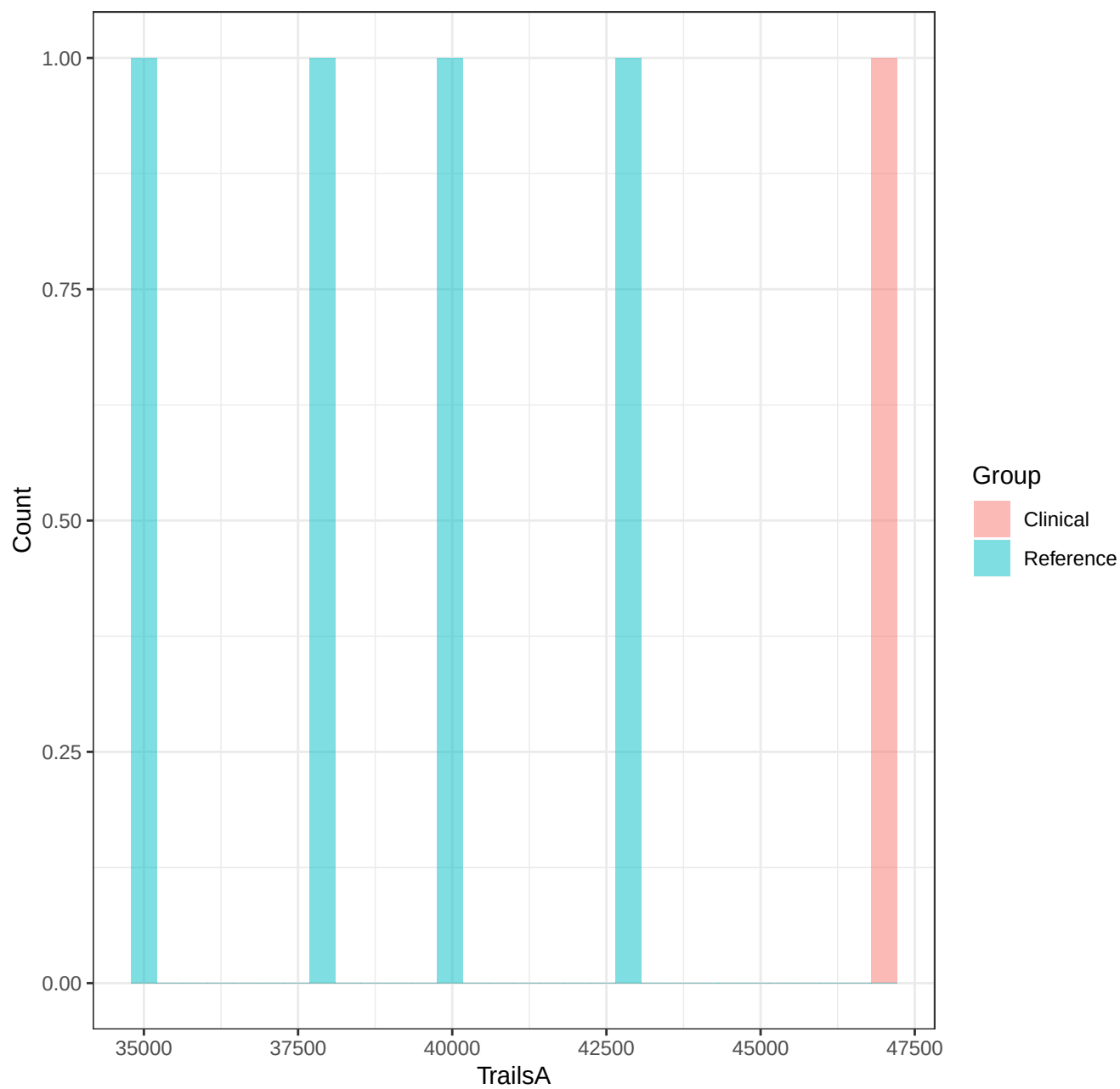


A**B**
 $t_{\text{Student}}(320) = 1.72, p = 0.09, \hat{r}_{\text{Pearson}} = 0.1$


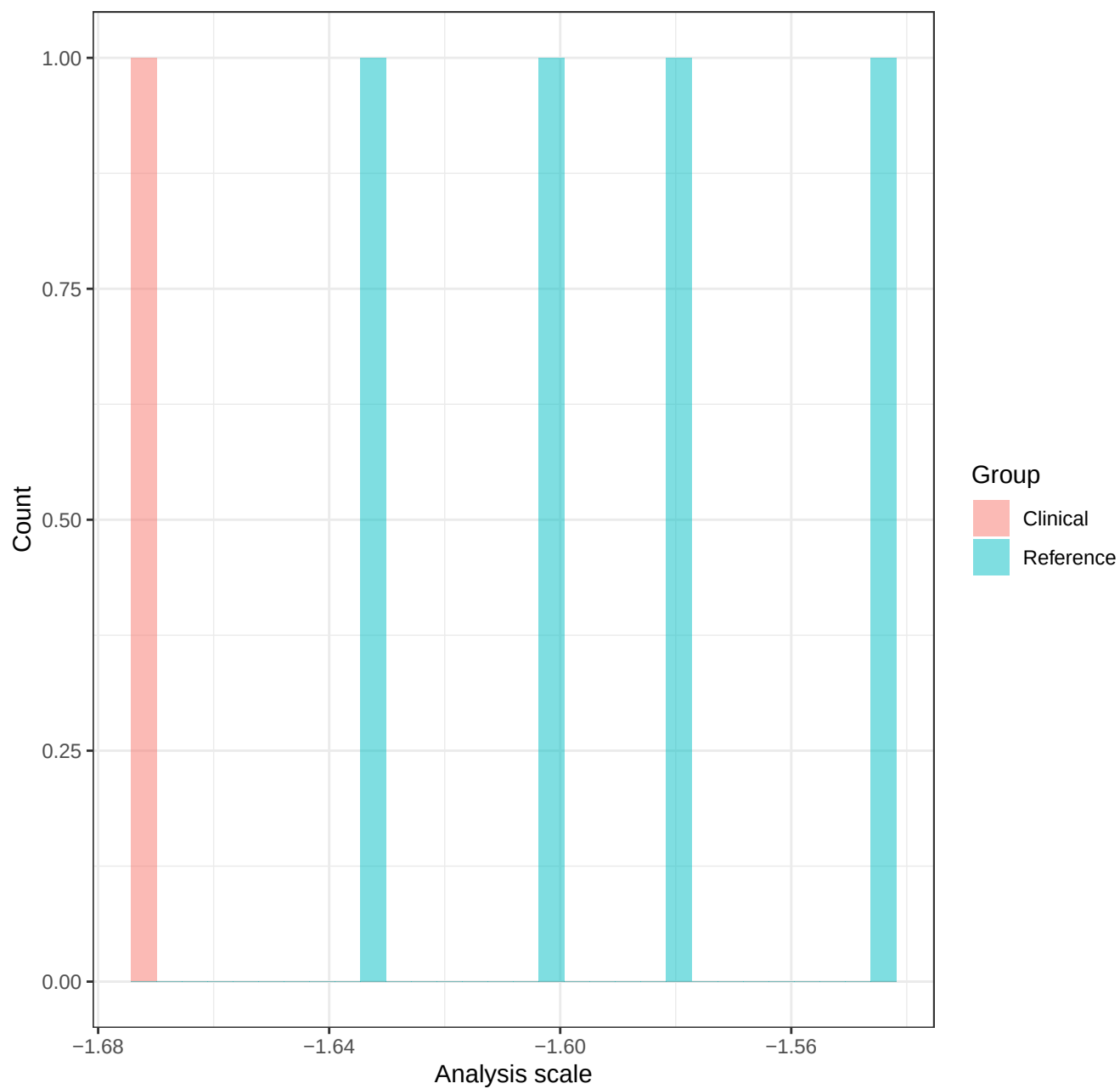




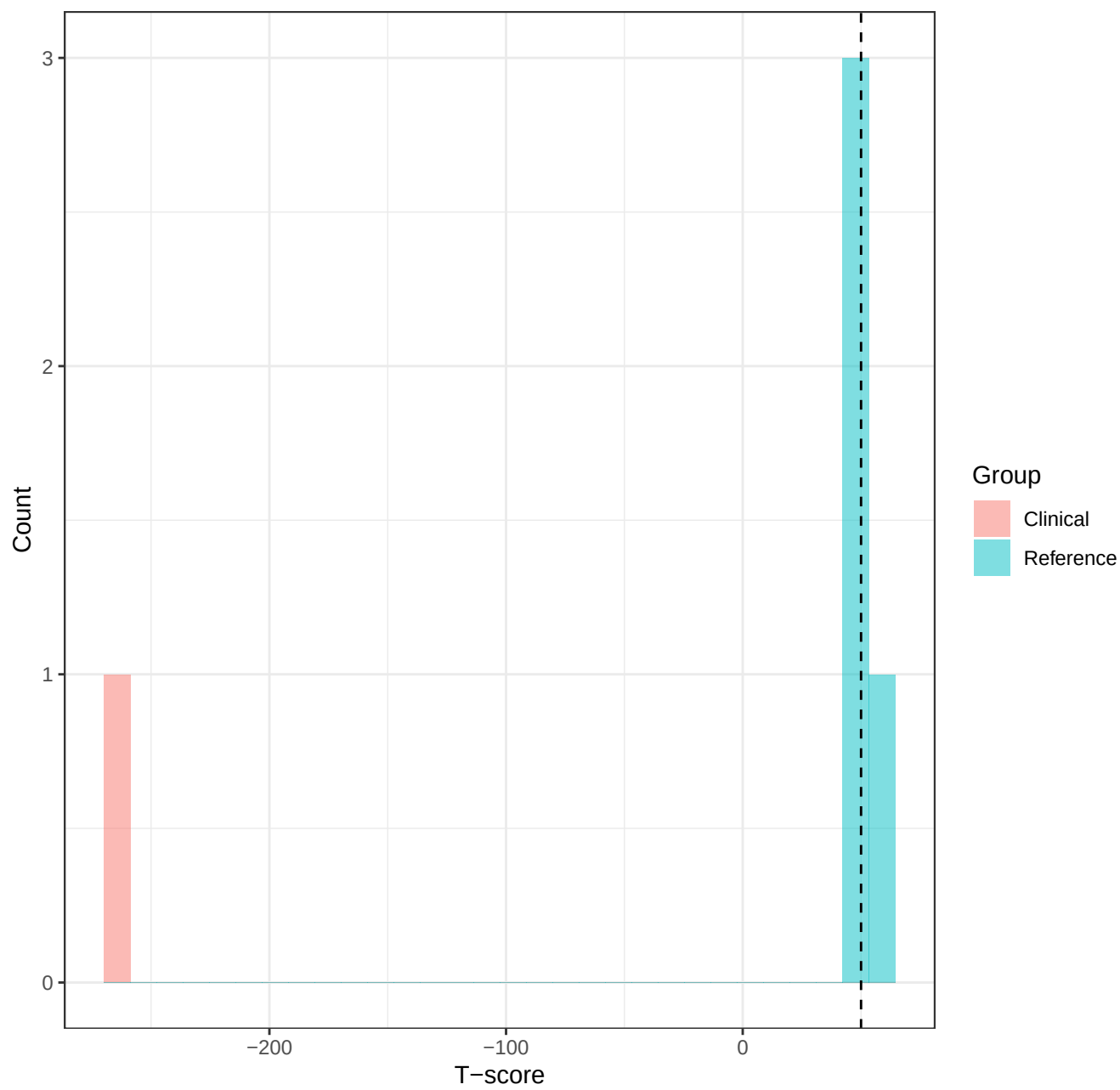
Raw score distribution



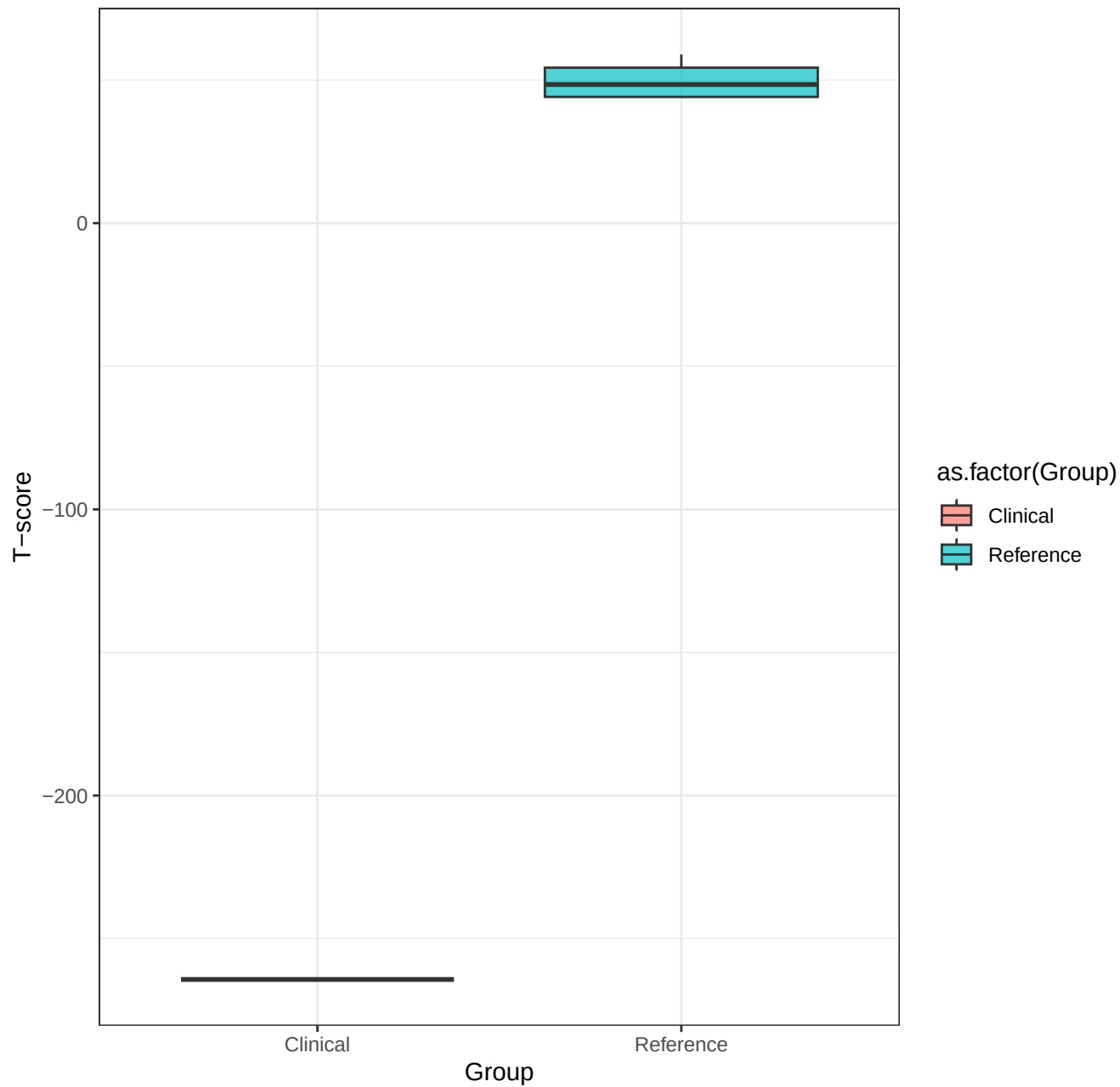
Scaled score distribution



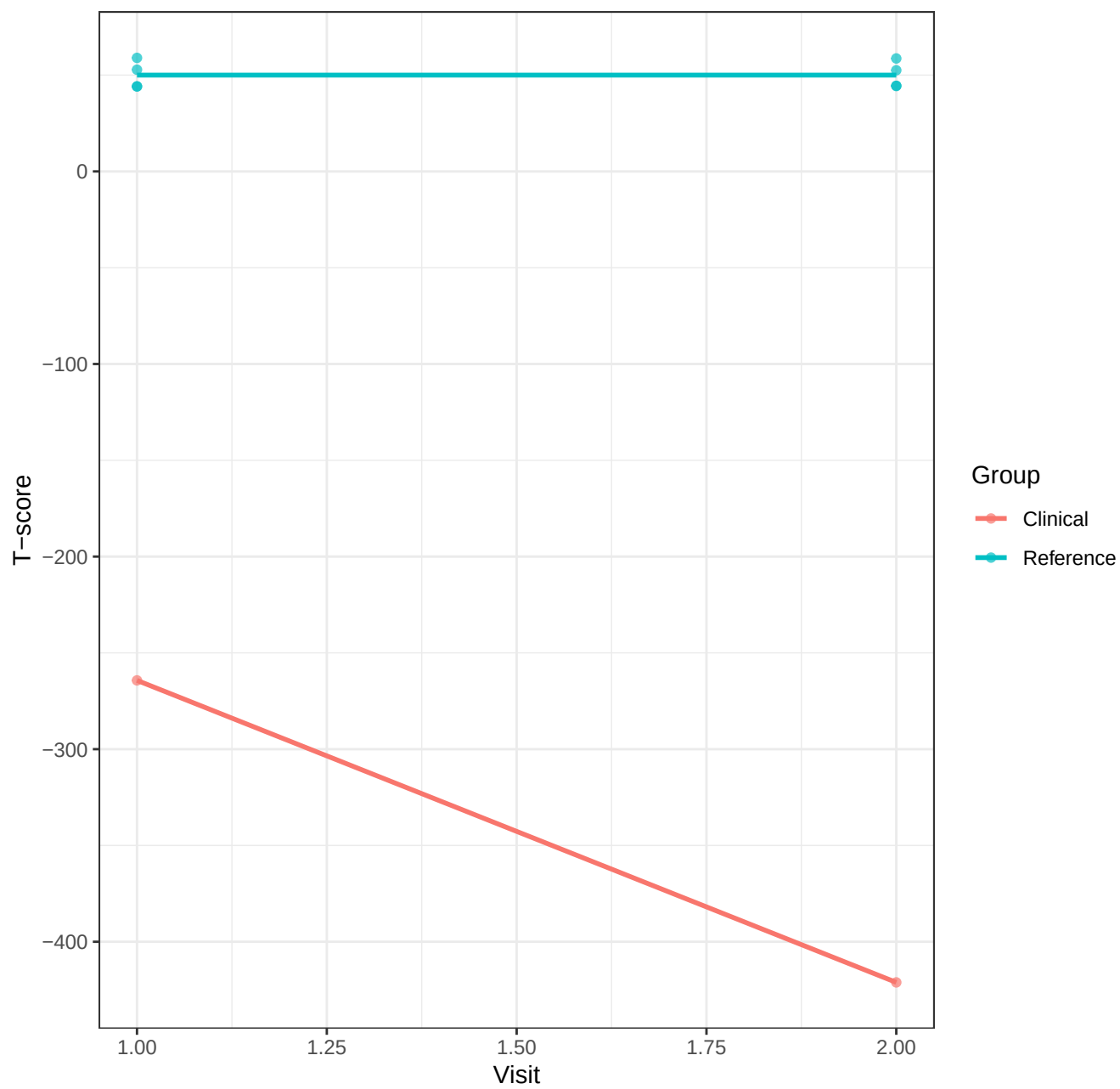
T-score distribution



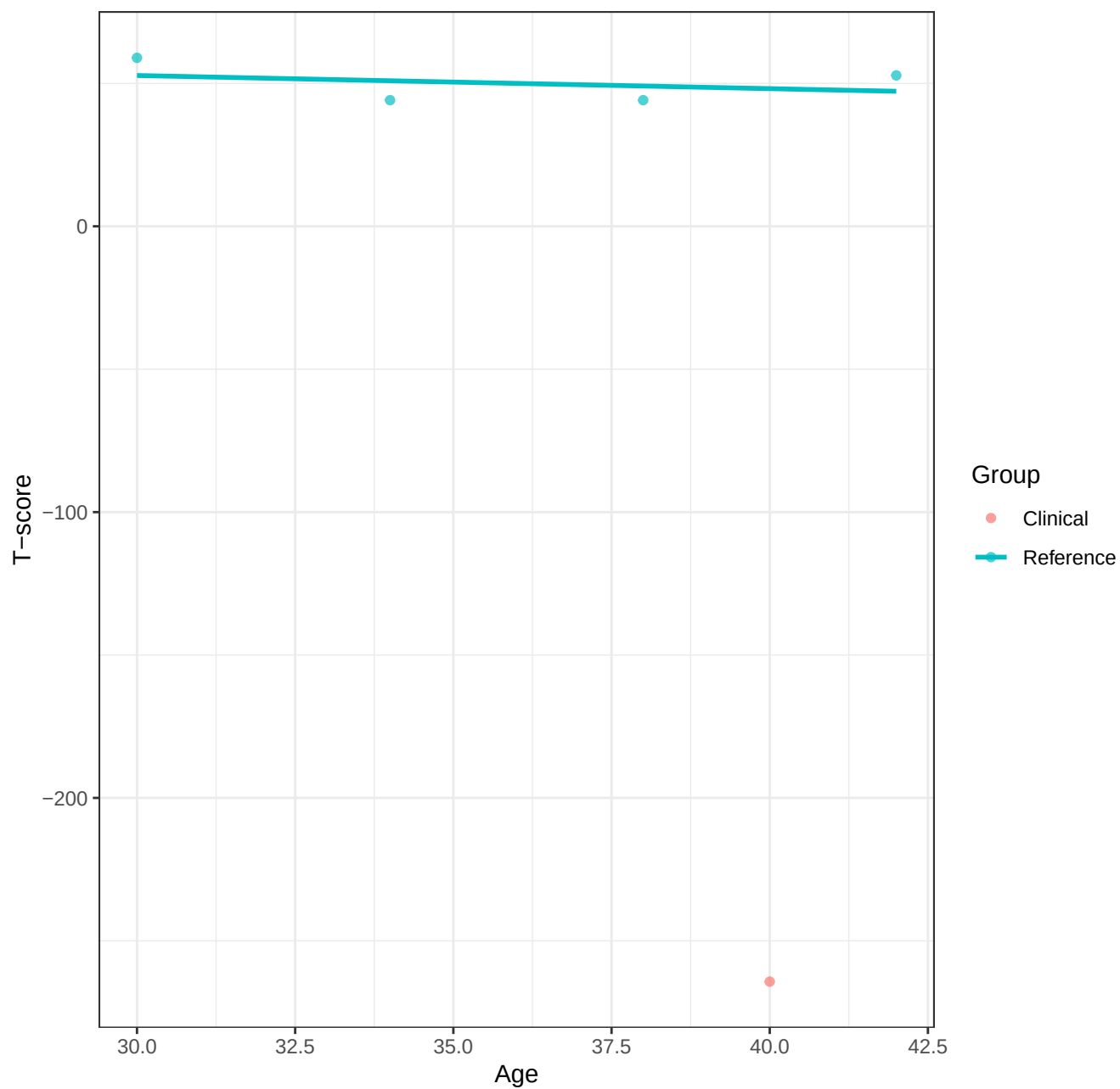
T-scores by reference group



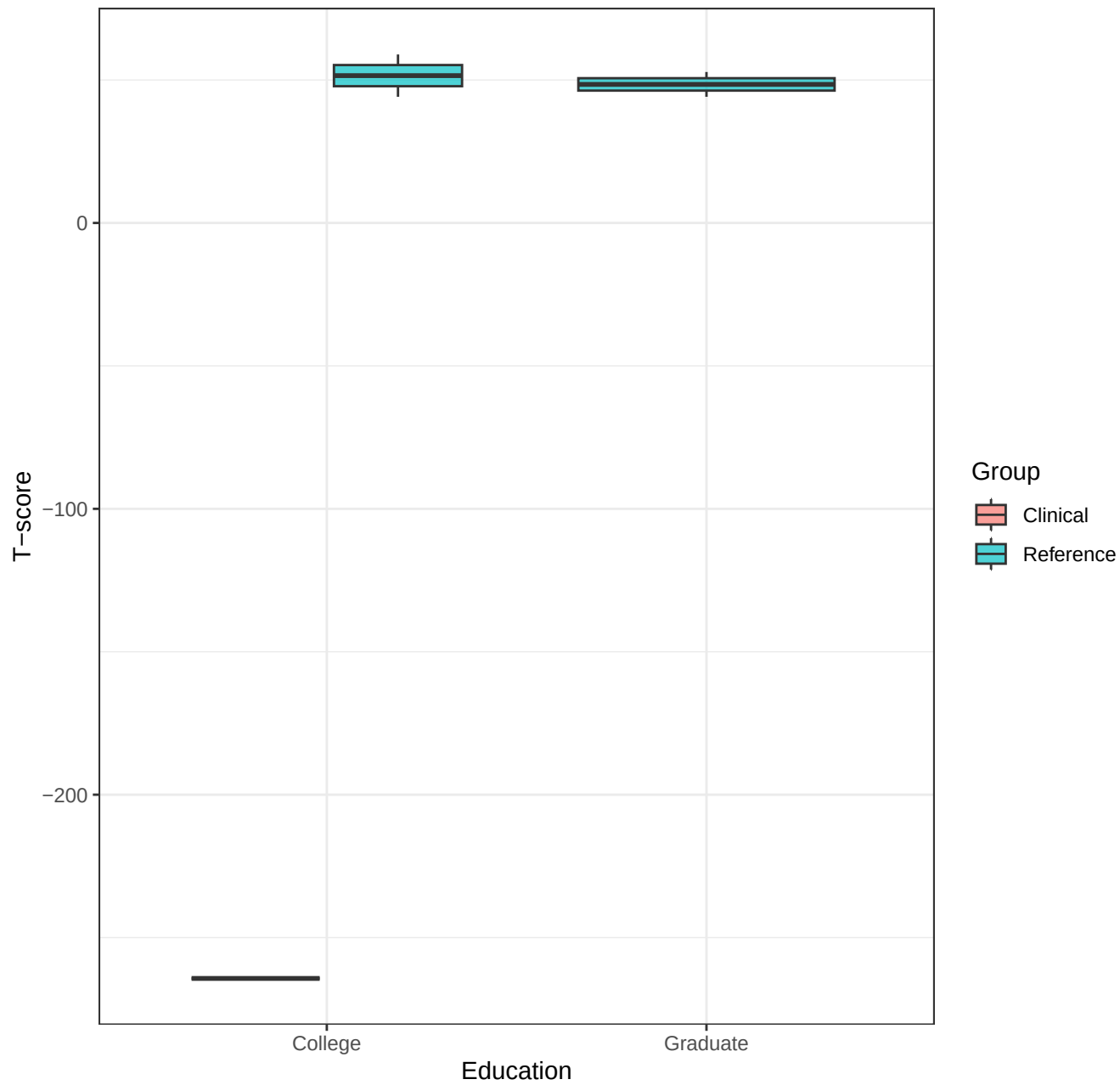
T-scores by visit count



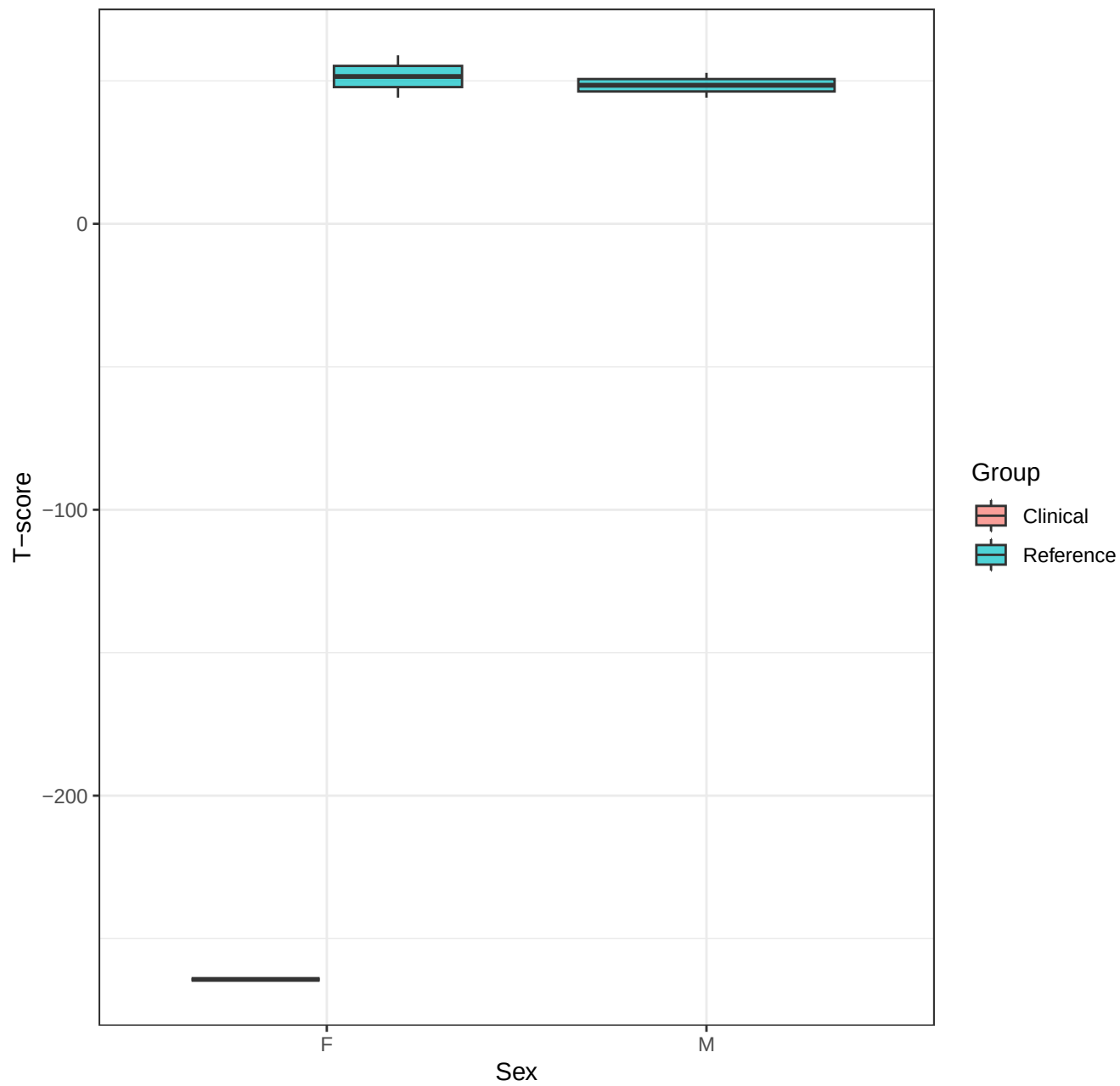
T-scores by Age

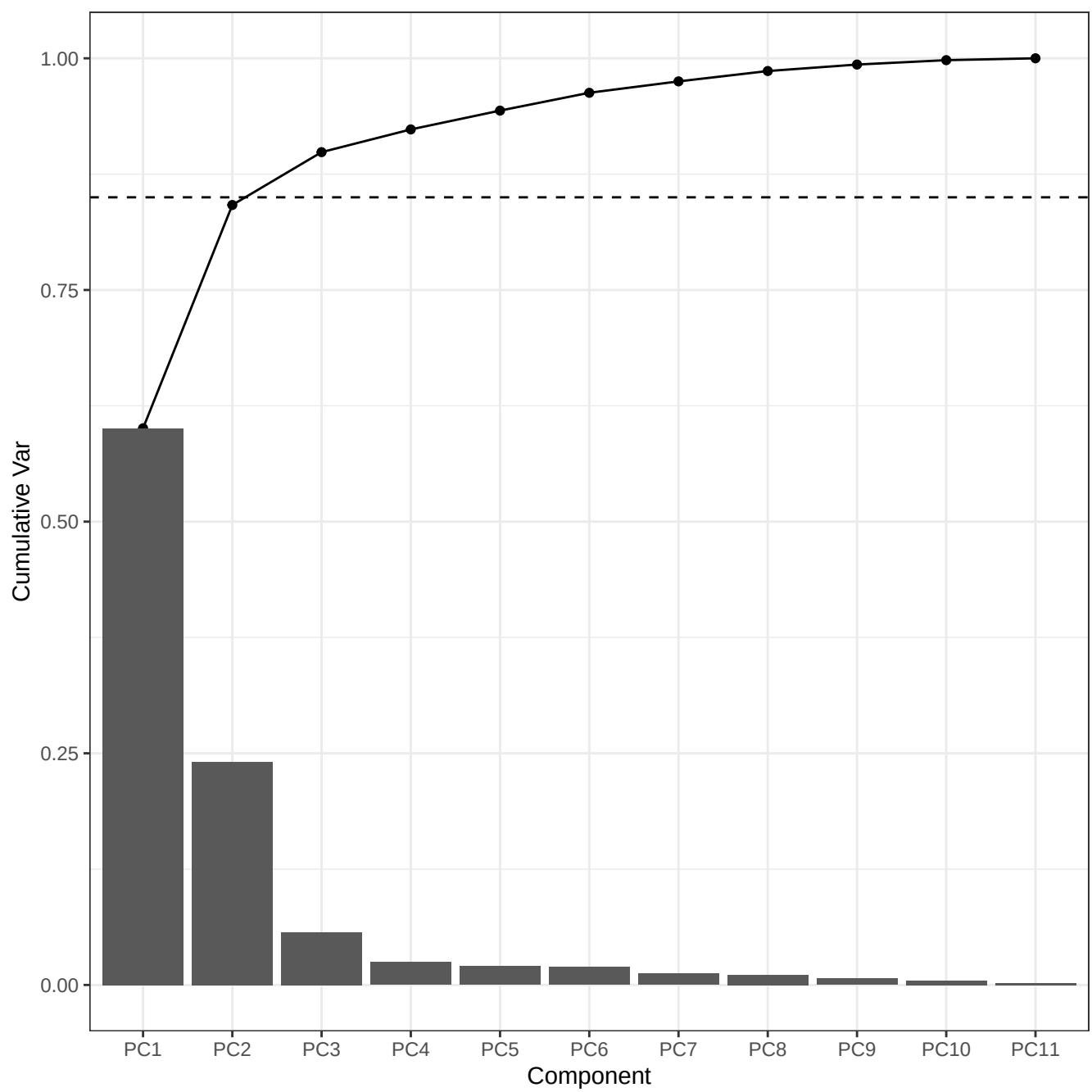


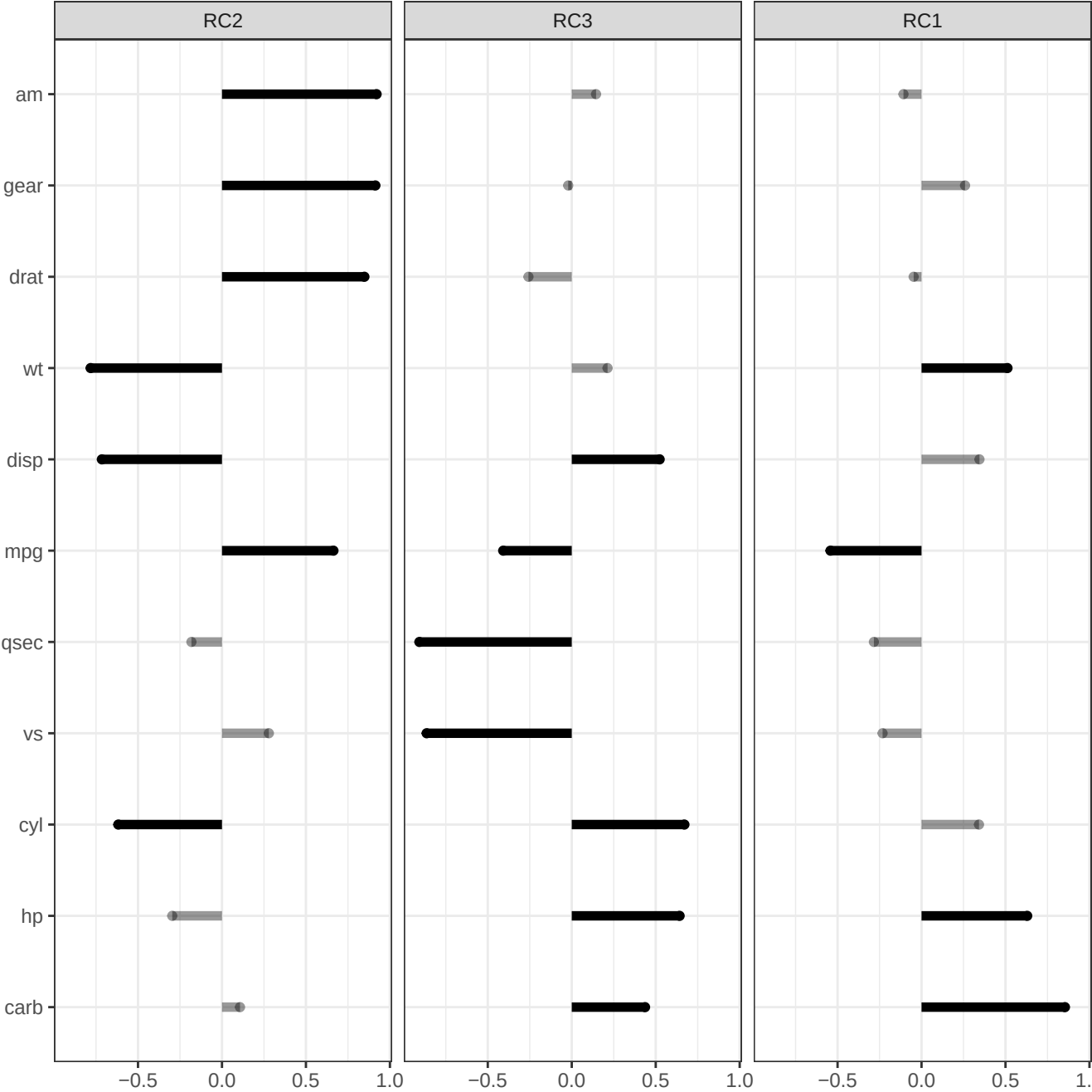
T-scores by Education

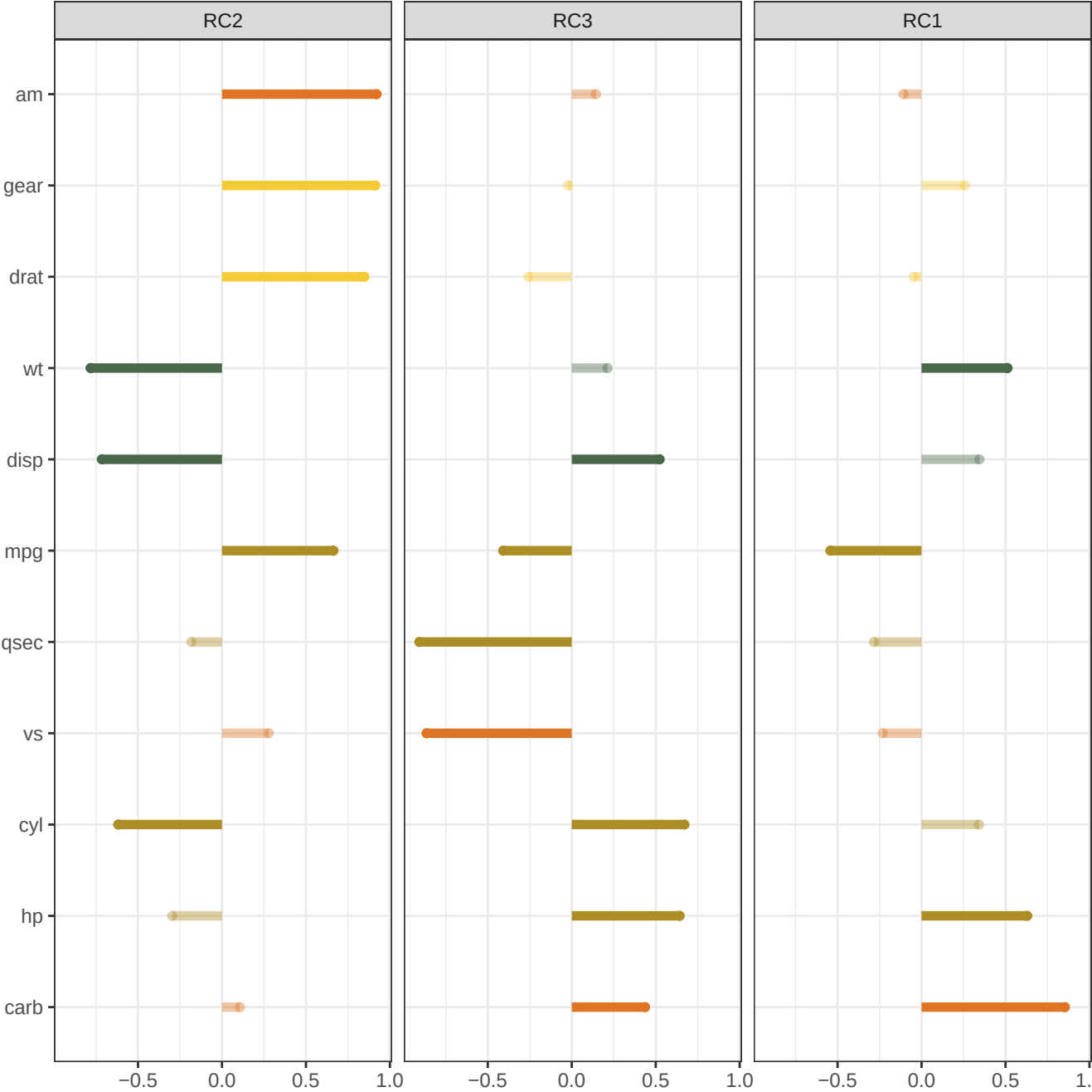


T-scores by Sex



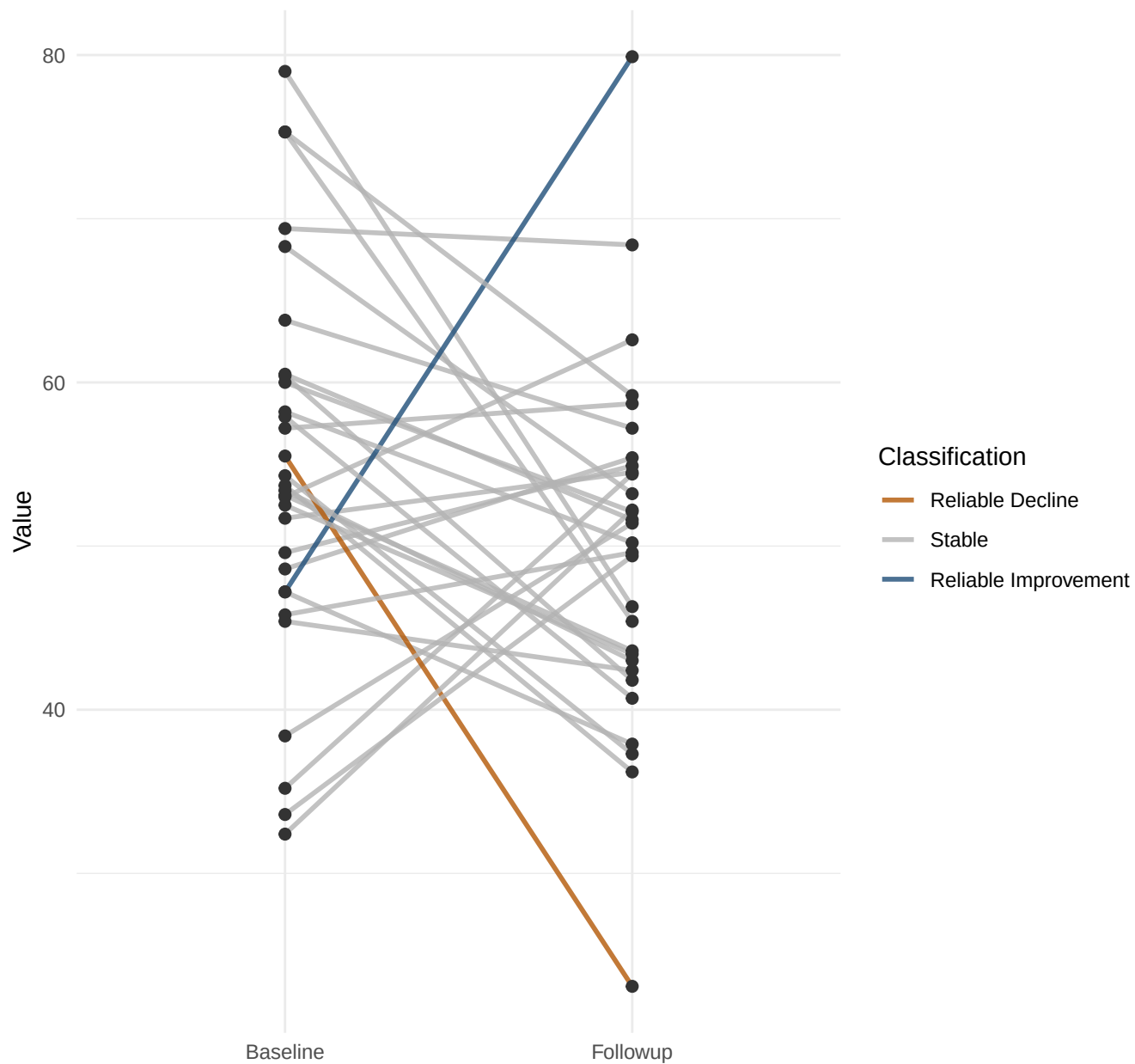






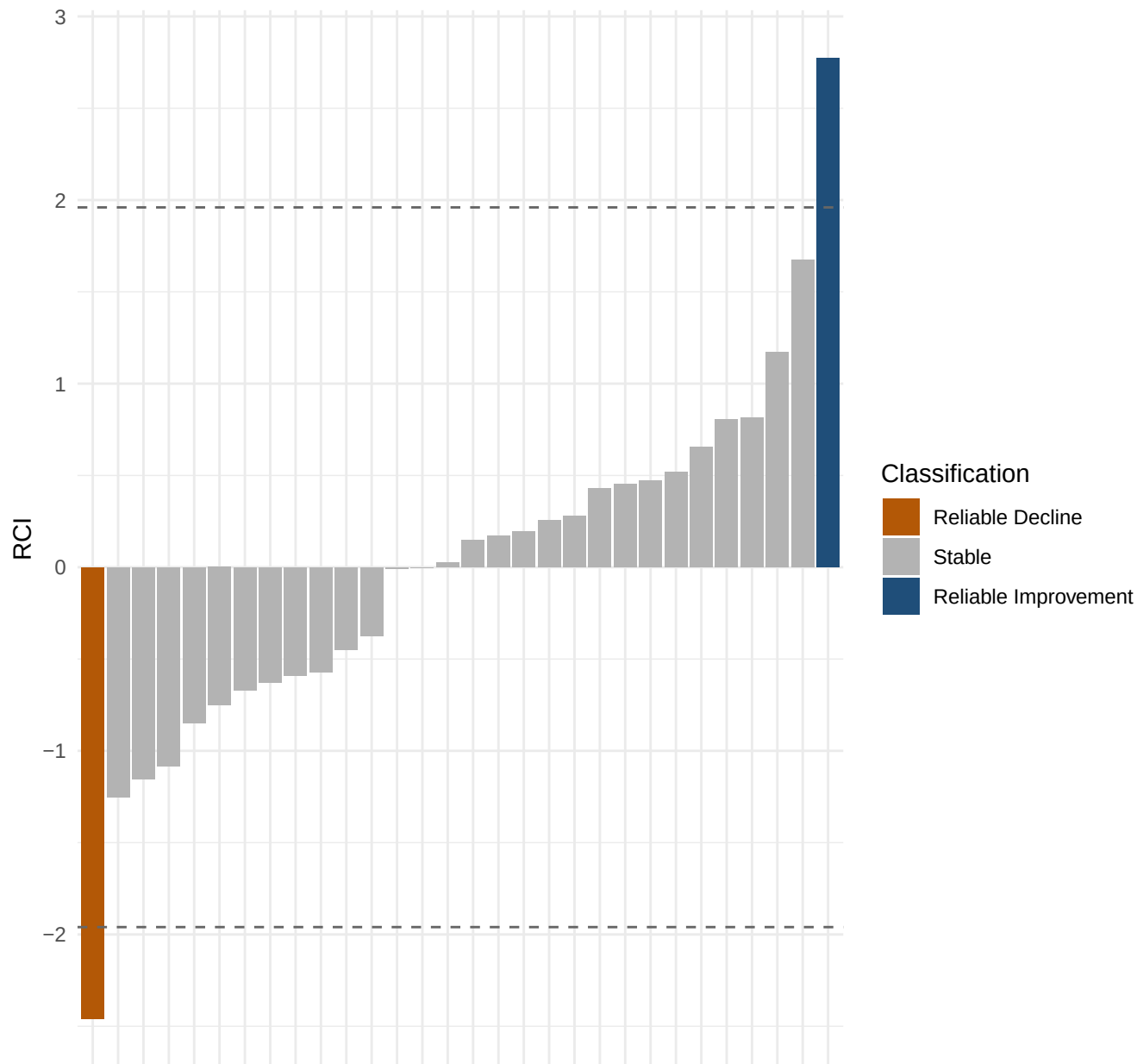
Score

Reference visit: Baseline



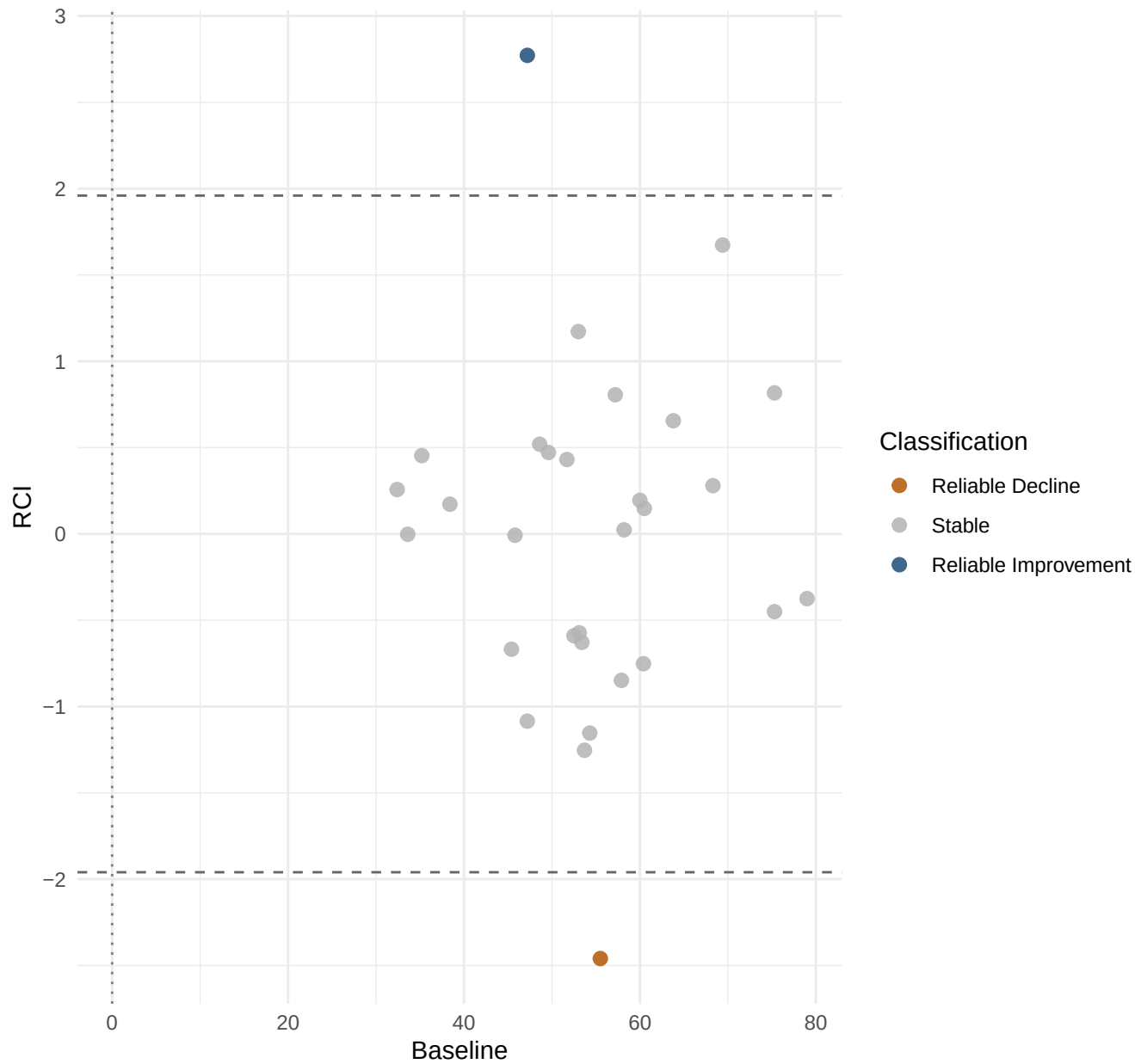
Score

Reference visit: Baseline

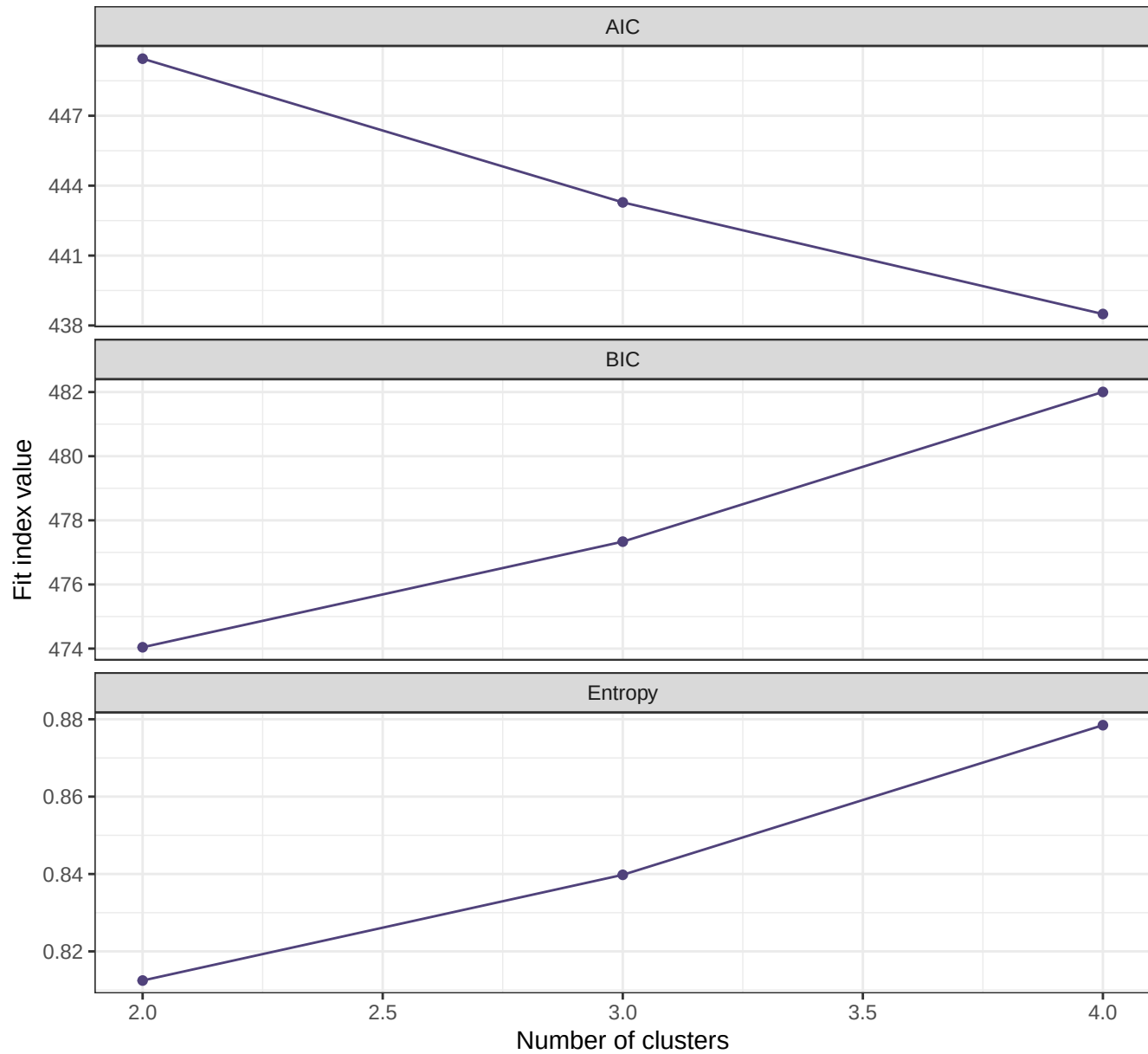


Score

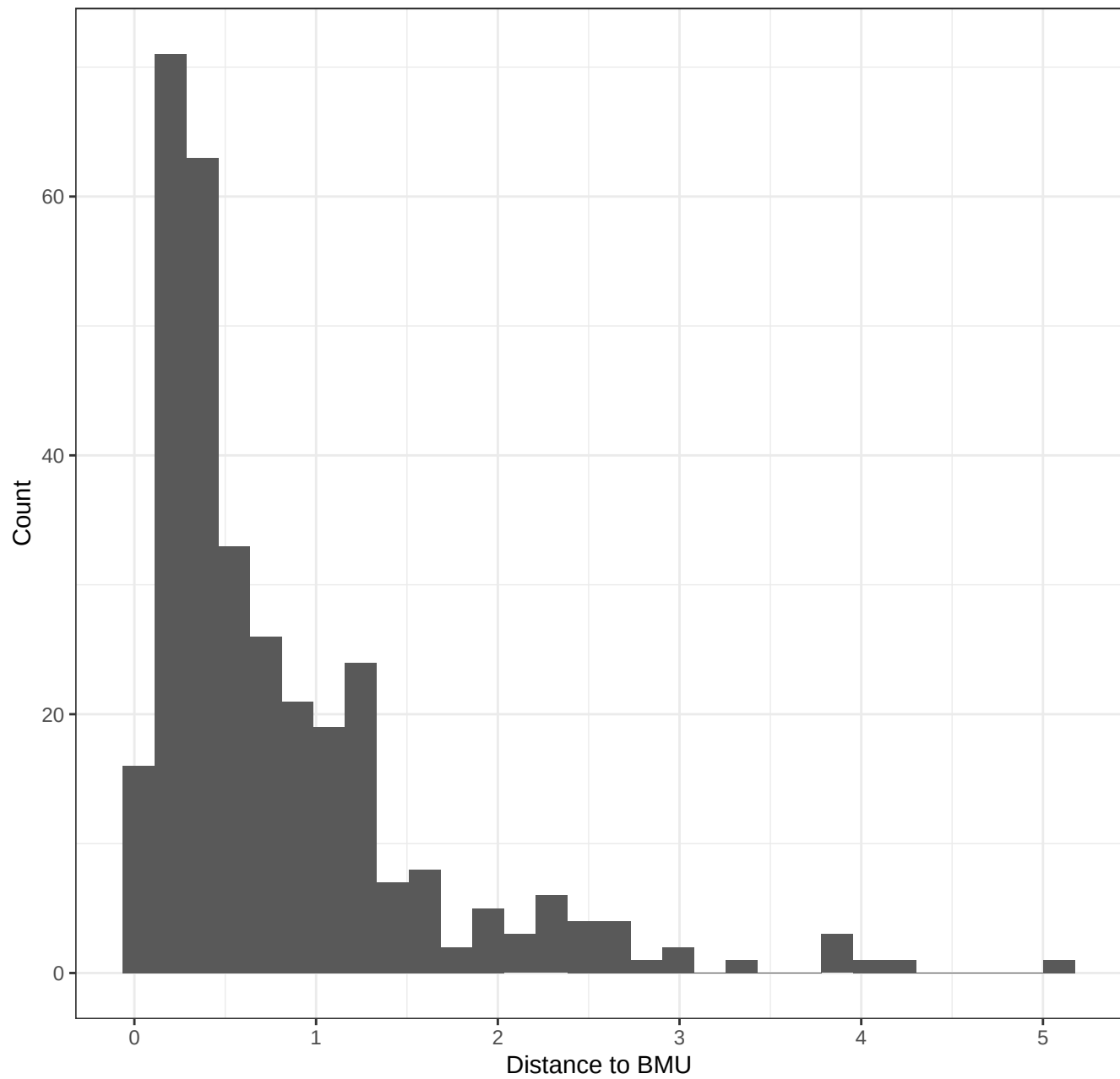
Reference visit: Baseline



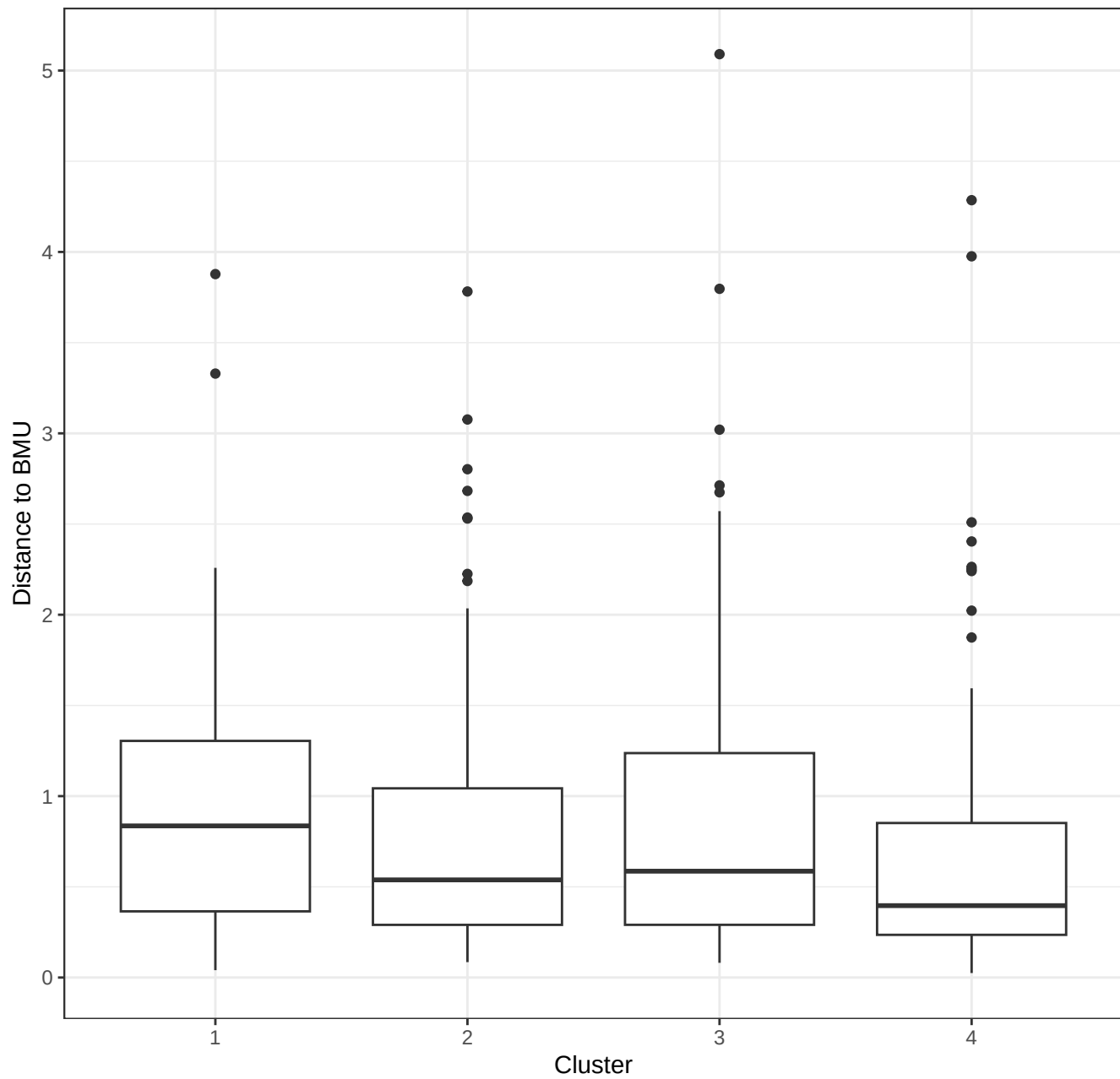
Model 1: Equal variance, cov = 0



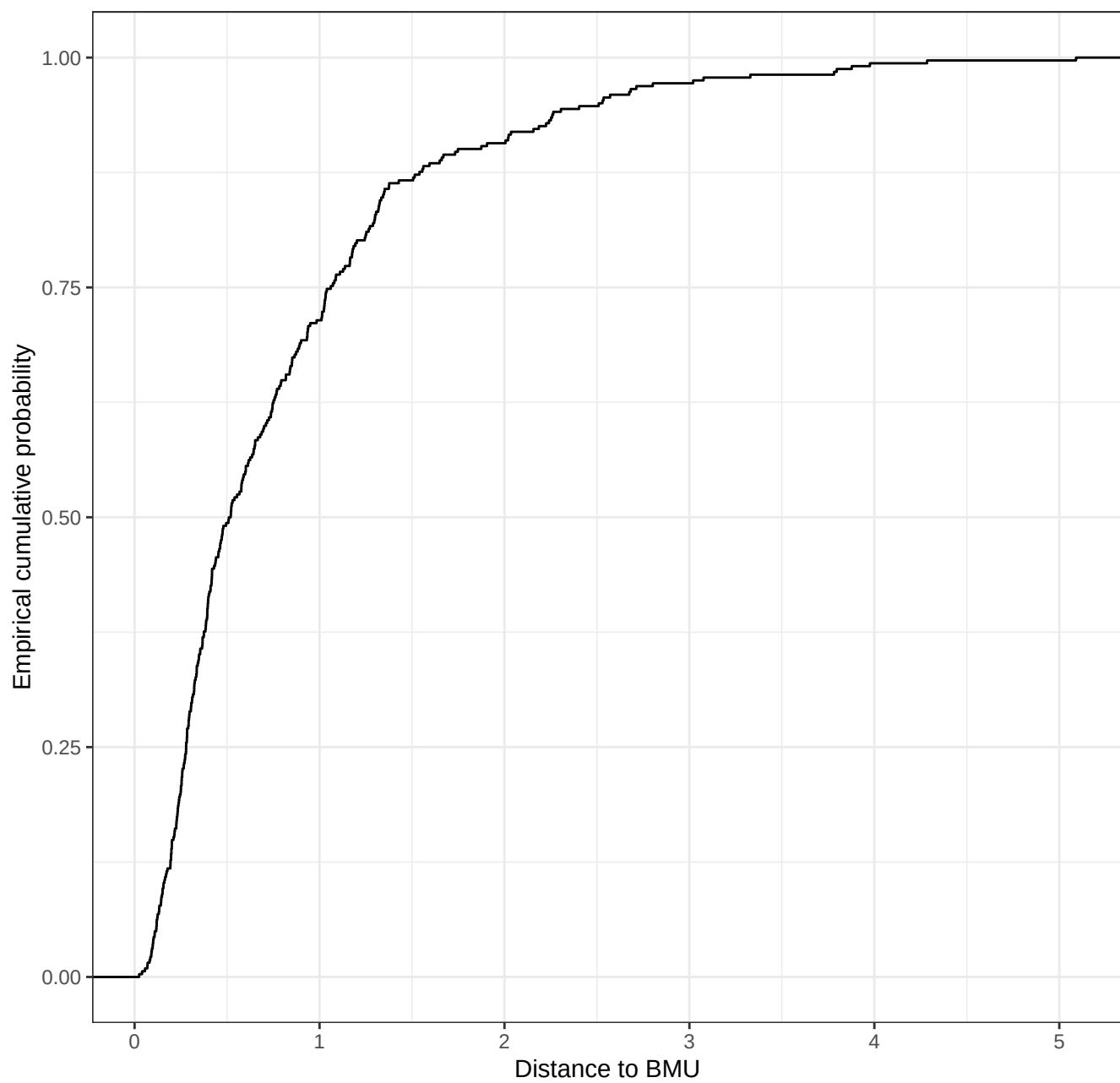
SOM distances (training data)



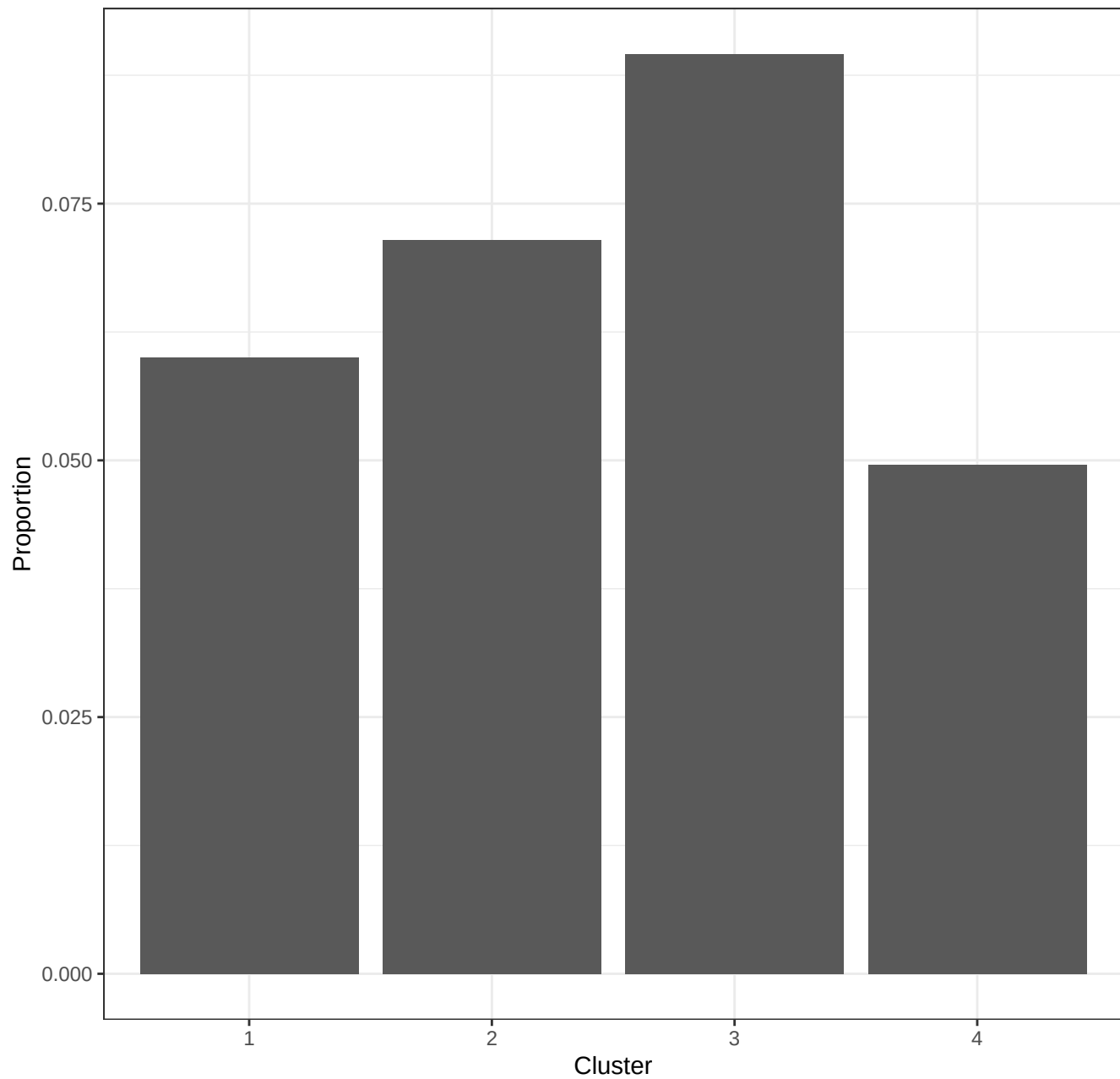
SOM distances by cluster (training data)



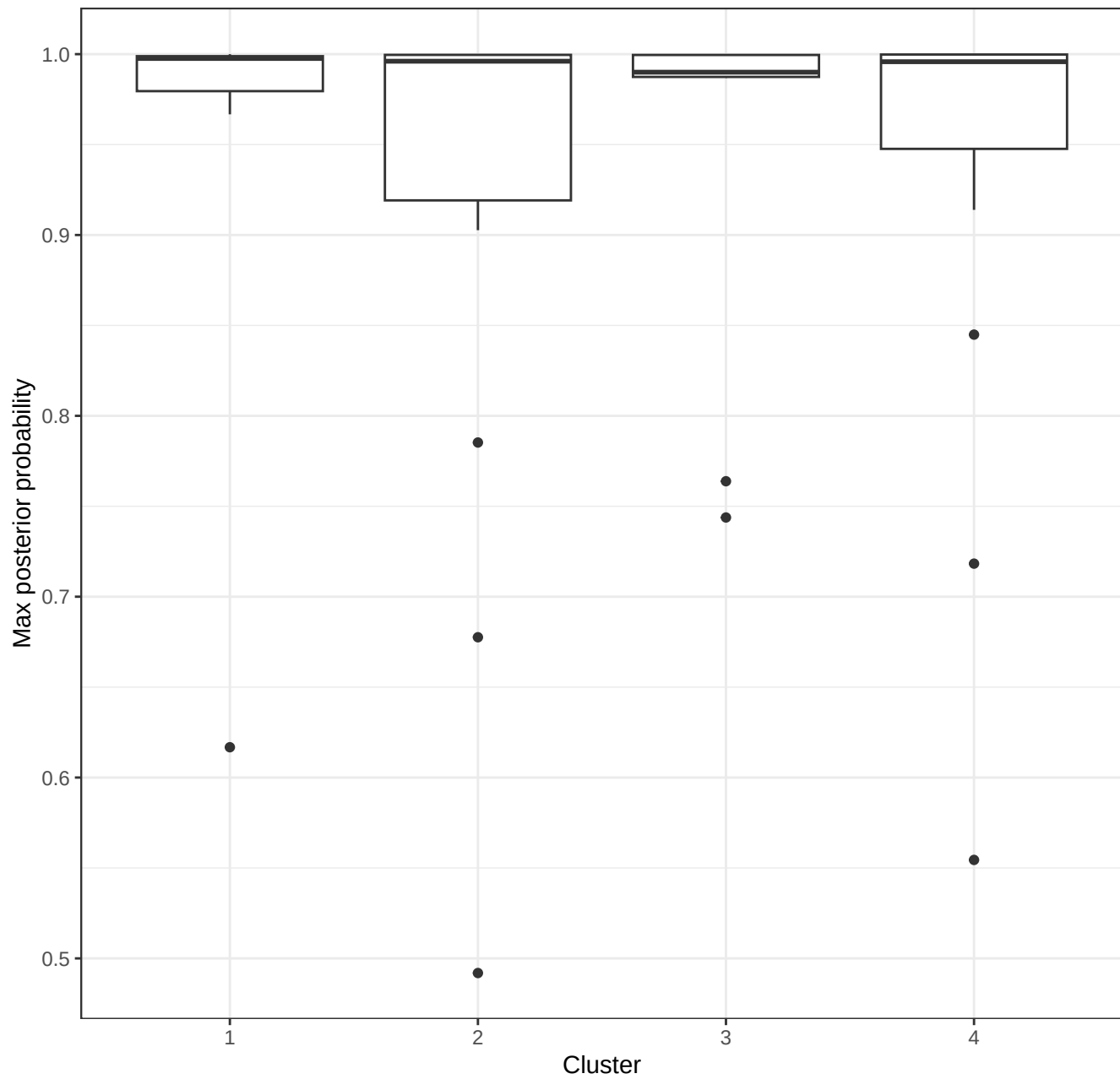
Training SOM distance ECDF



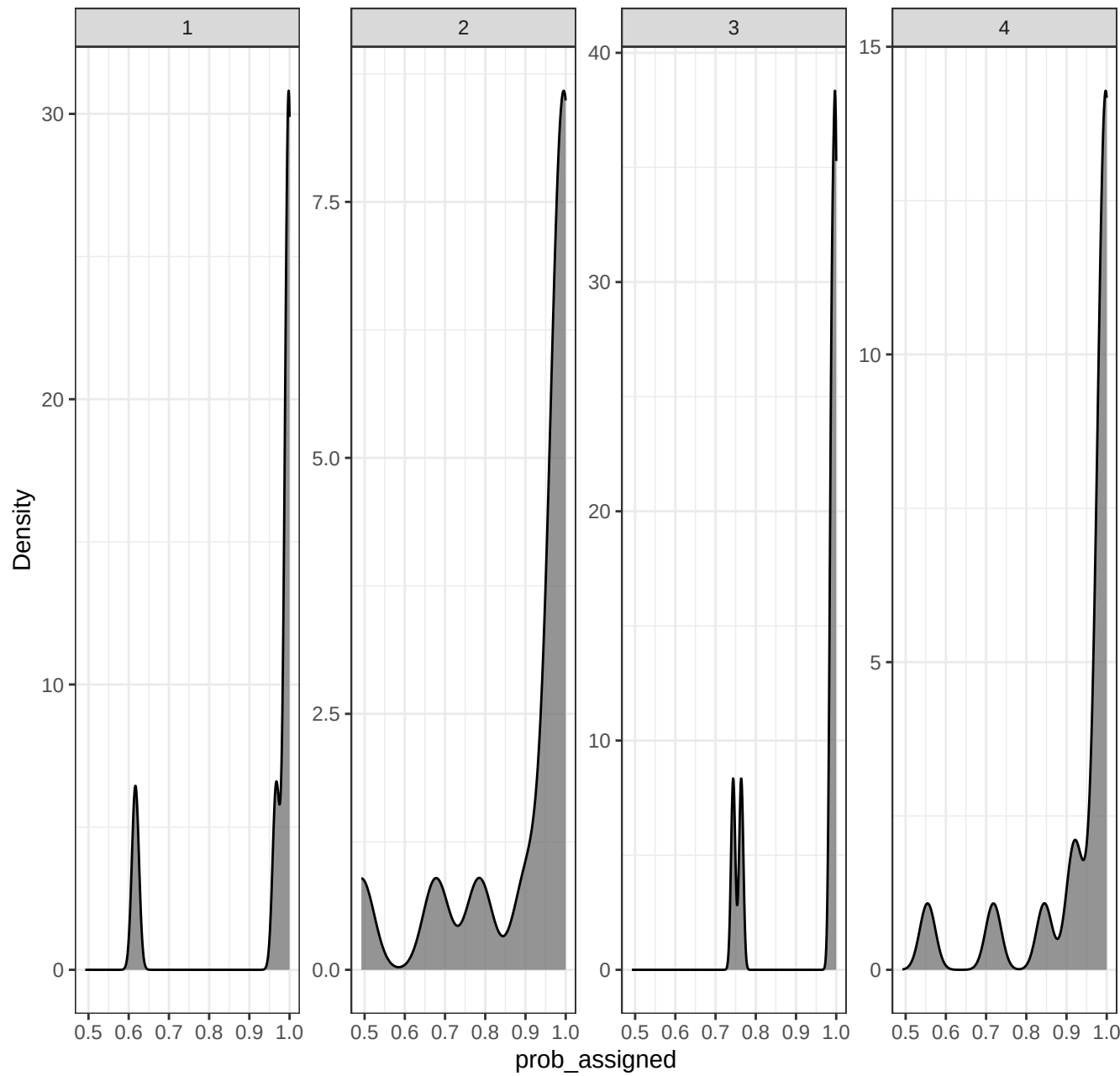
High-distance training cases by cluster



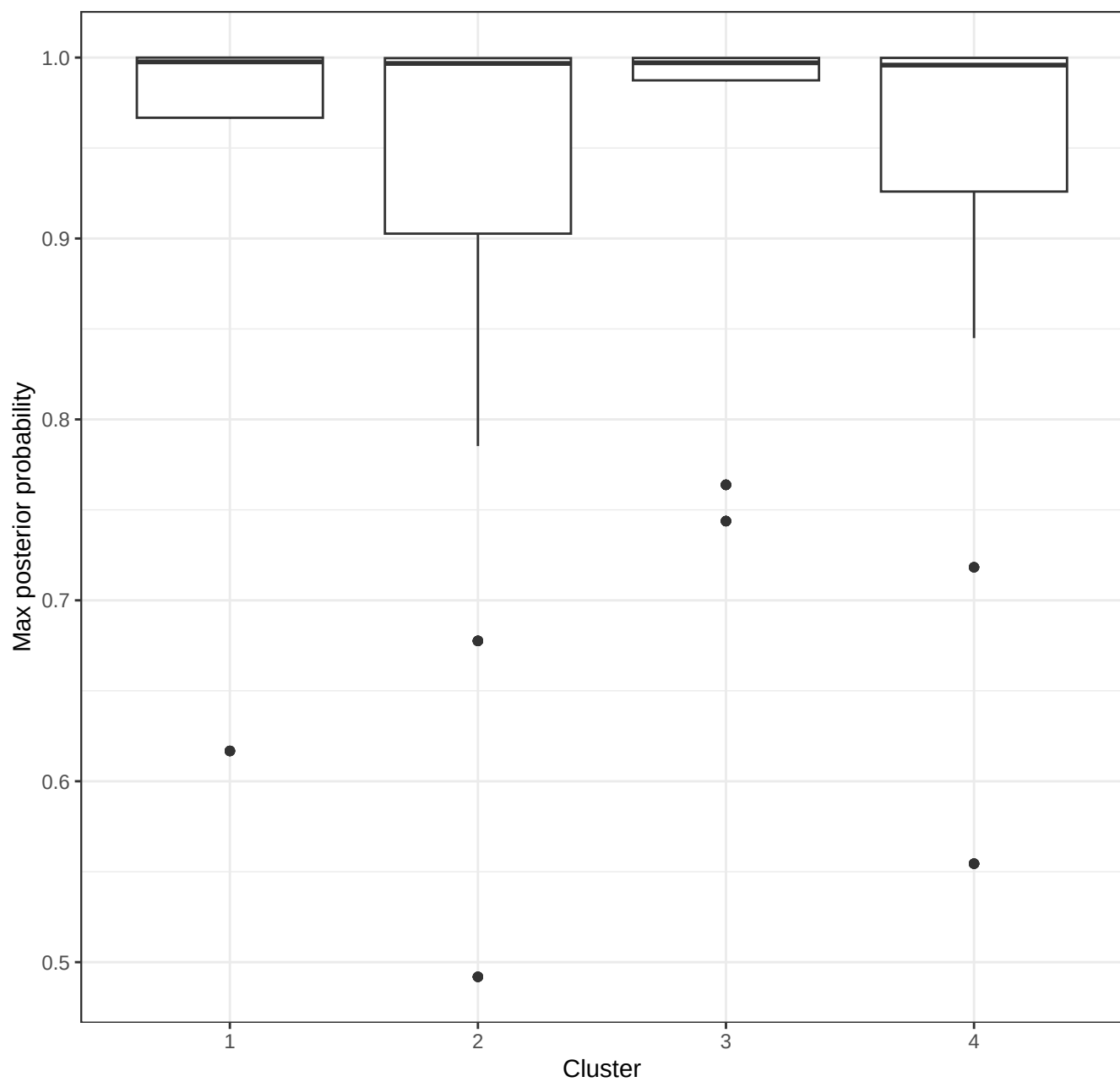
Node-level max posterior probability by class



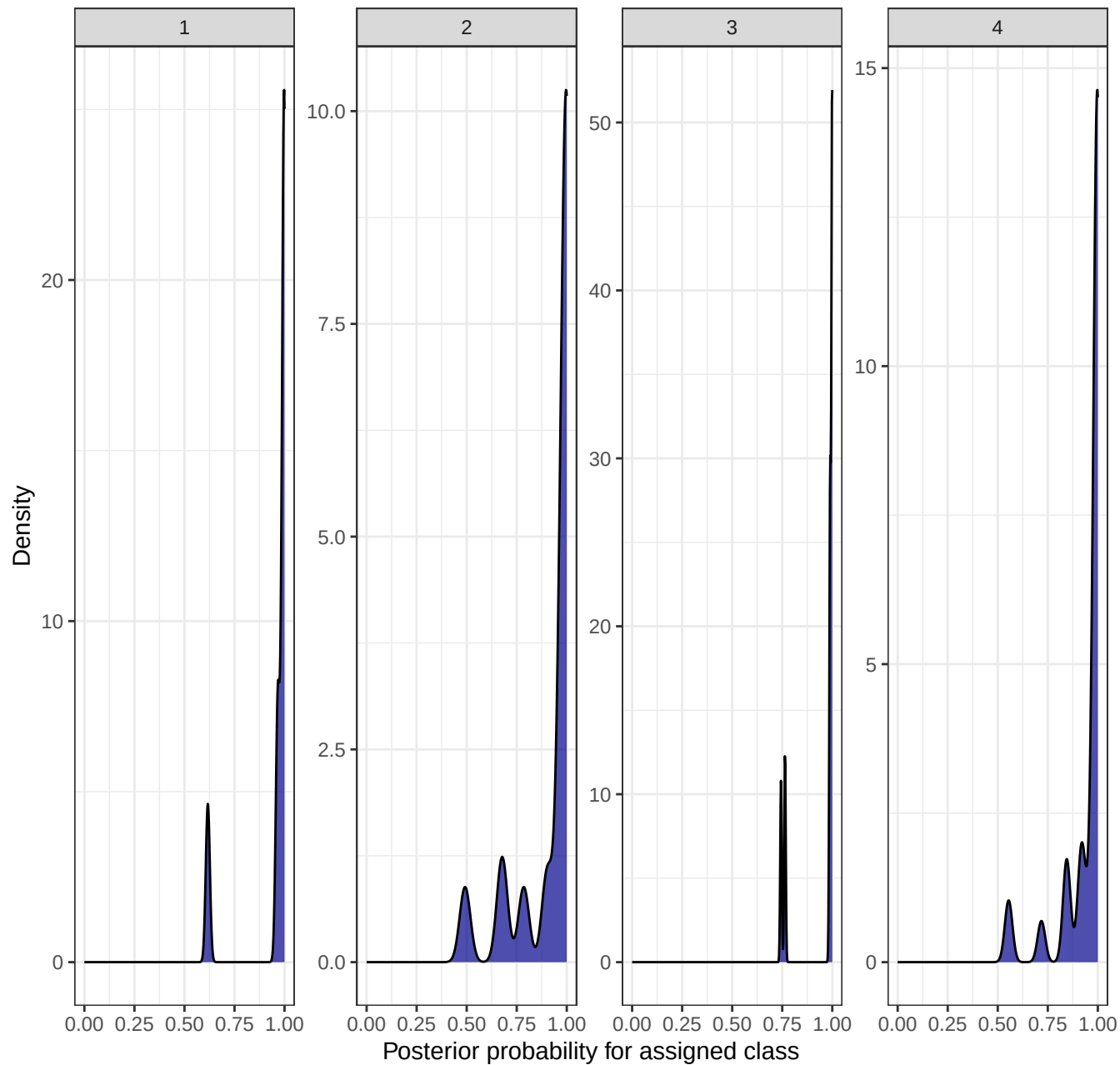
Density of node prob_assigned by class

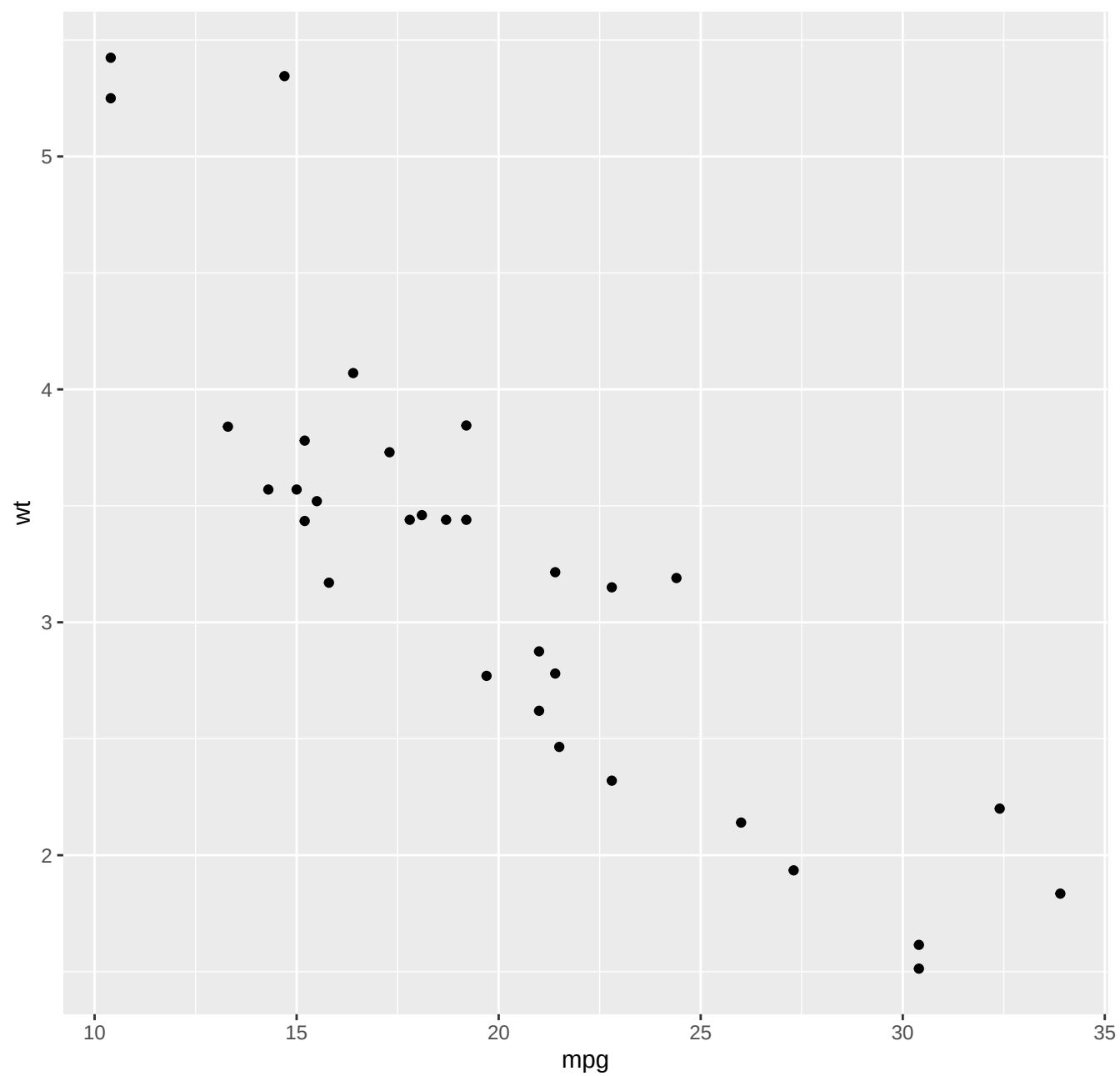


Individual max posterior probability (training)

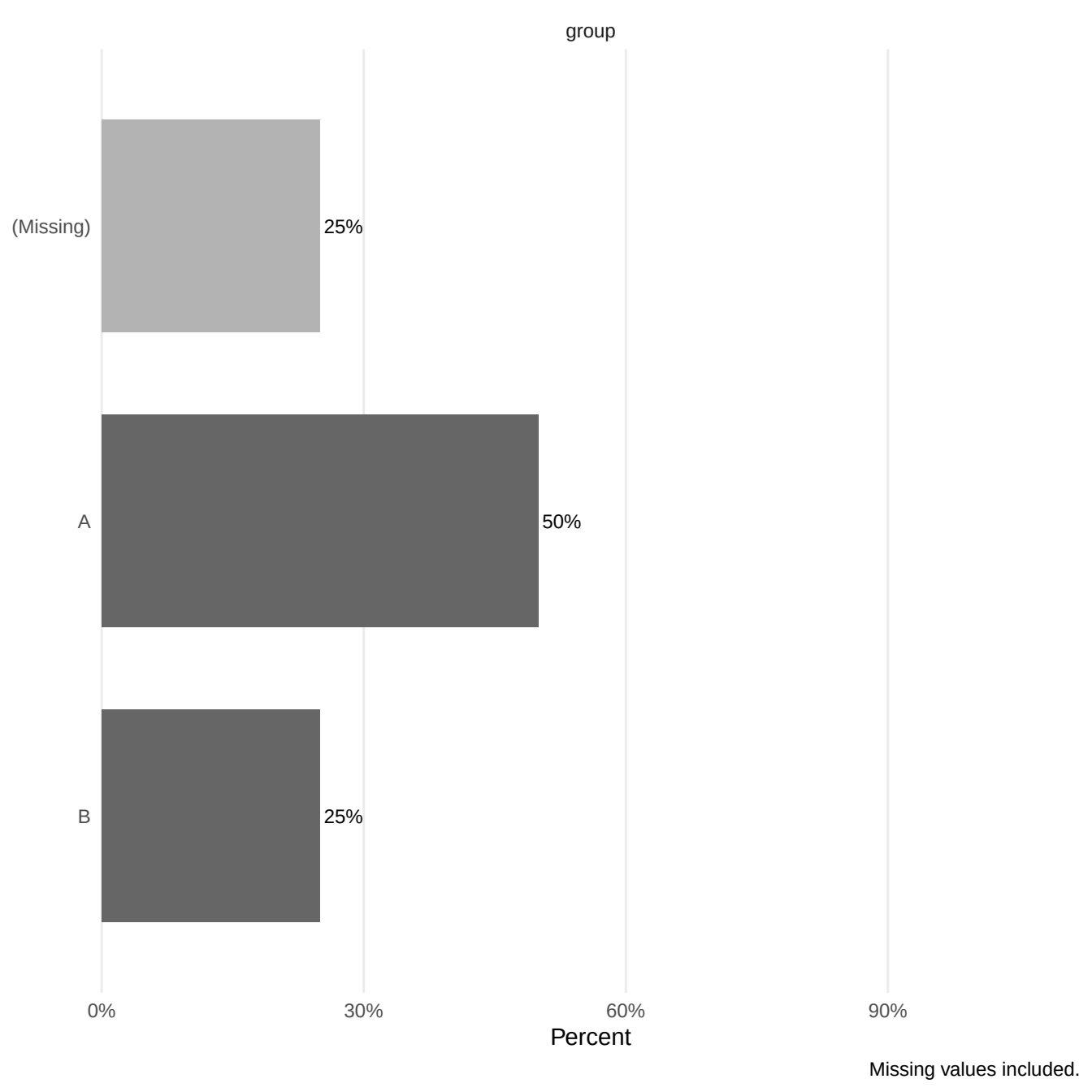


Density of prob_assigned by class (training)

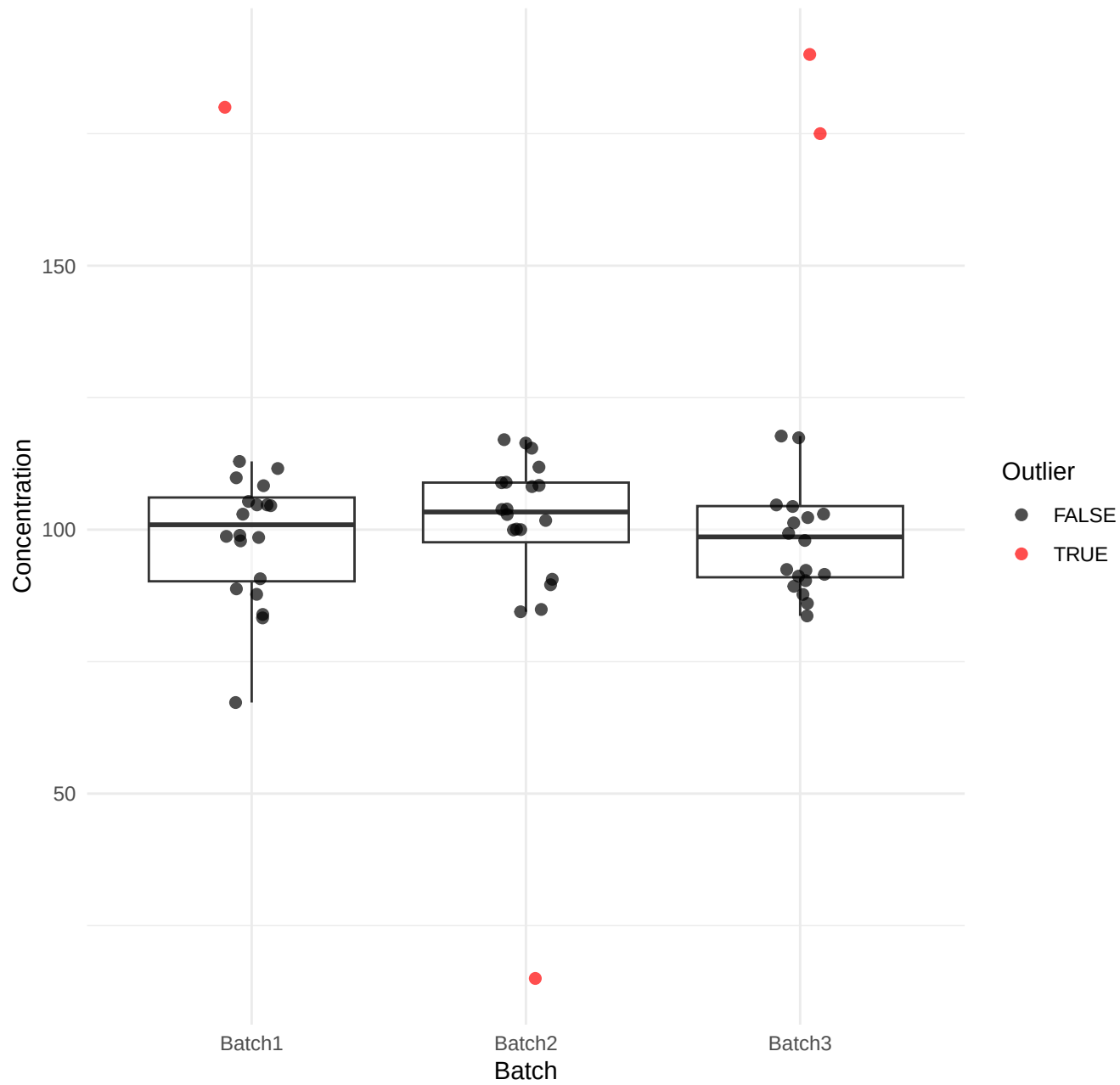




* = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.001$



IQR Outlier Detection for Concentration



AXL receptor tyrosine kinase

Adiponectin

Cortisol

Impaired

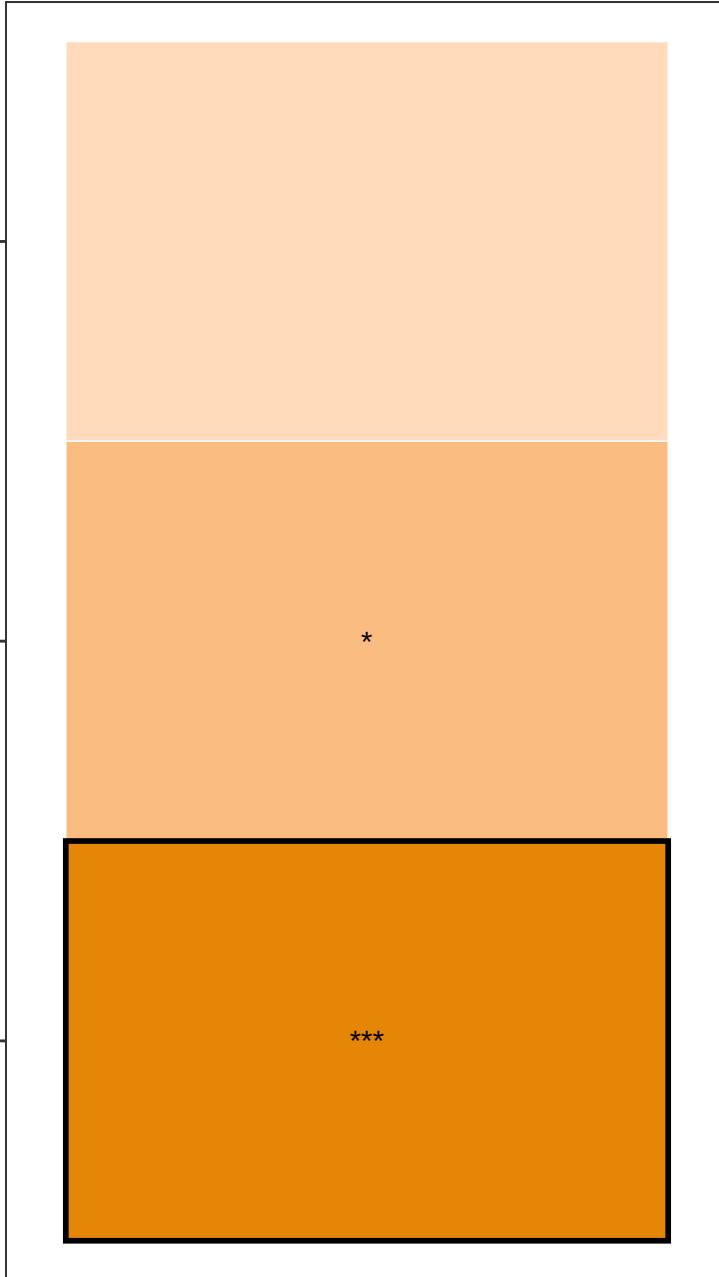
Referent-scaled
Z-score

0.25

0.00

-0.25

*



$p < 0.001$

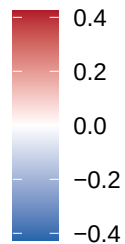
Adiponectin

Alpha-1-antitrypsin

Alpha-2-macroglobulin

AXL receptor tyrosine kinase

Standardized beta



p < 0.001

Adiponectin

Alpha-1-antitrypsin

Alpha-2-macroglobulin

Partial R2

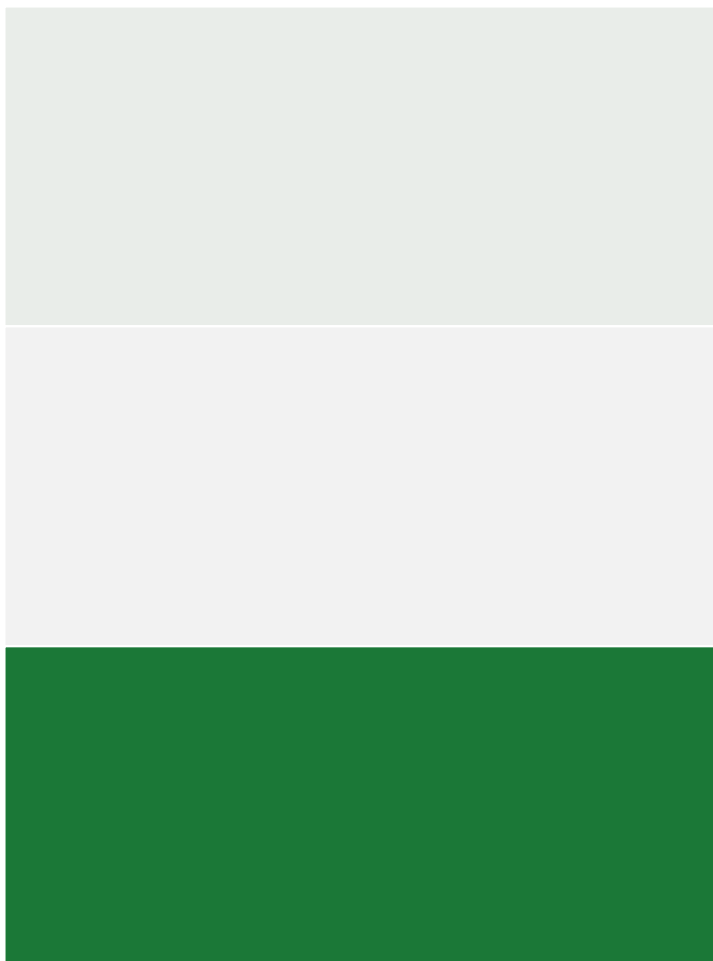
0.12

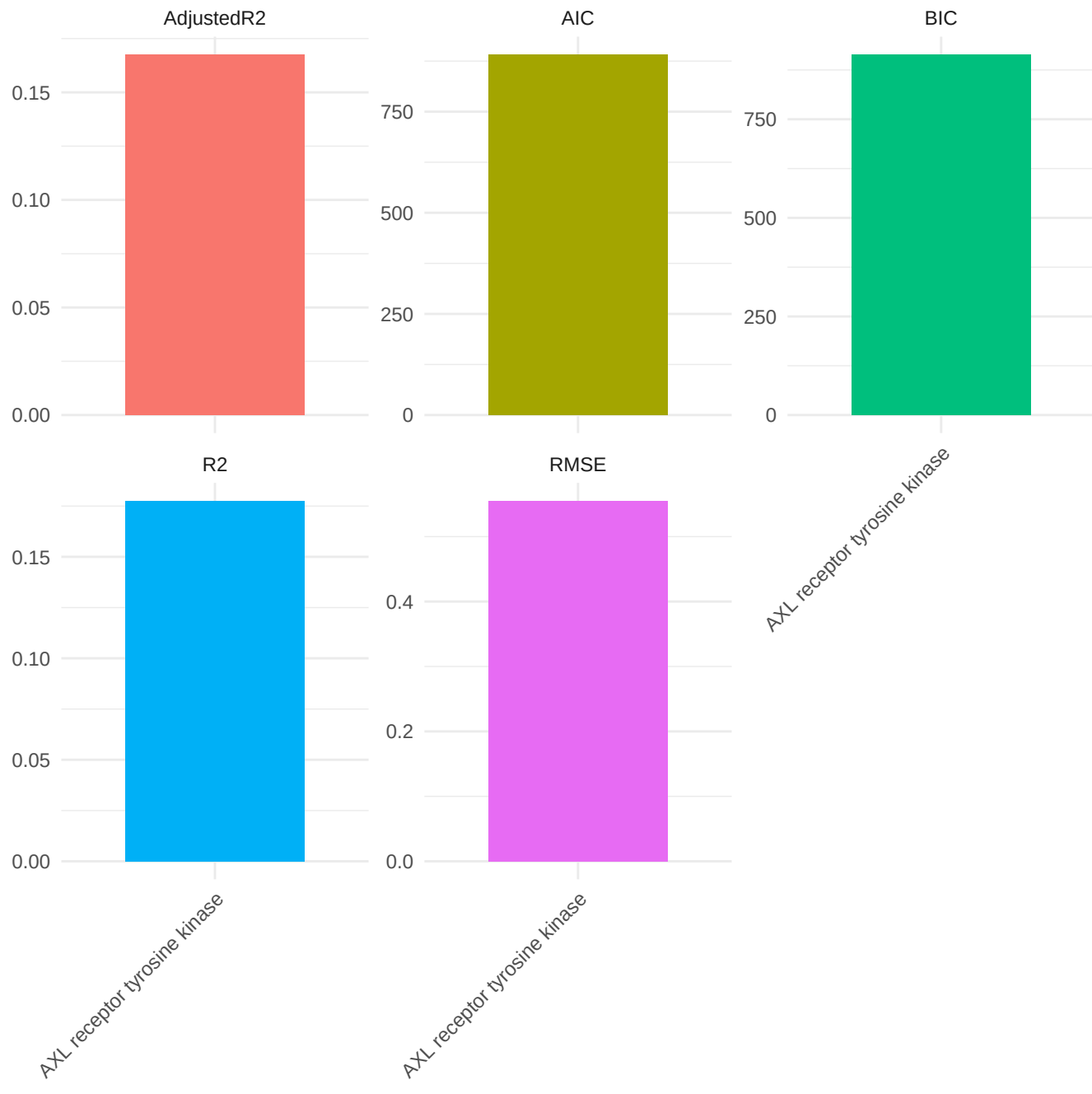
0.08

0.04

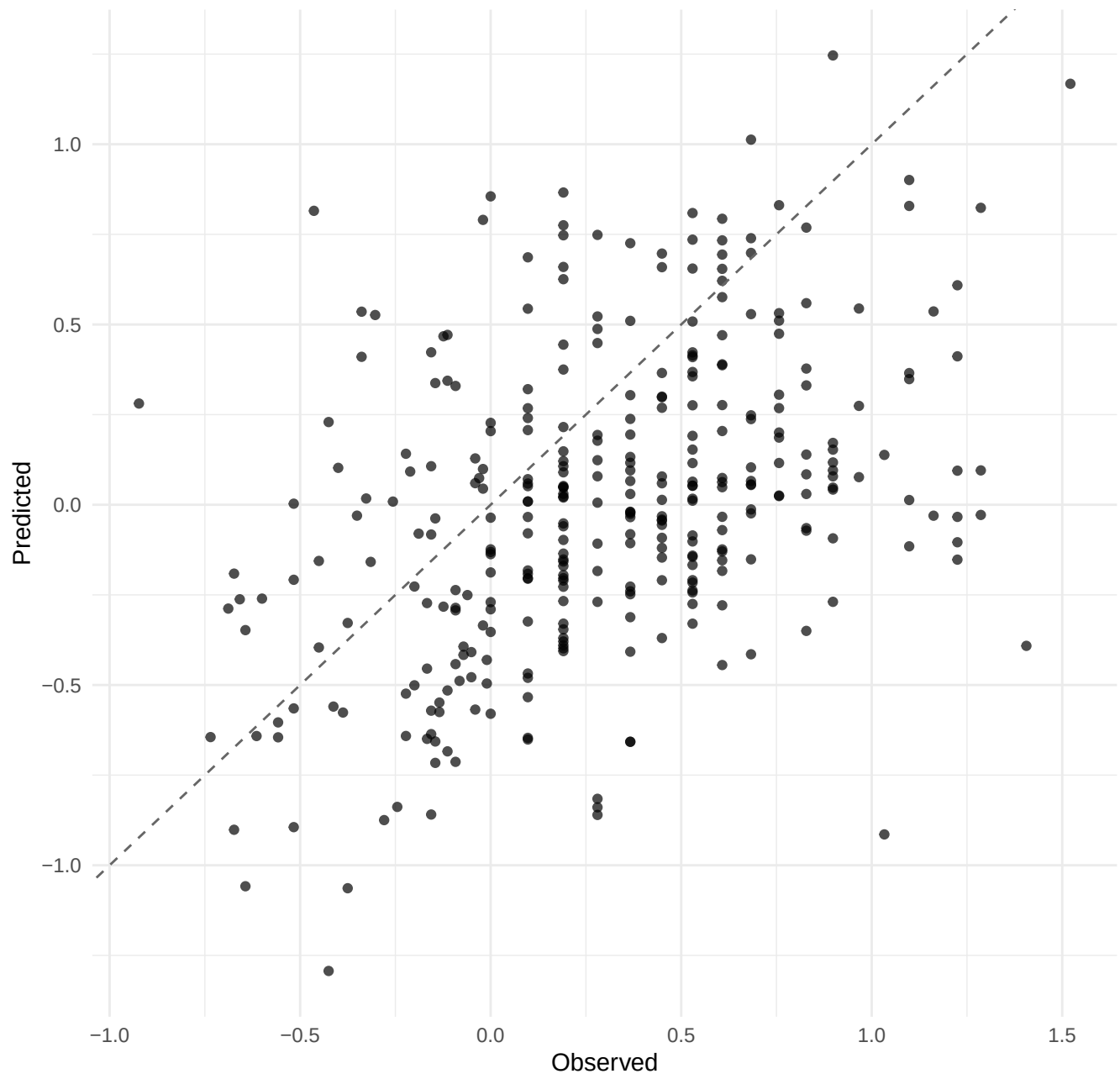
0.00

AXL receptor tyrosine kinase

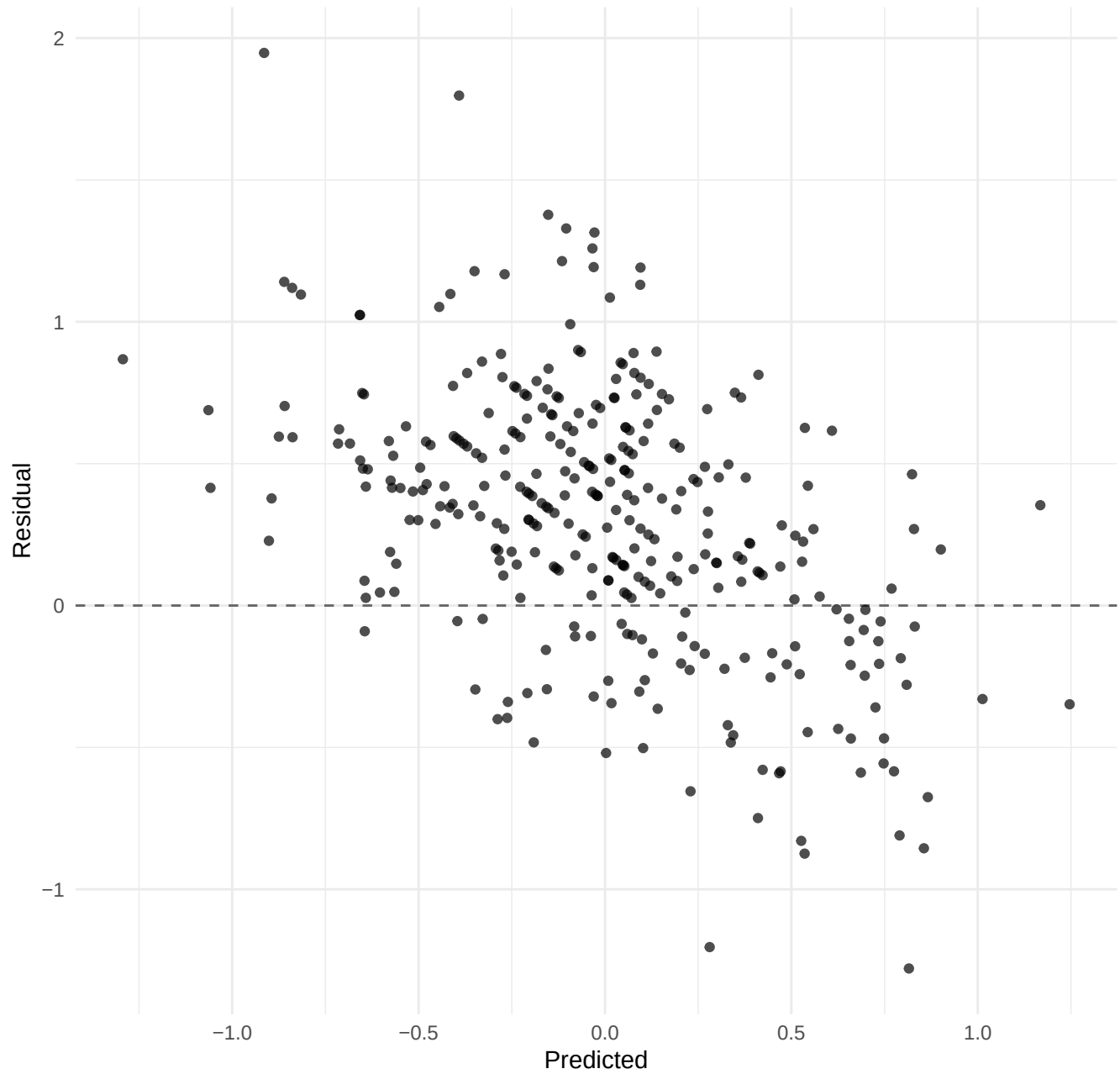




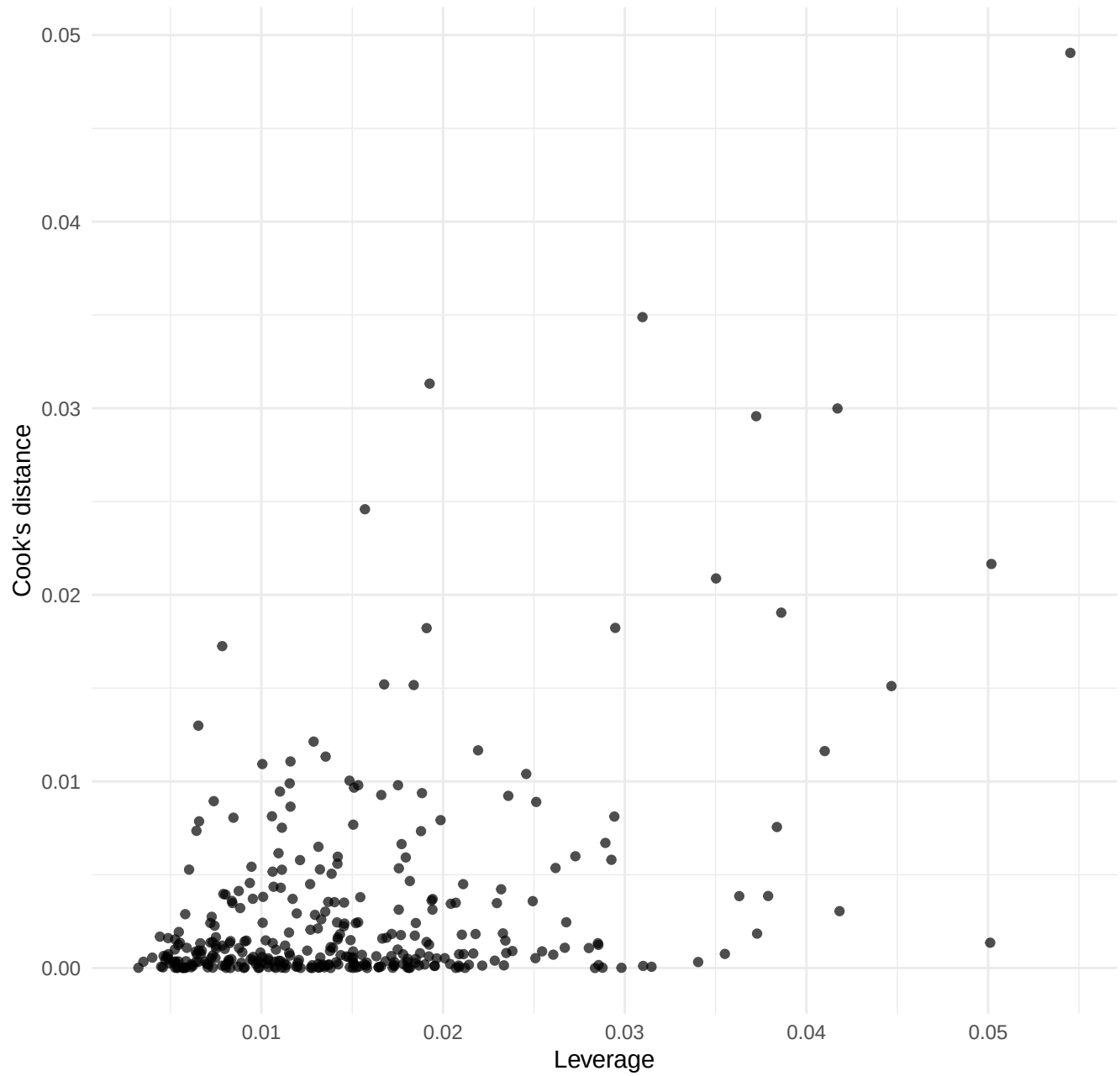
AXL



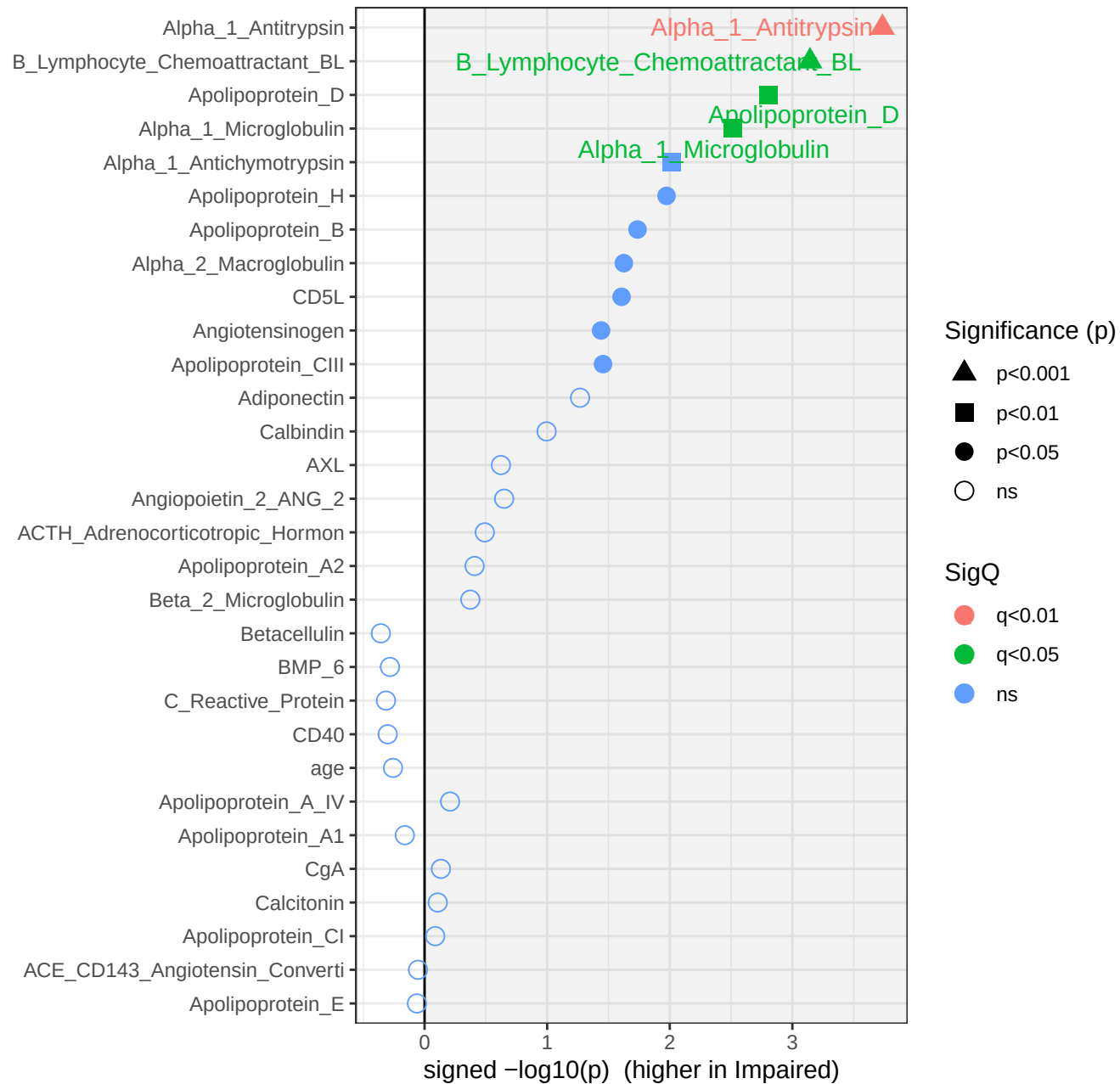
AXL



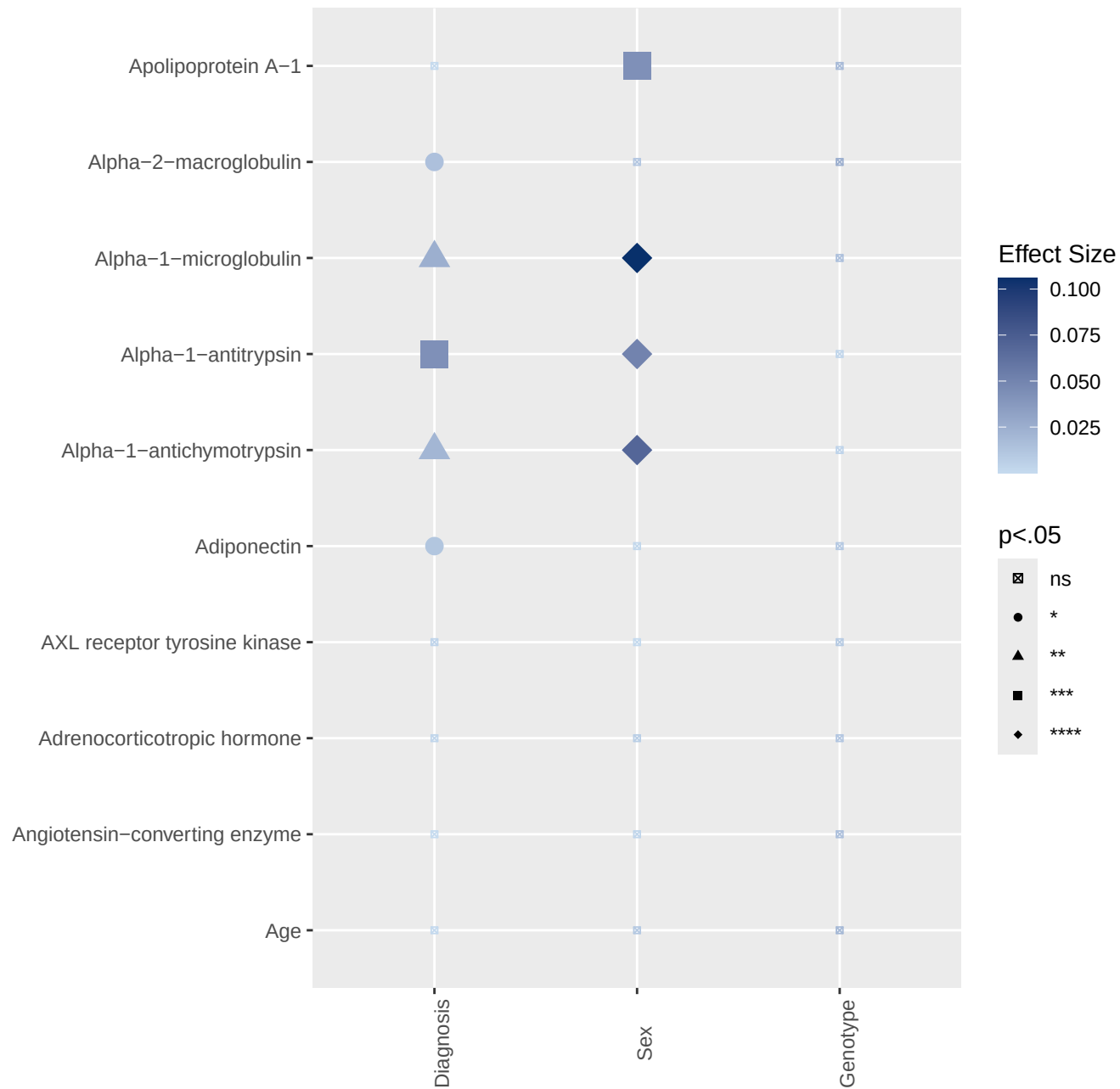
AXL



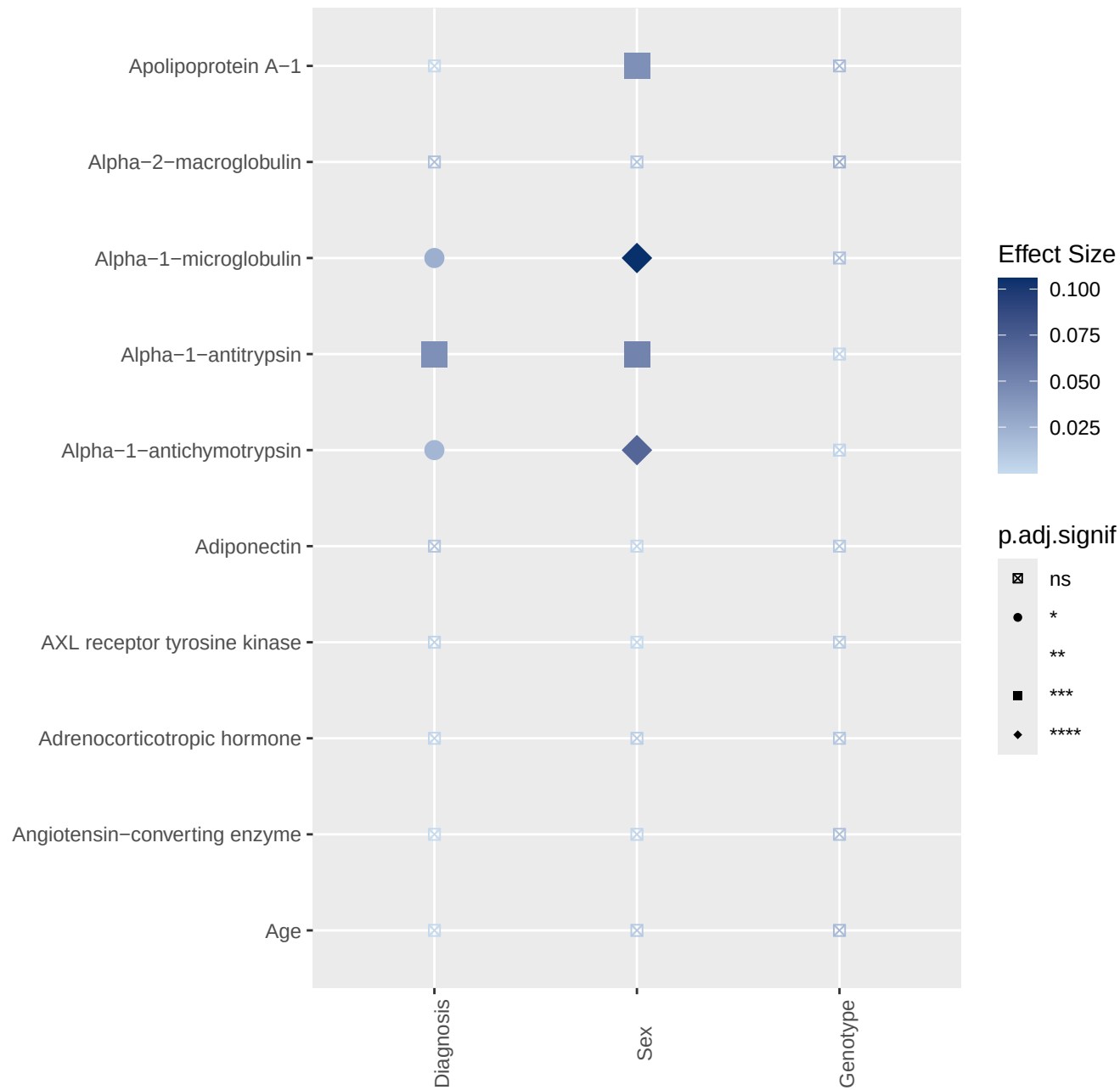
---> Higher in Impaired

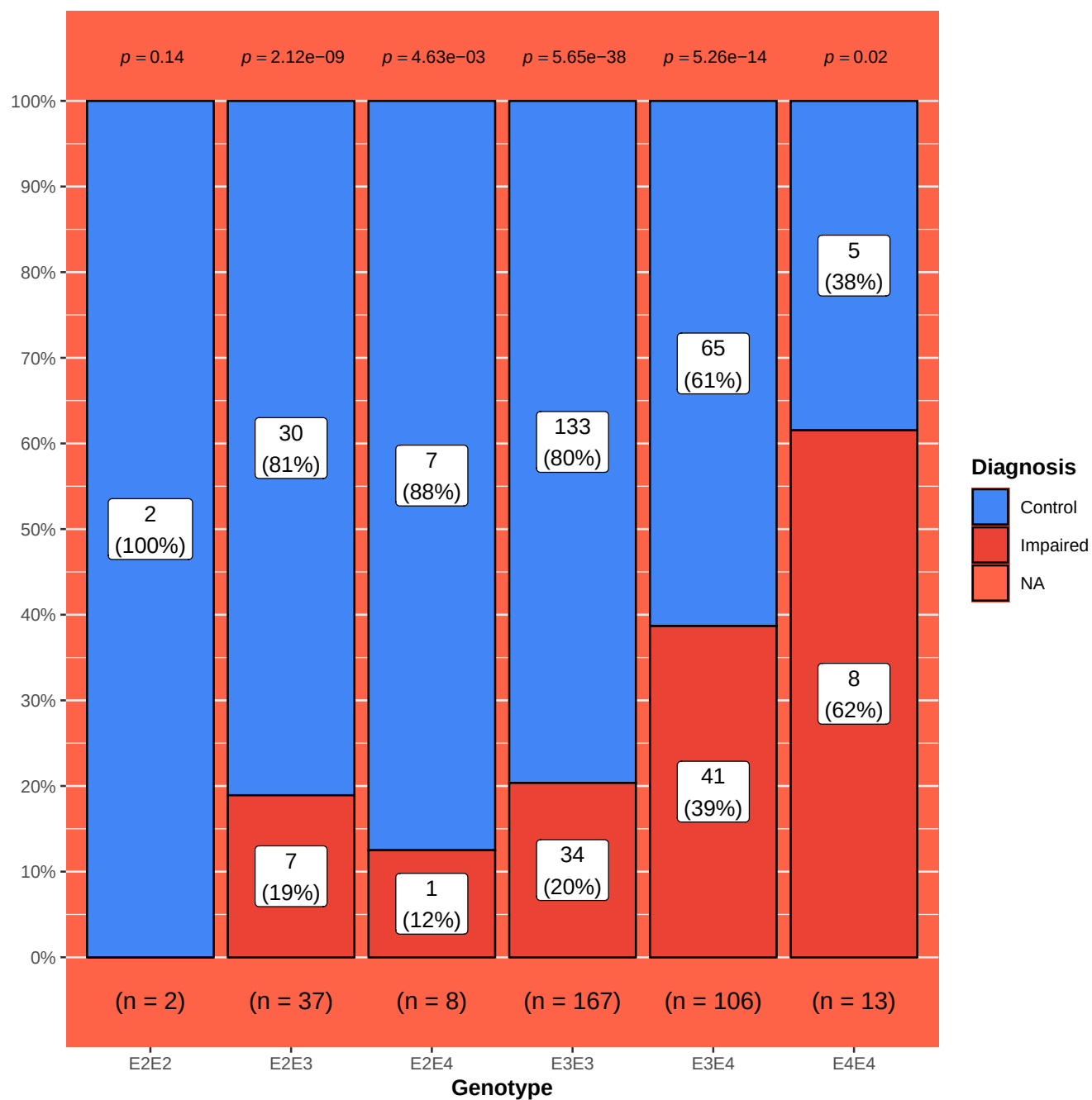


No Multiple Comparison Correction

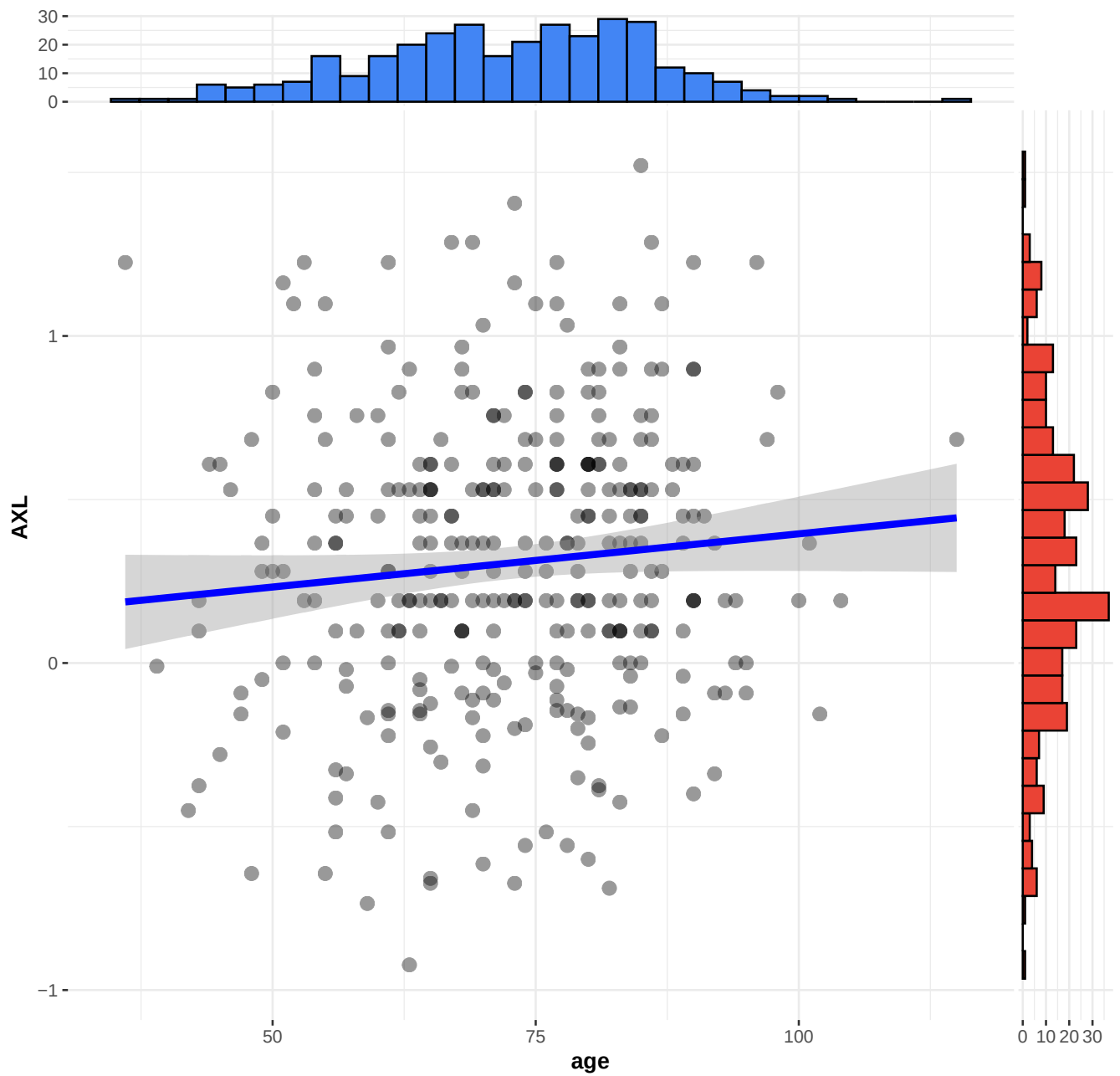


FDR Correction

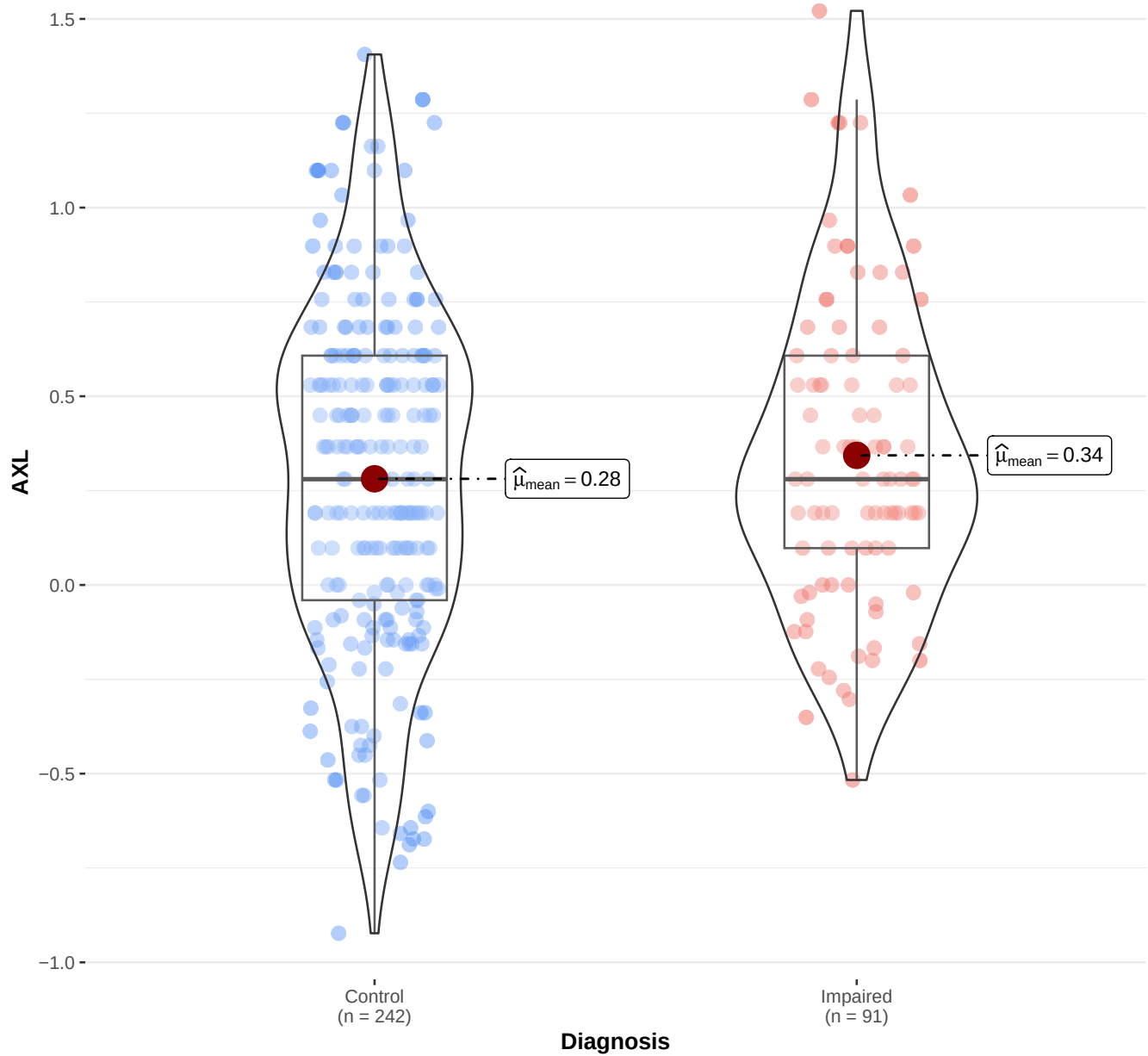




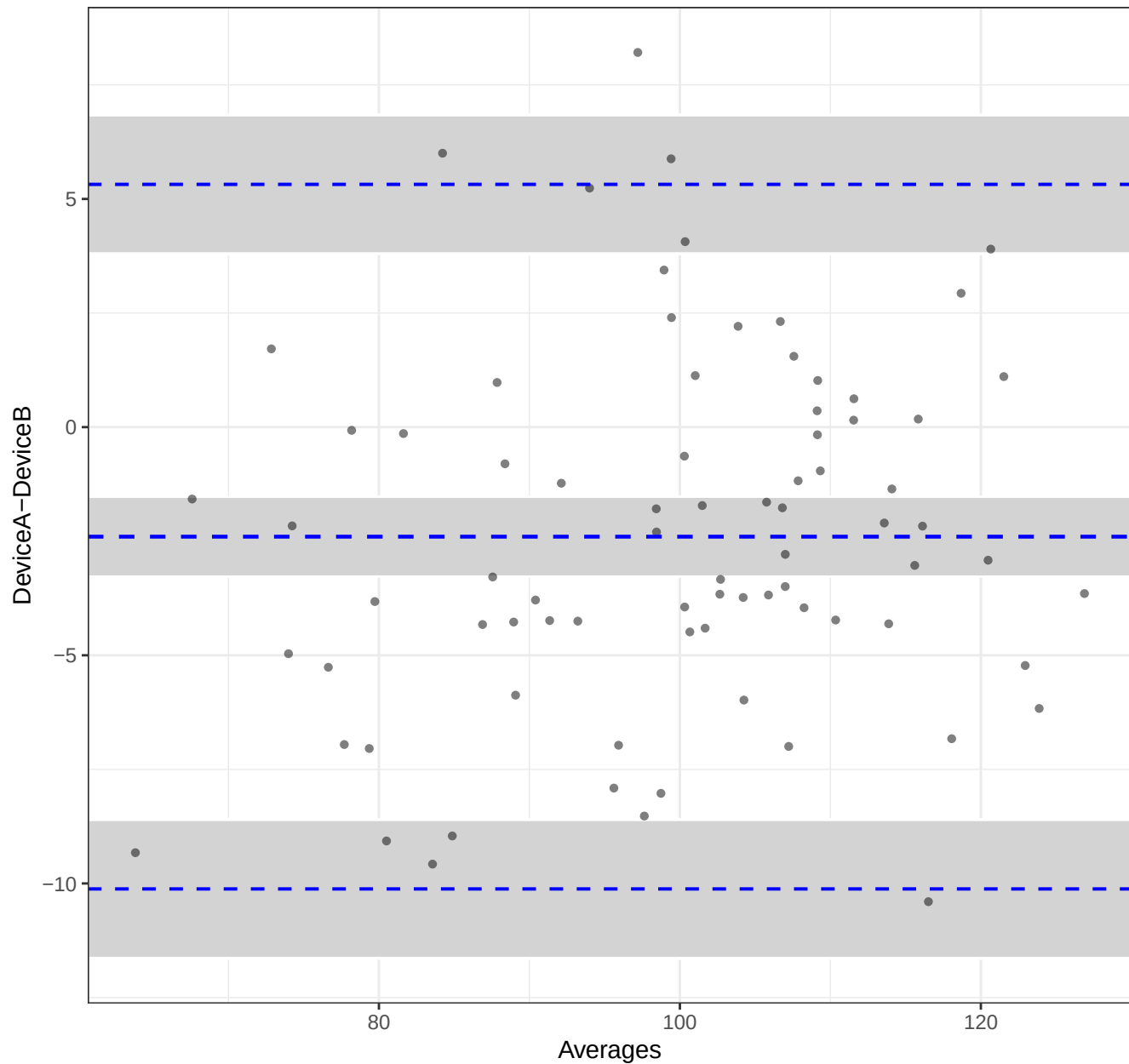
$t_{\text{Student}}(320) = 1.72, p = 0.09, \hat{r}_{\text{Pearson}} = 0.10, \text{CI}_{95\%} [-0.01, 0.20], n_{\text{pairs}} = 322$



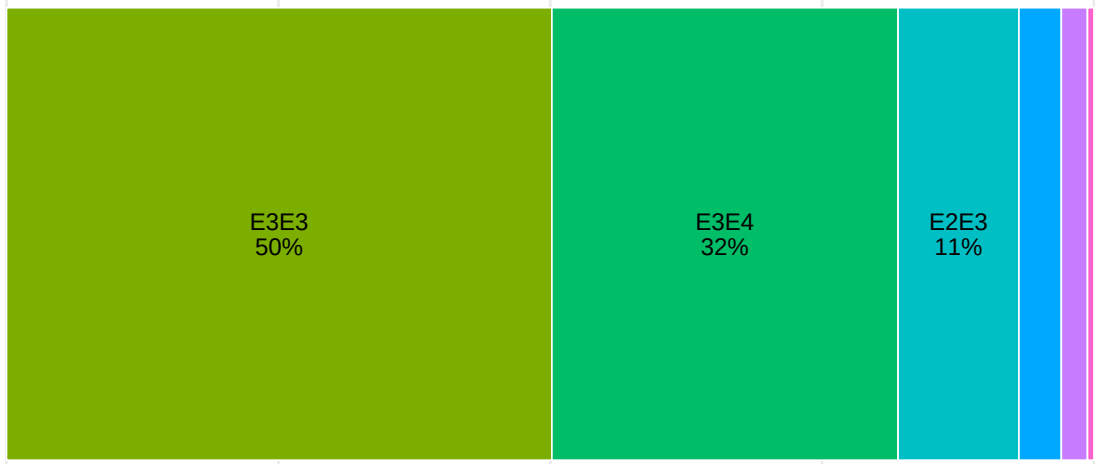
$t_{\text{Welch}}(179.77) = -1.18, p = 0.24, \hat{g}_{\text{Hedges}} = -0.14, \text{CI}_{95\%} [-0.38, 0.09], n_{\text{obs}} = 333$



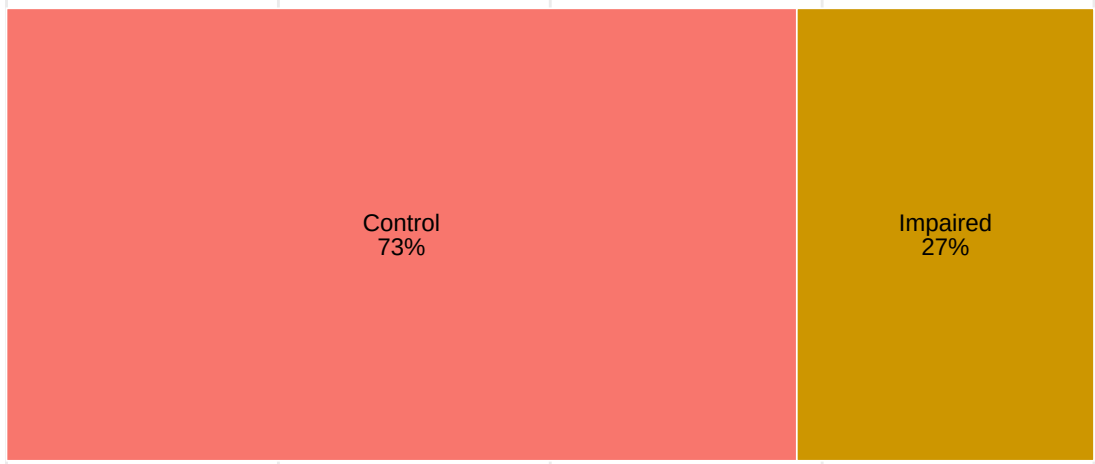
Mean Difference = -2.4
Limits of Agreement: [-10.12 , 5.32]



Genotype

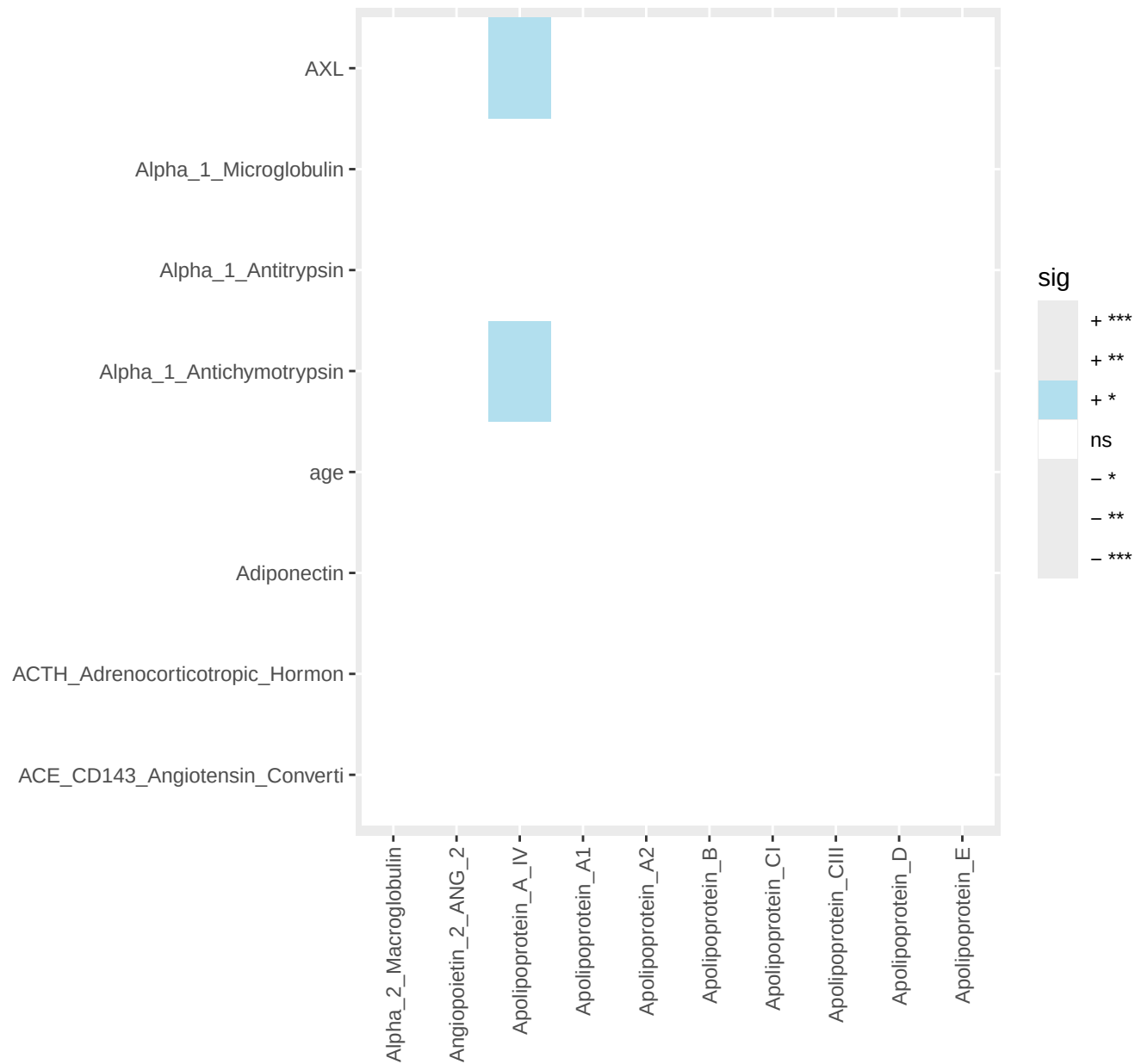


Diagnosis

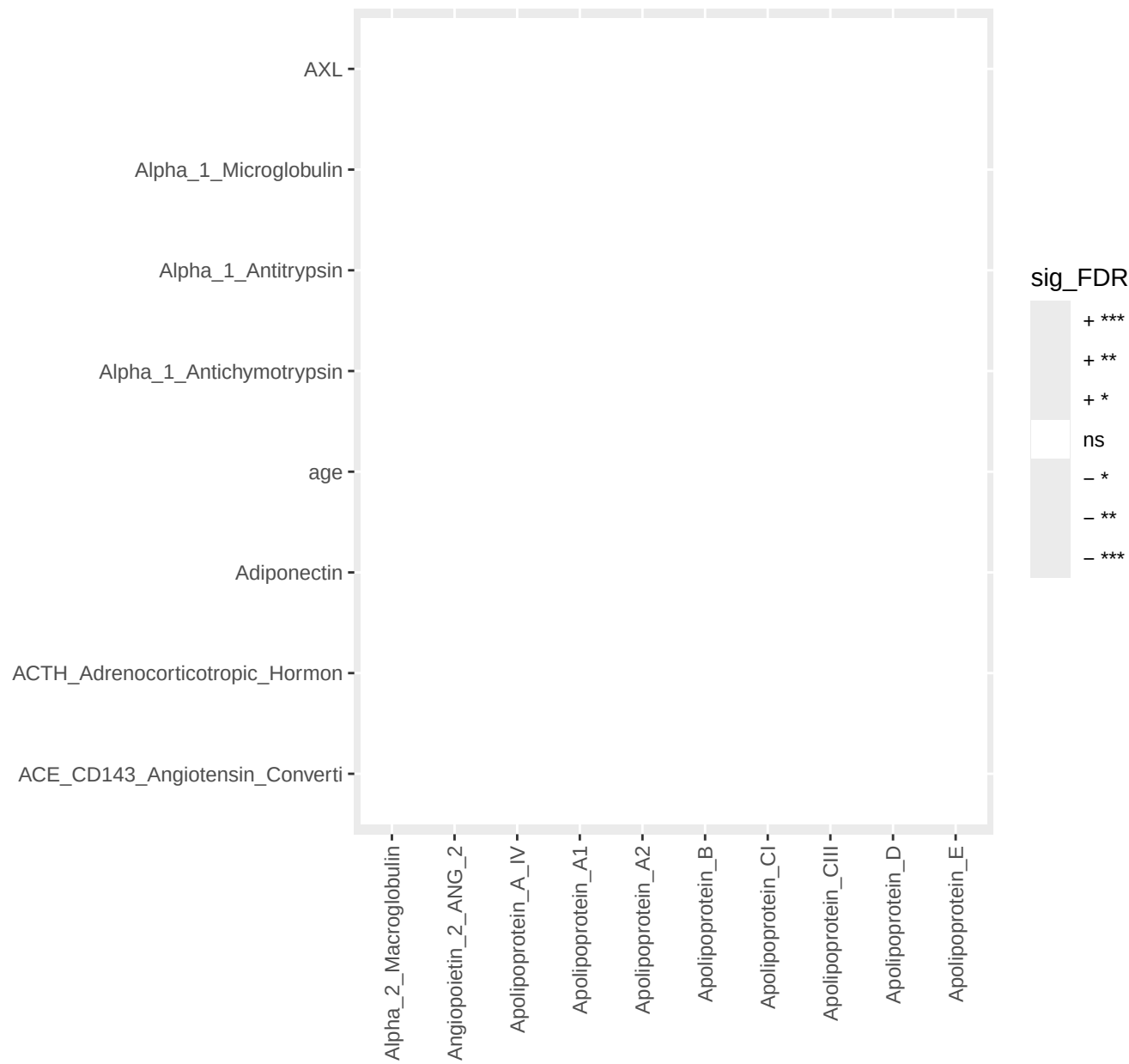


Diagnosis

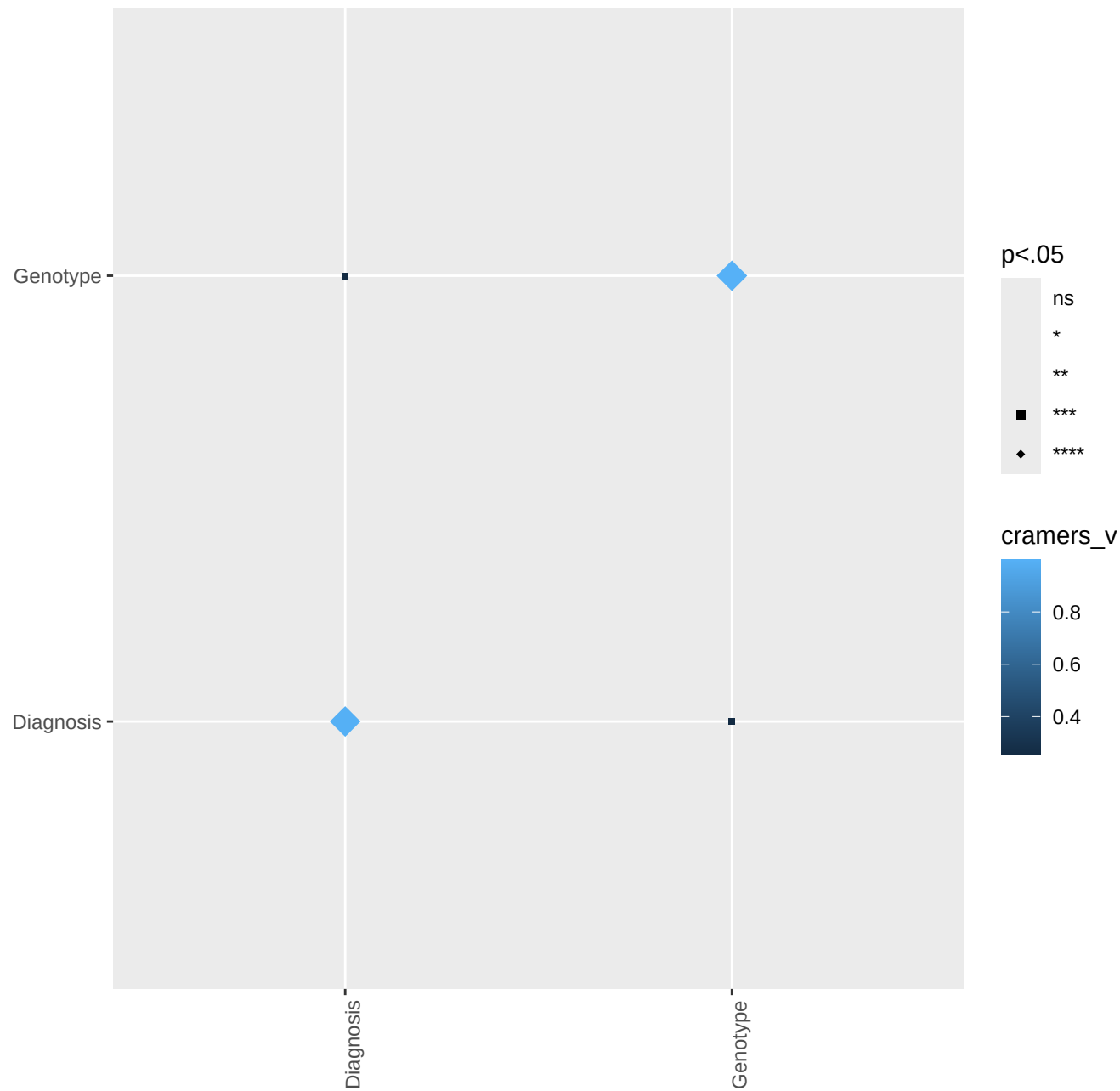
No Multiple Comparison Correction



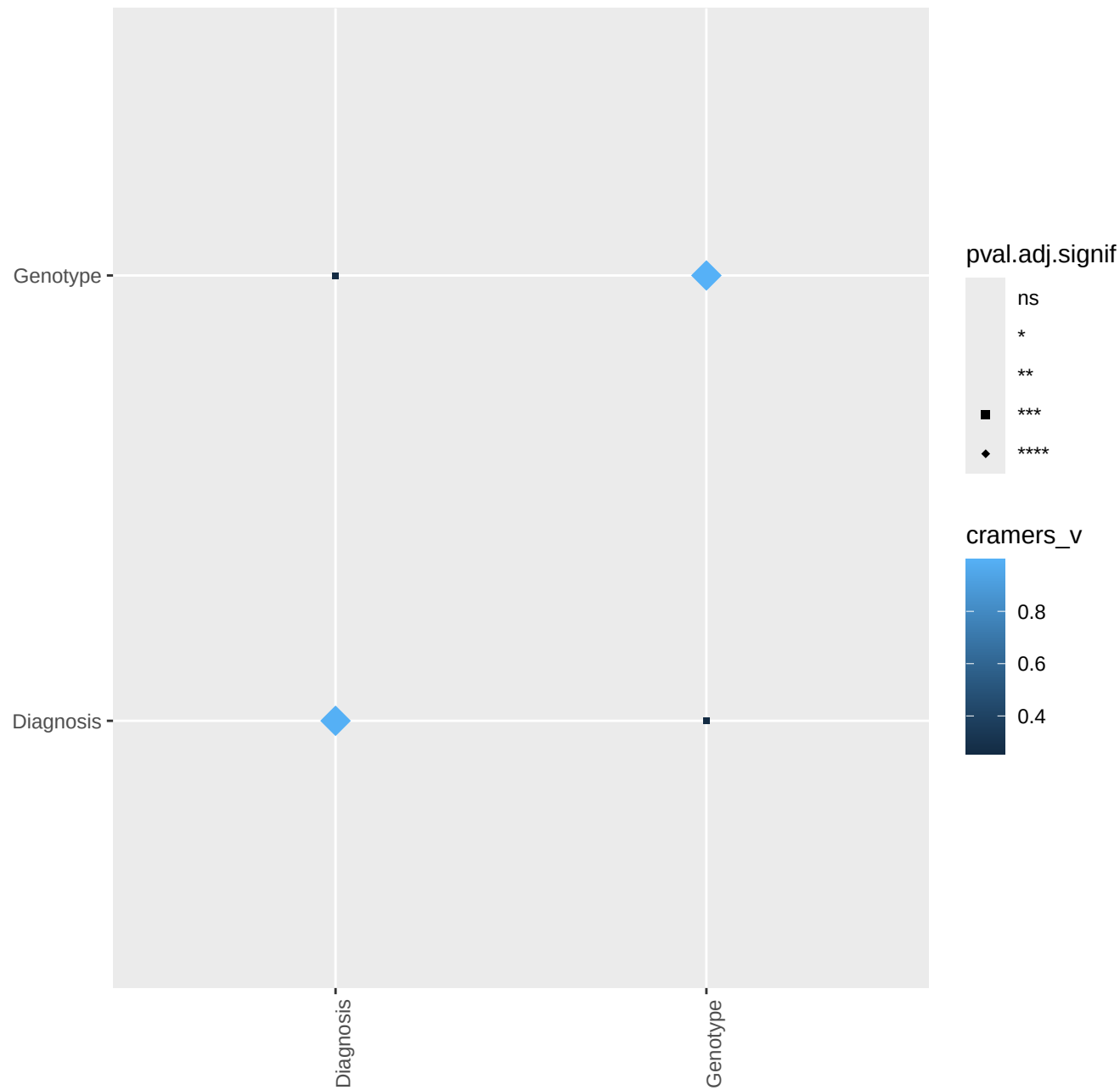
FDR Correction

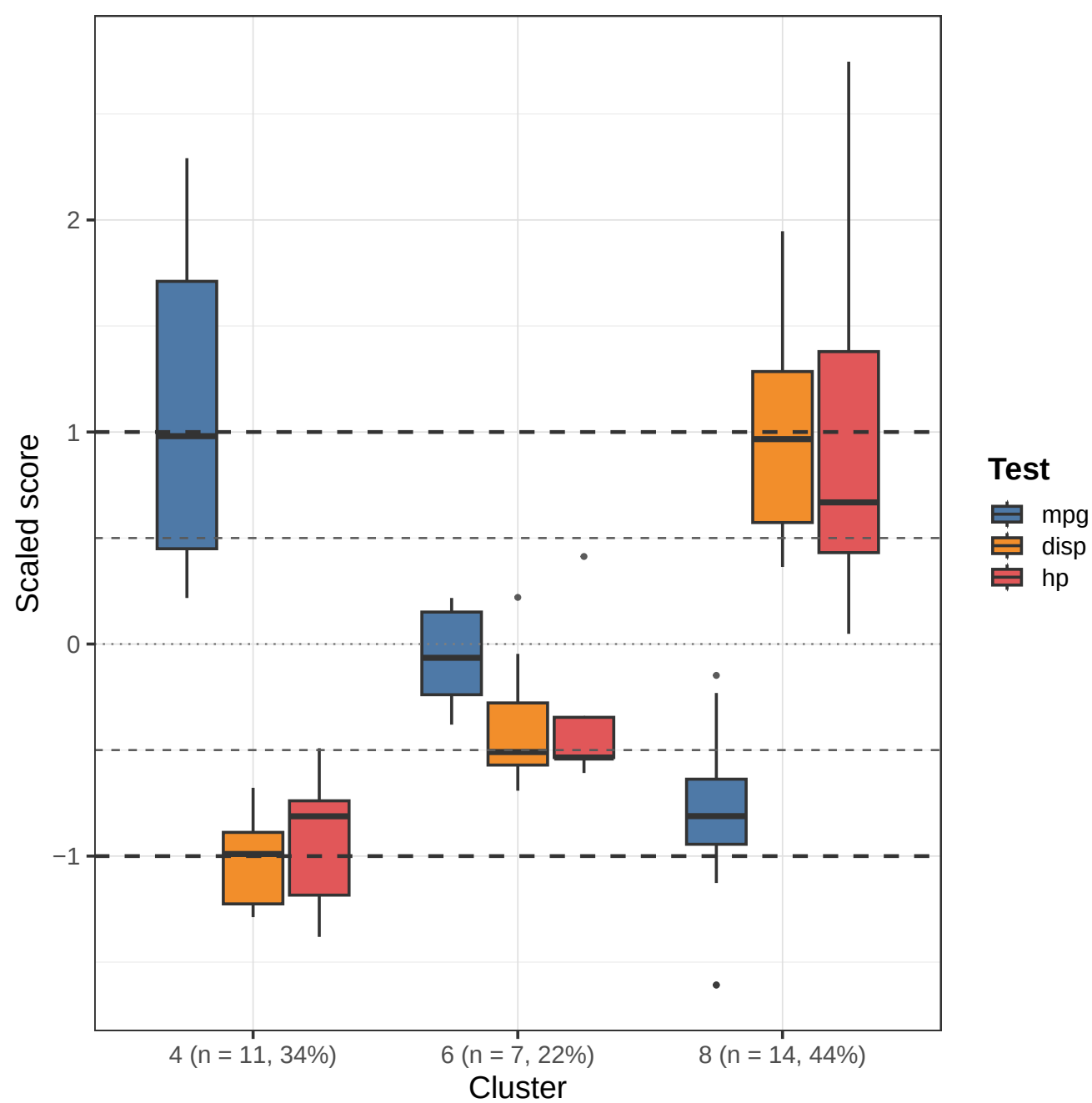


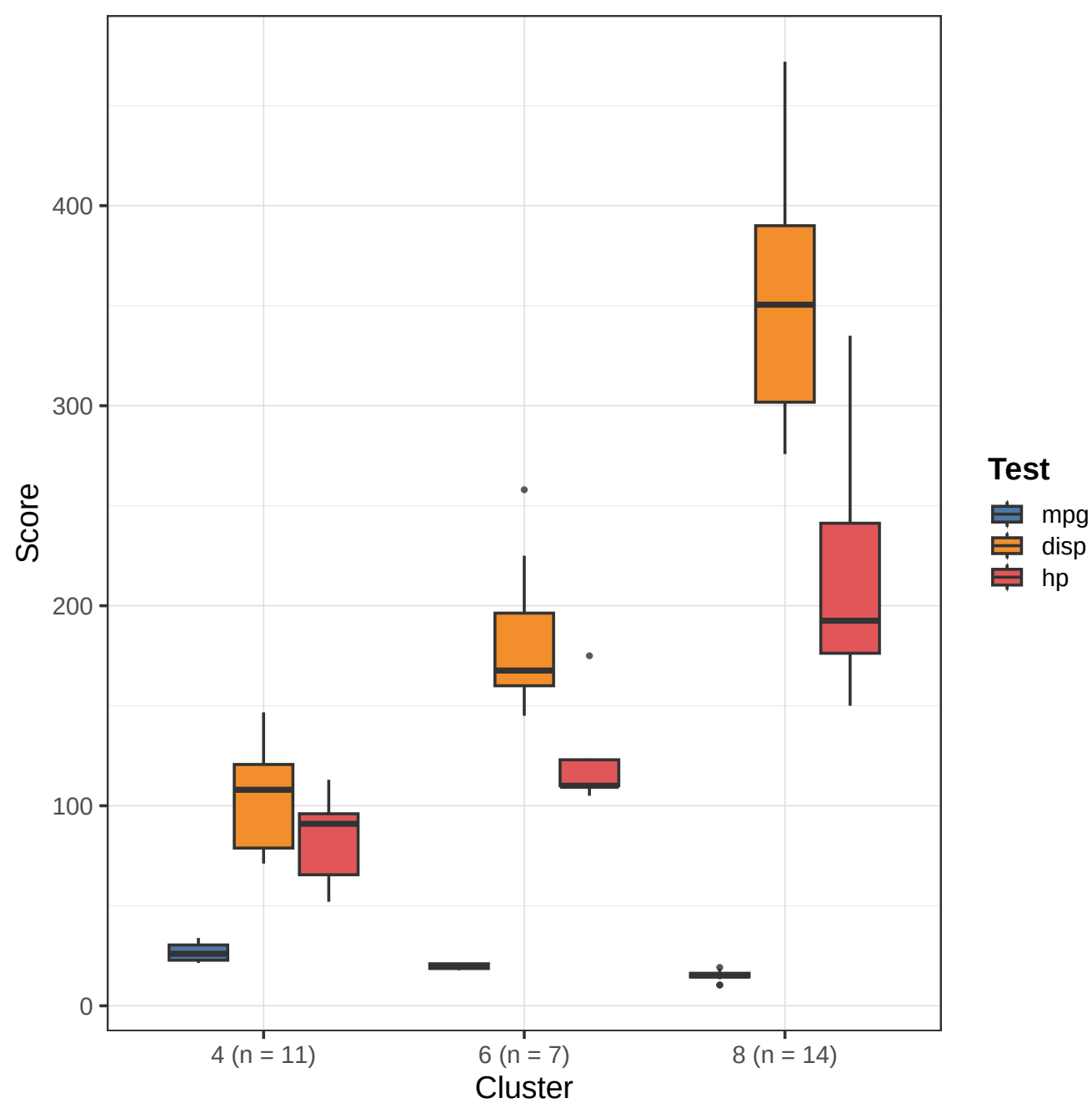
No Multiple Comparison Correction

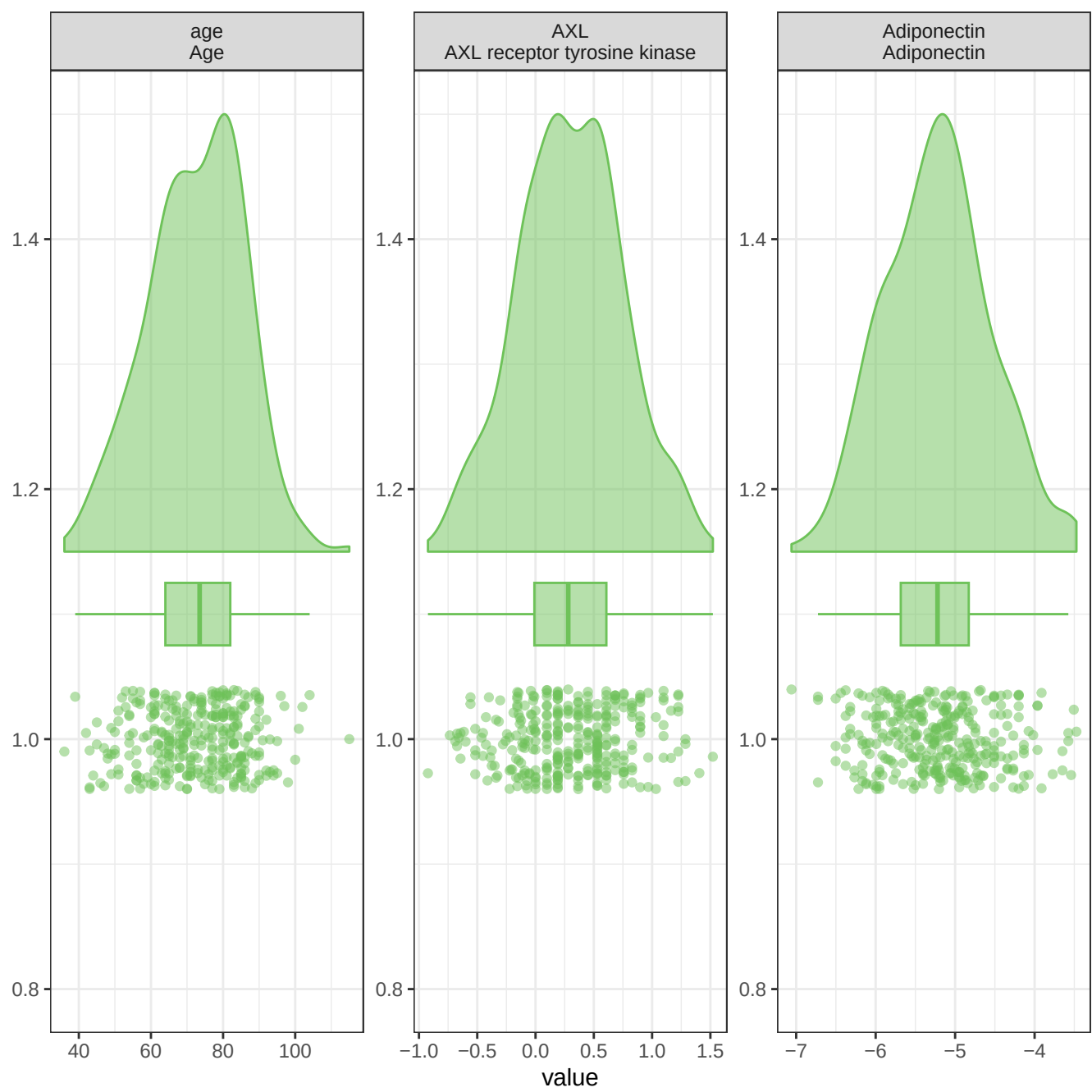


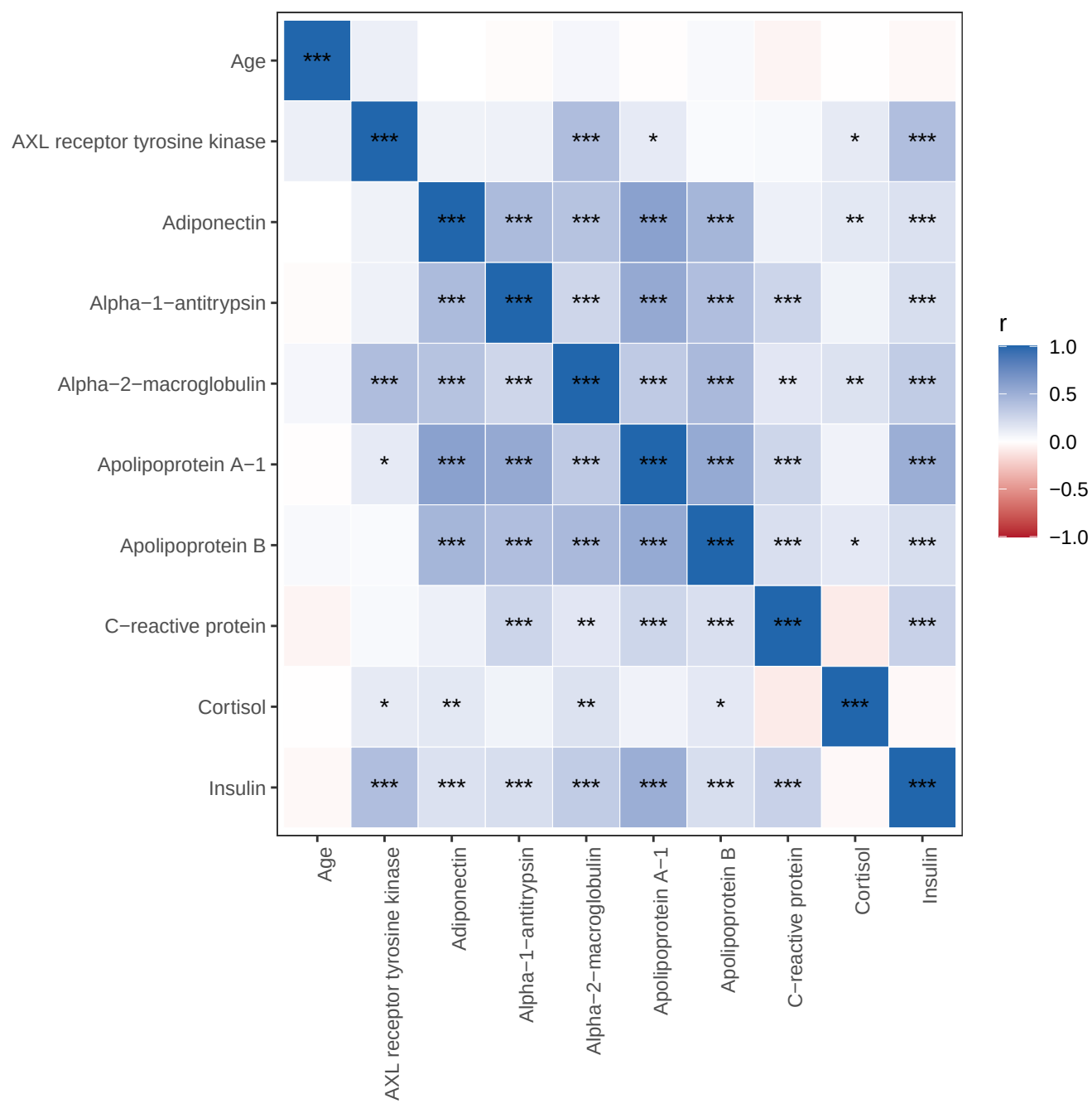
FDR Correction

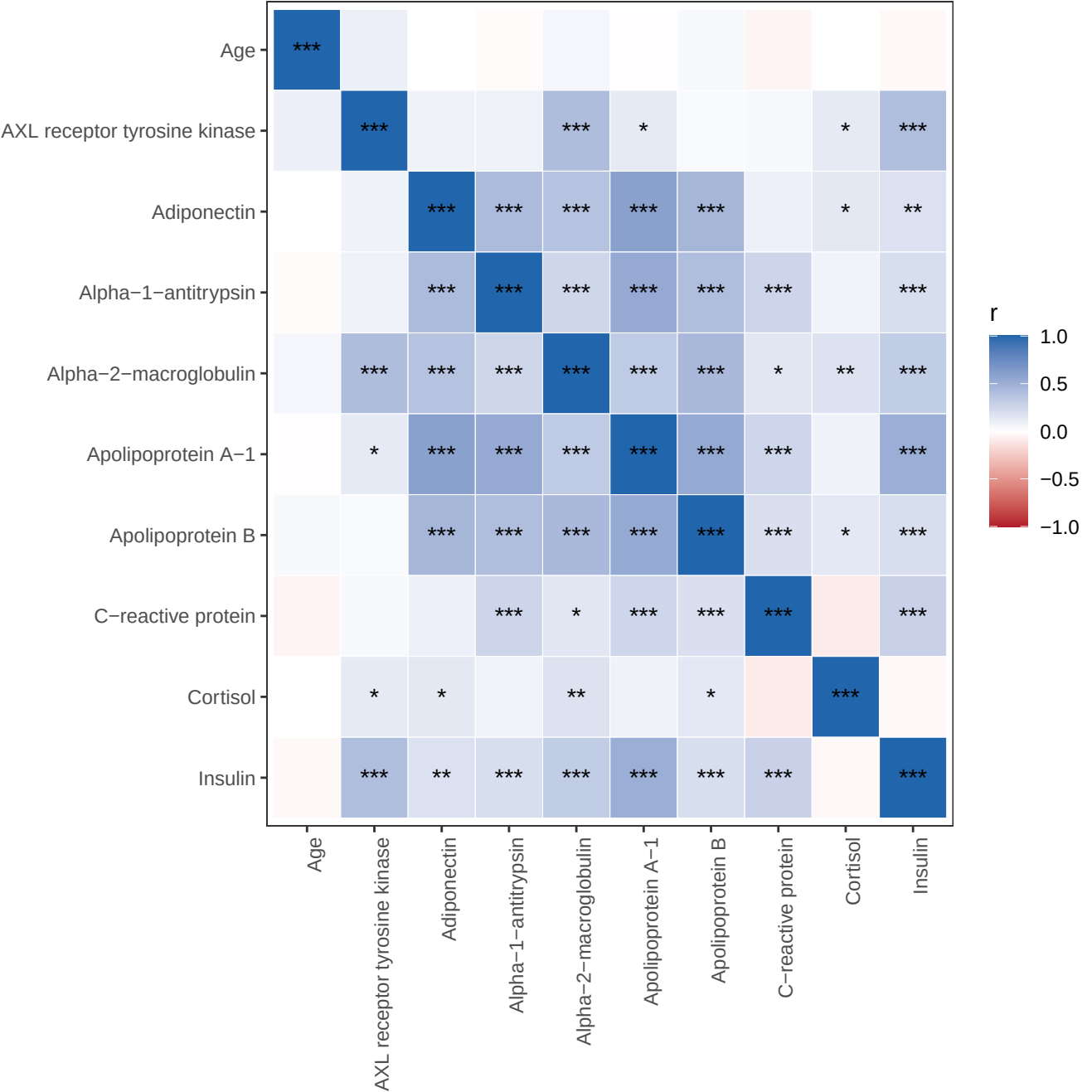


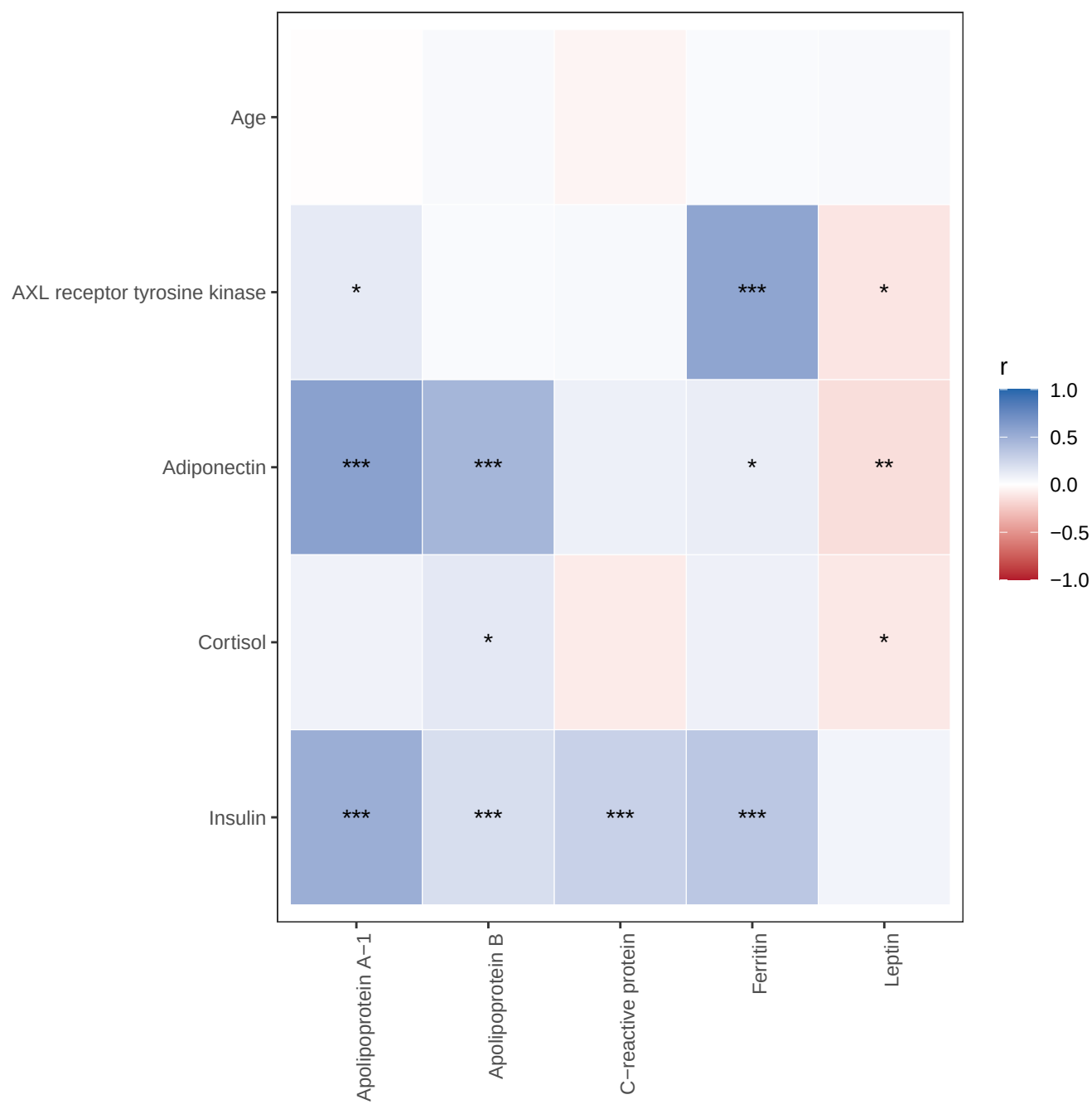






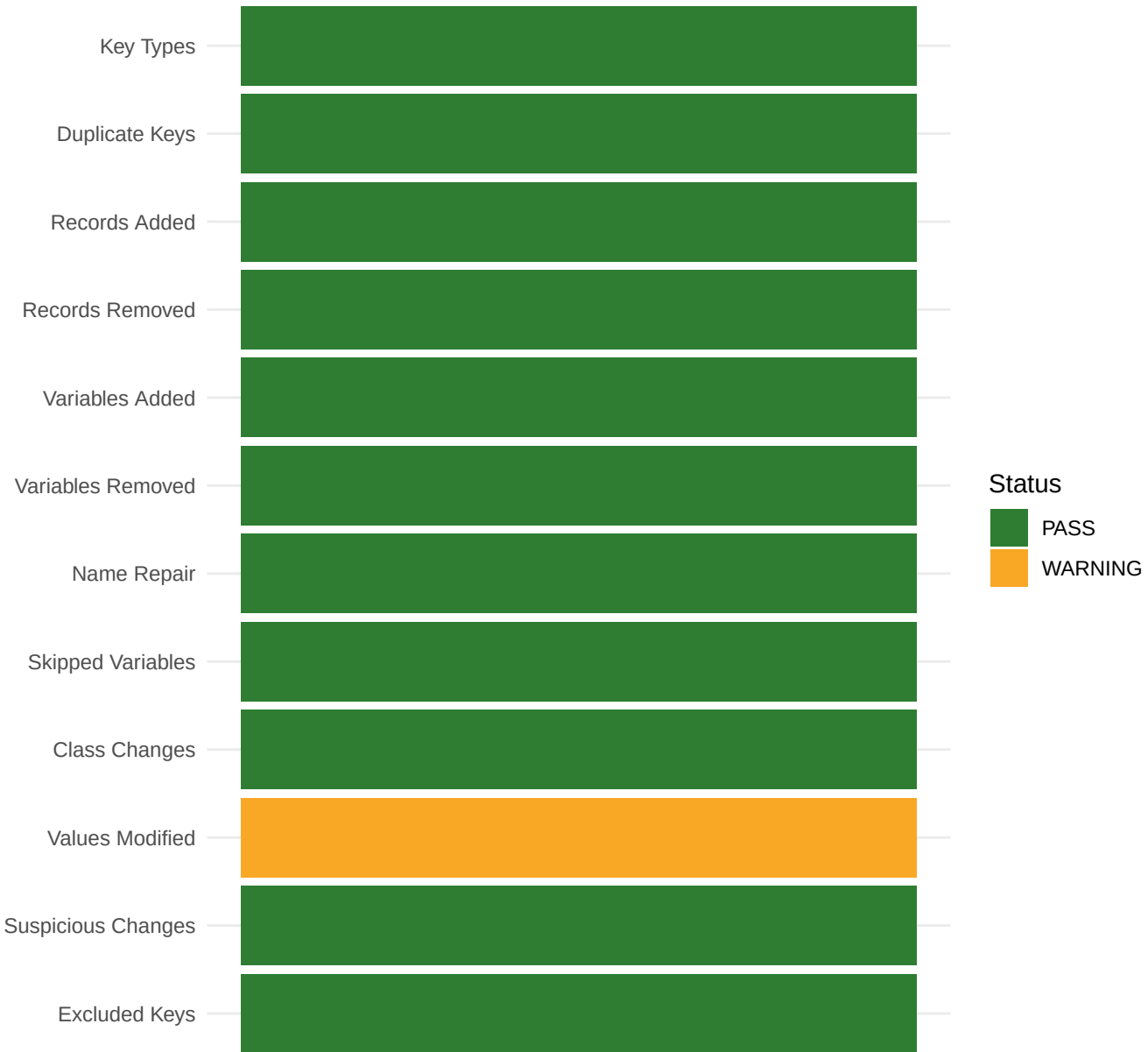






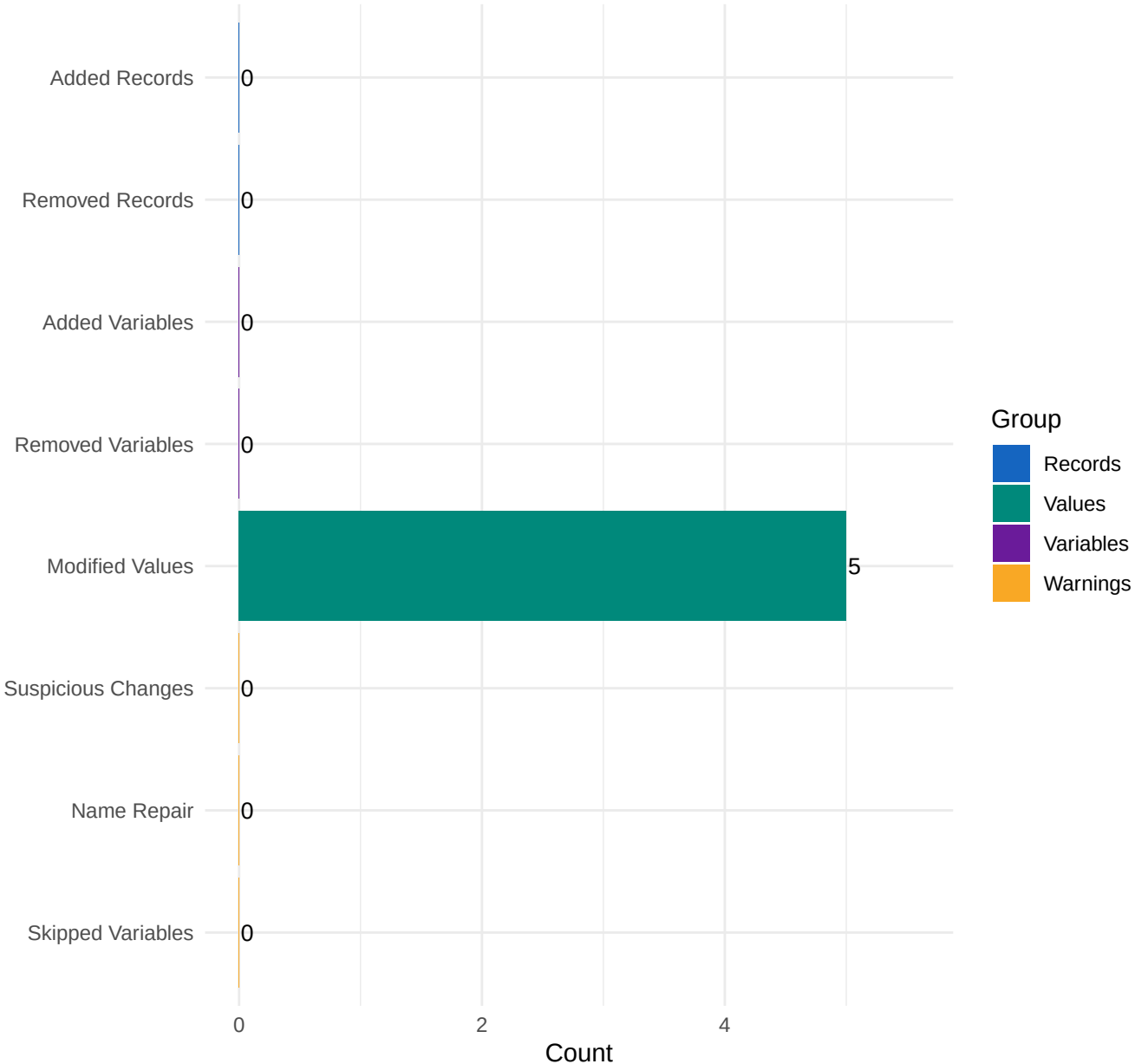
Dataset comparison checks

PASS, WARNING, and FAIL status by comparison check



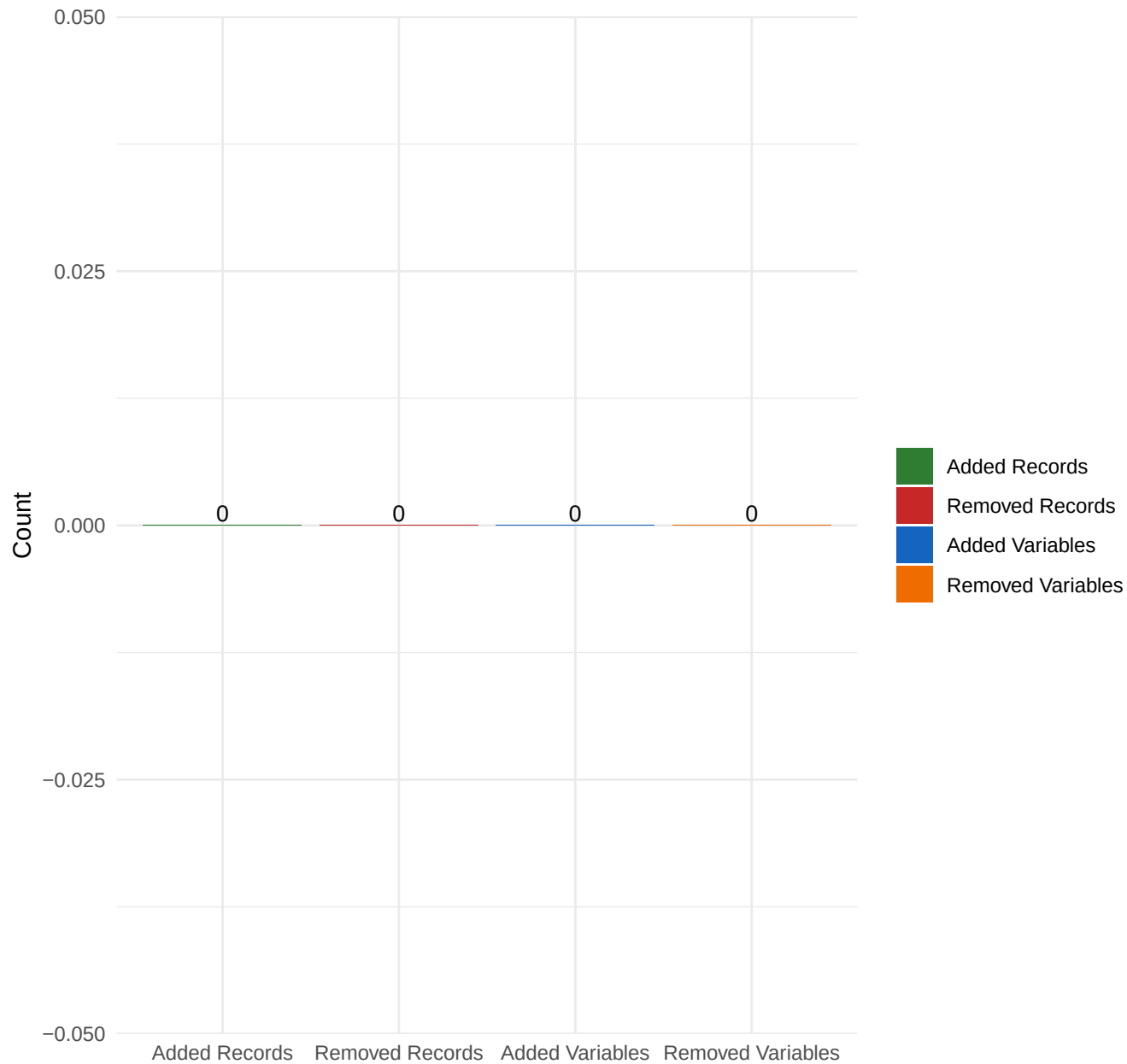
Dataset comparison summary

High-level record, variable, value, and warning counts



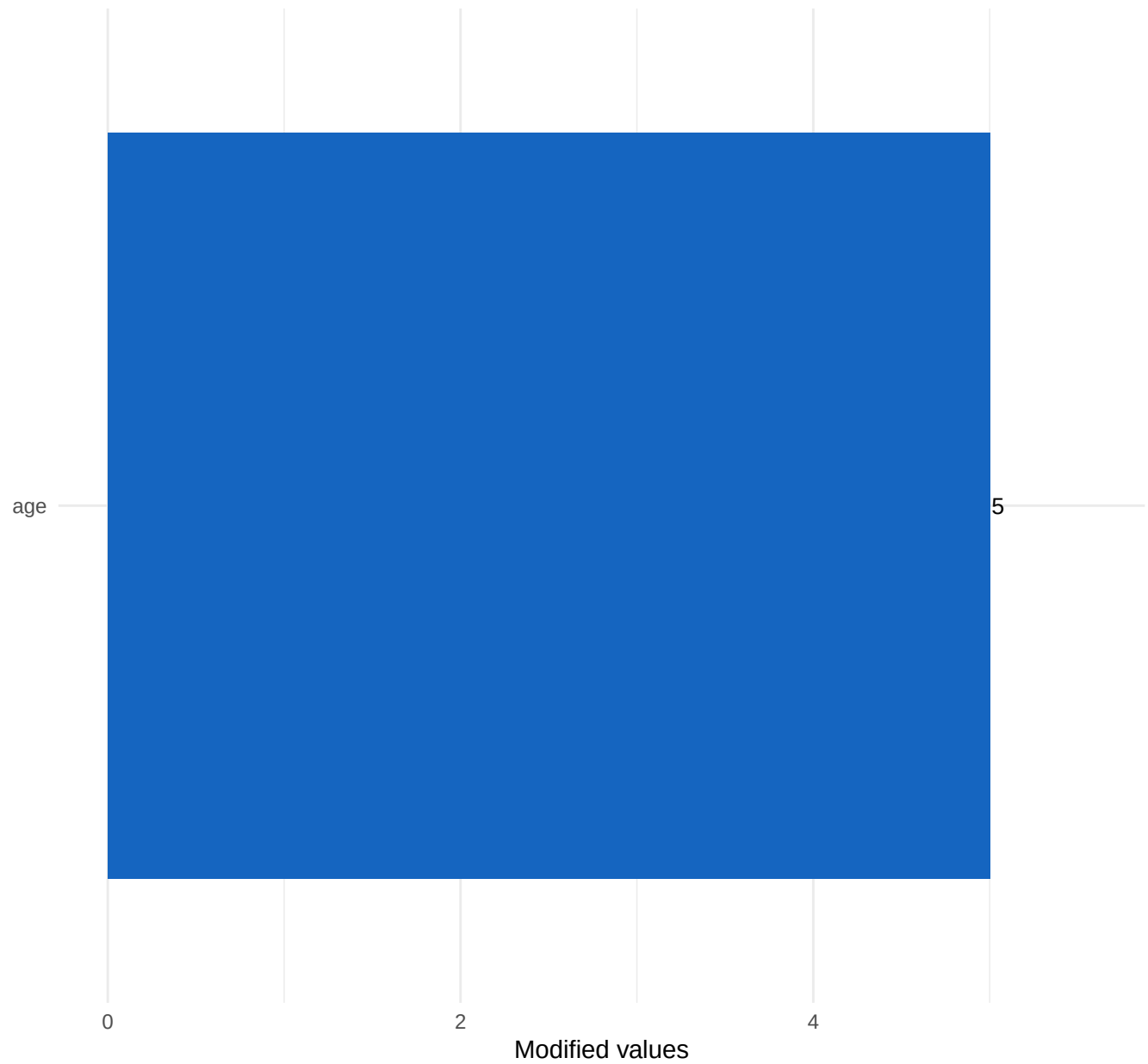
Record and structure changes

Records and variables added or removed



Variable-level changes

Top 1 variables by modified values



Top changed variables

Top 1 variables by percent changed

age

1.5%

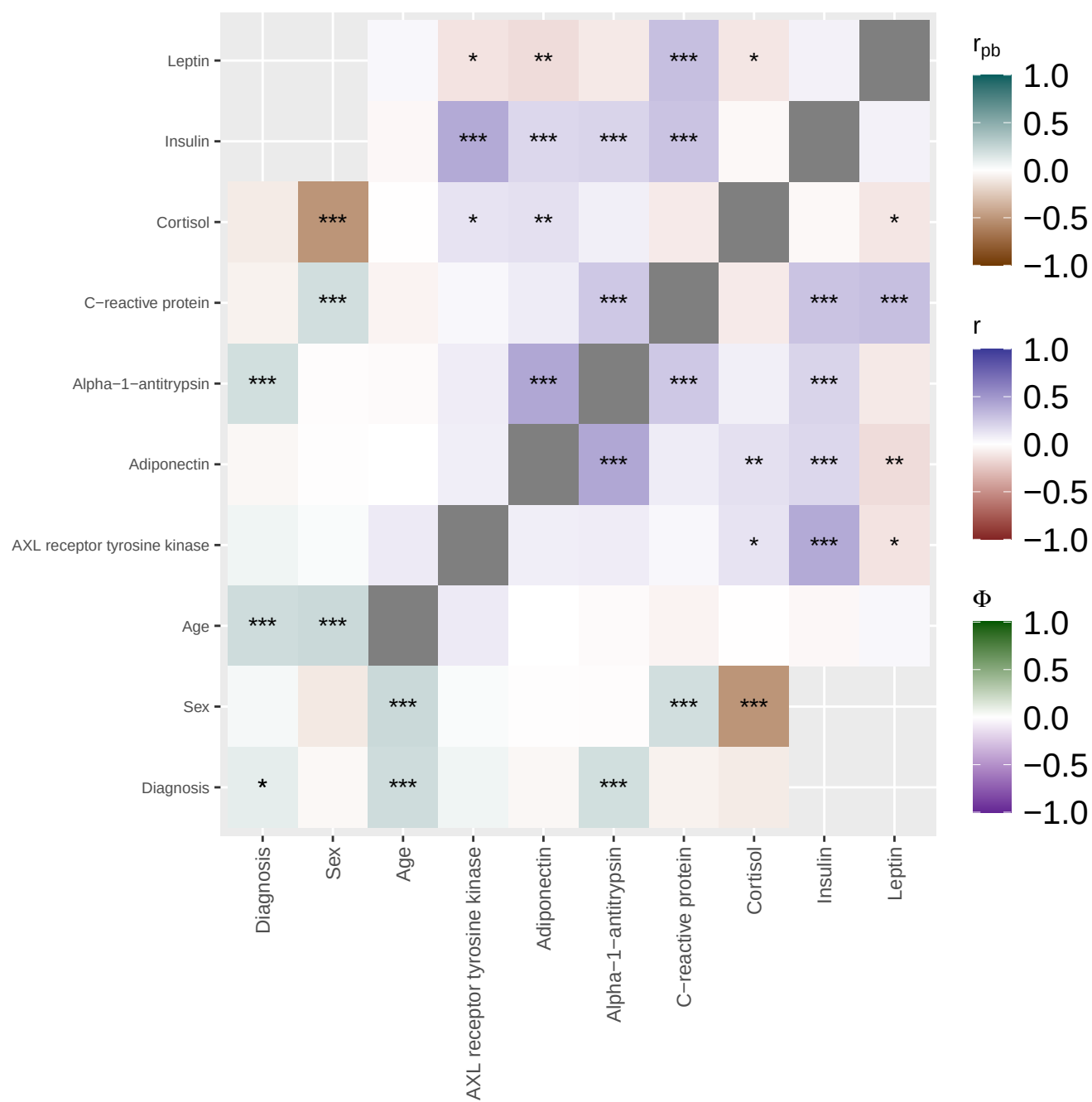
0.0

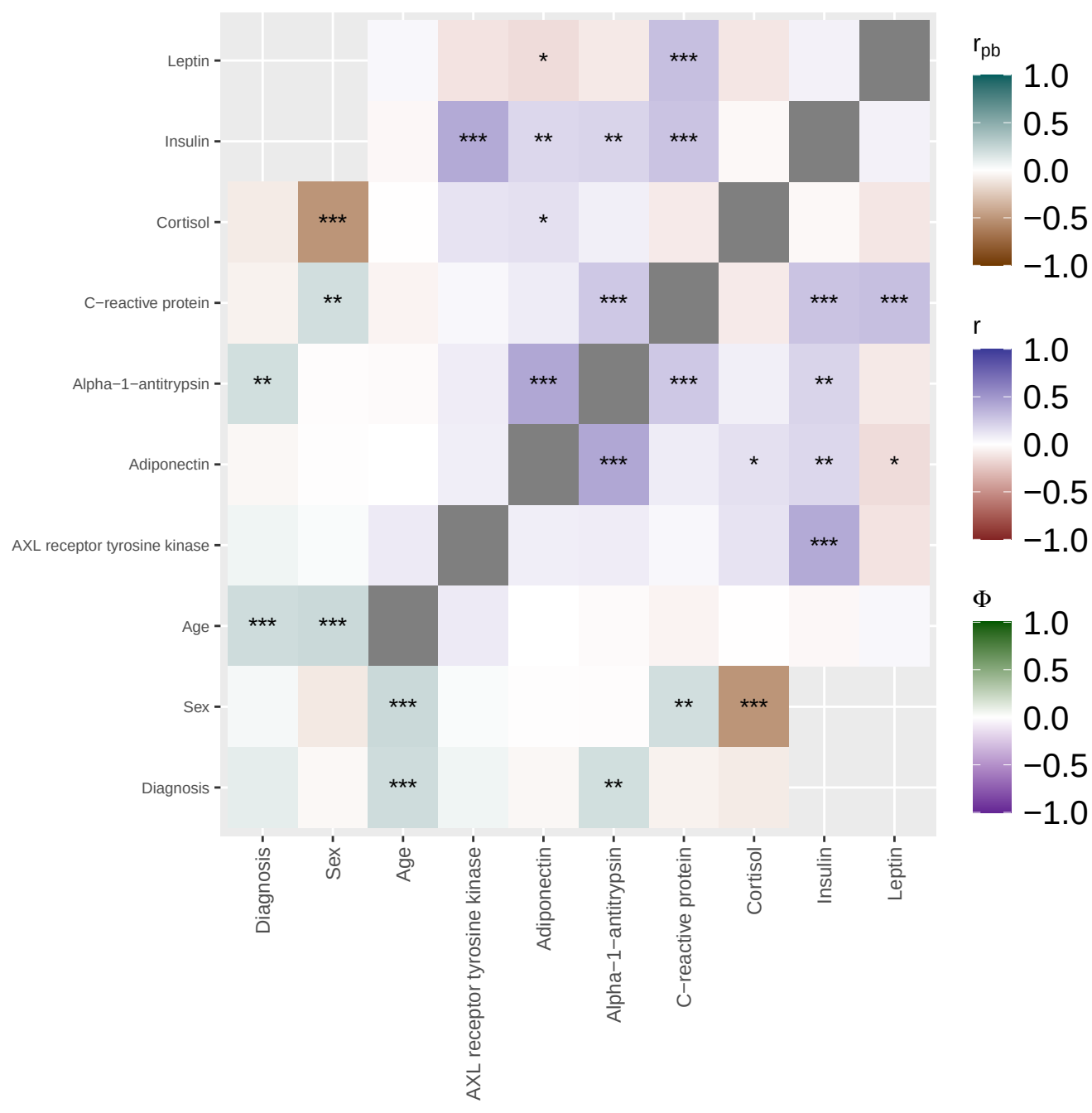
0.5

1.0

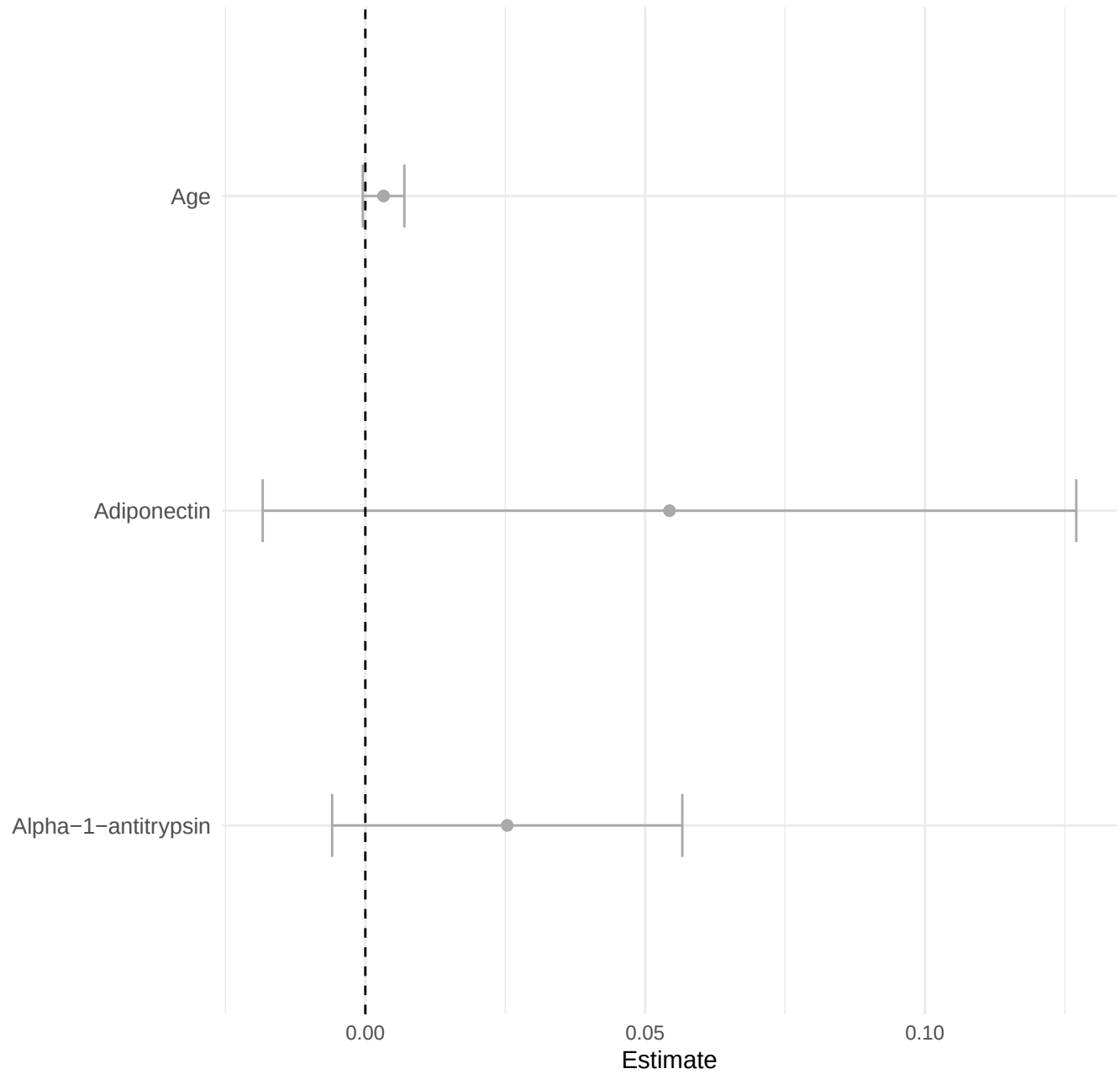
1.5

Percent changed





AXL receptor tyrosine kinase



AXL receptor tyrosine kinase

Adiponectin

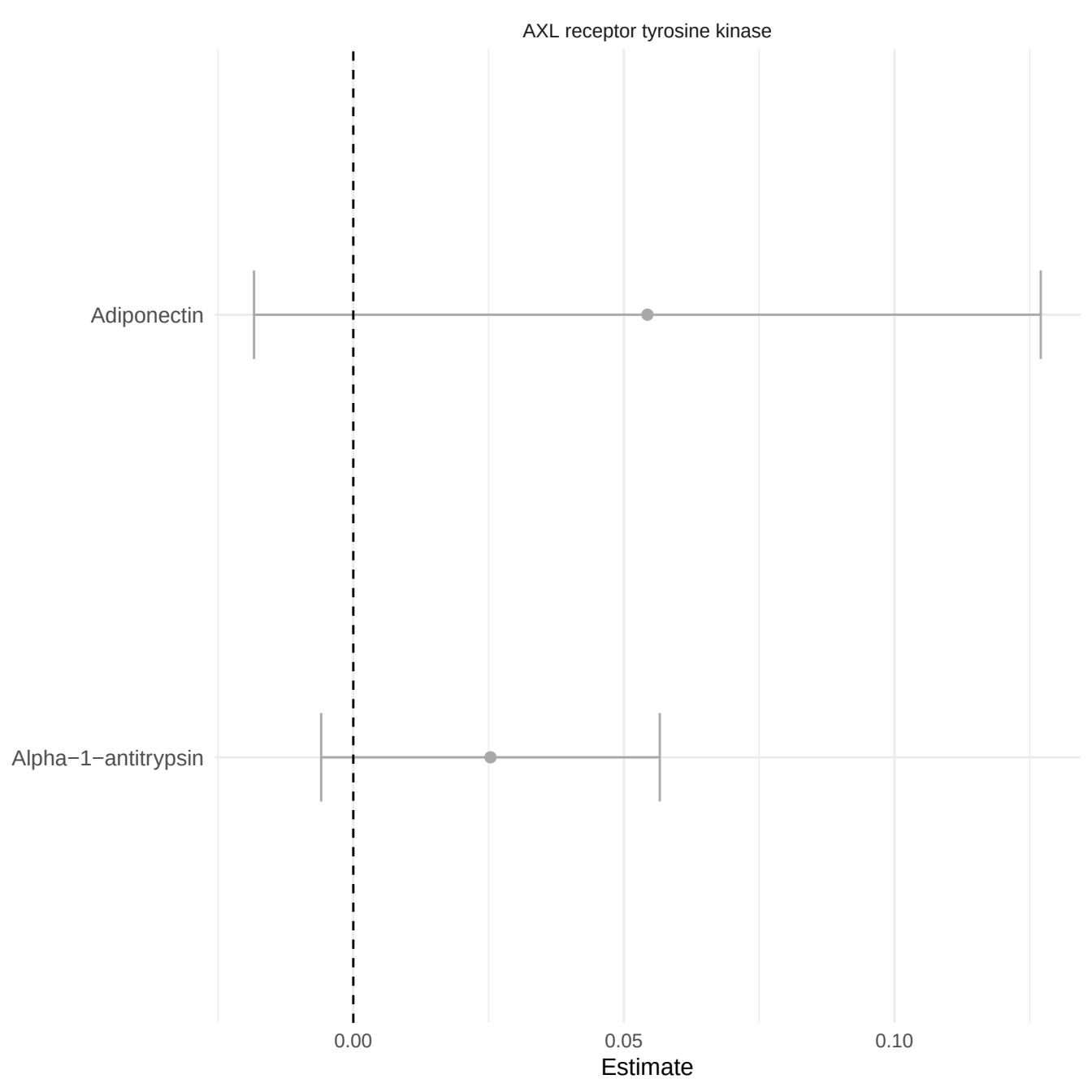
Alpha-1-antitrypsin

0.00

0.05

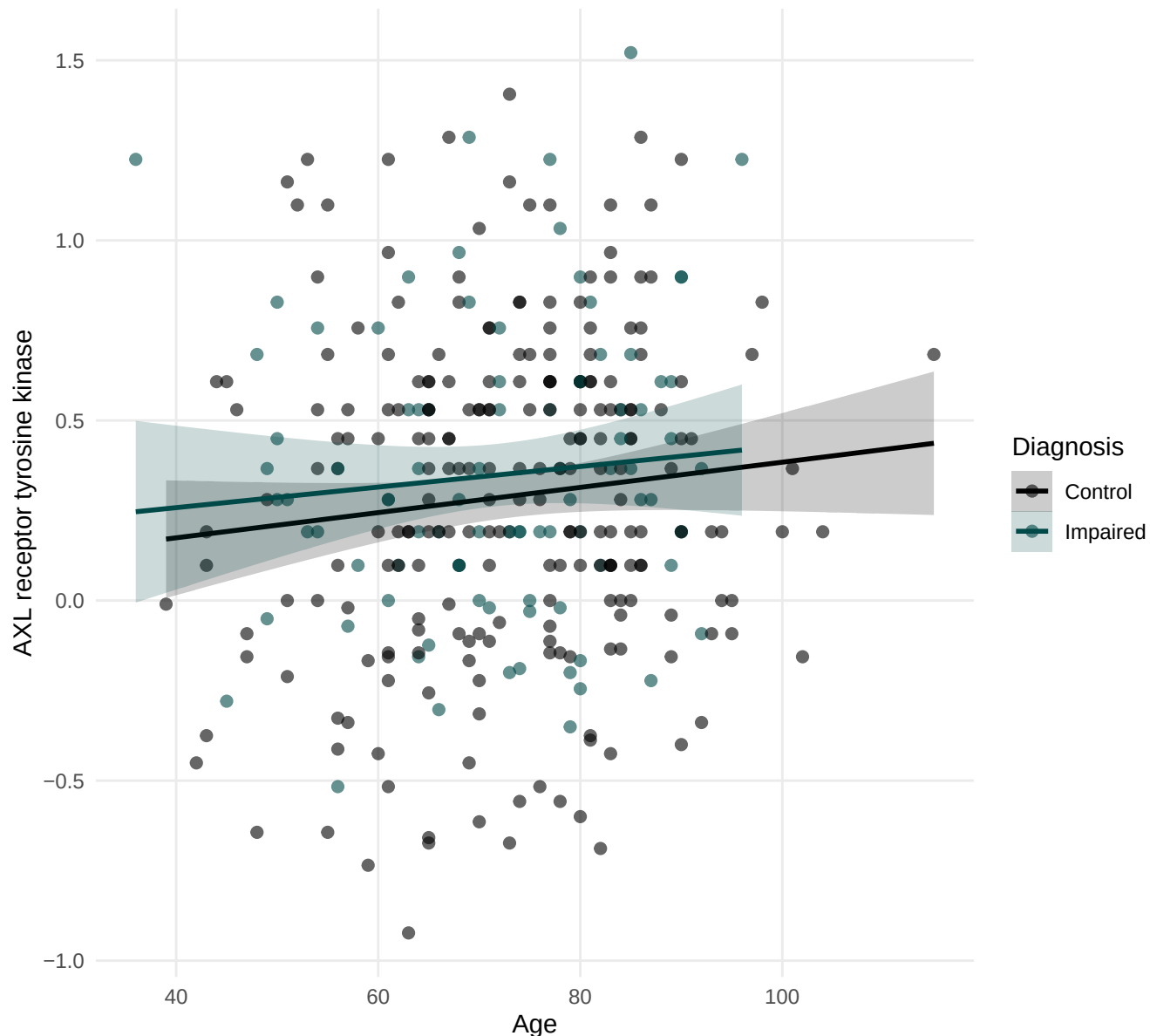
0.10

Estimate



Interaction with Diagnosis

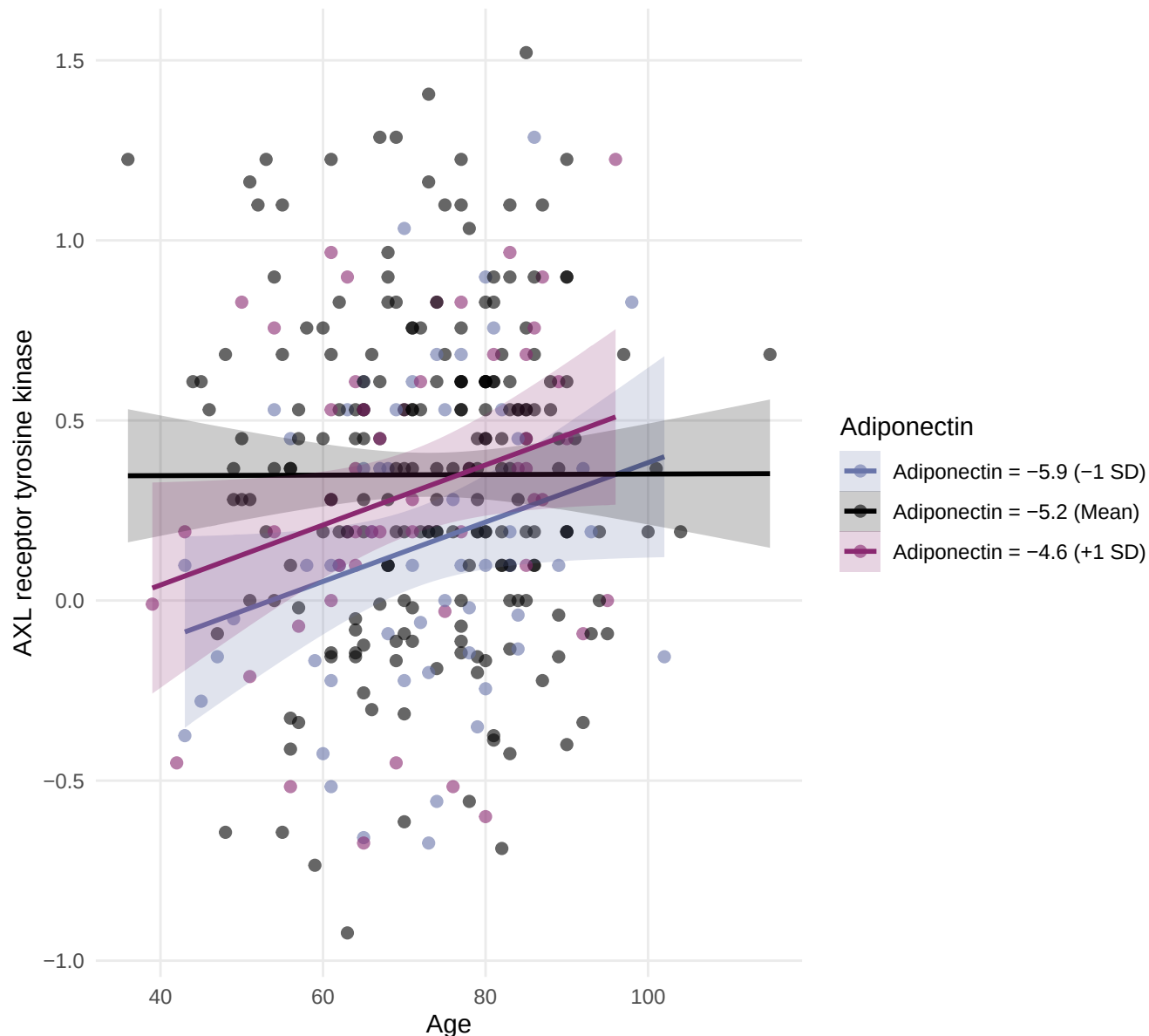
Interaction p-value: 0.8789



No covariates included in the model

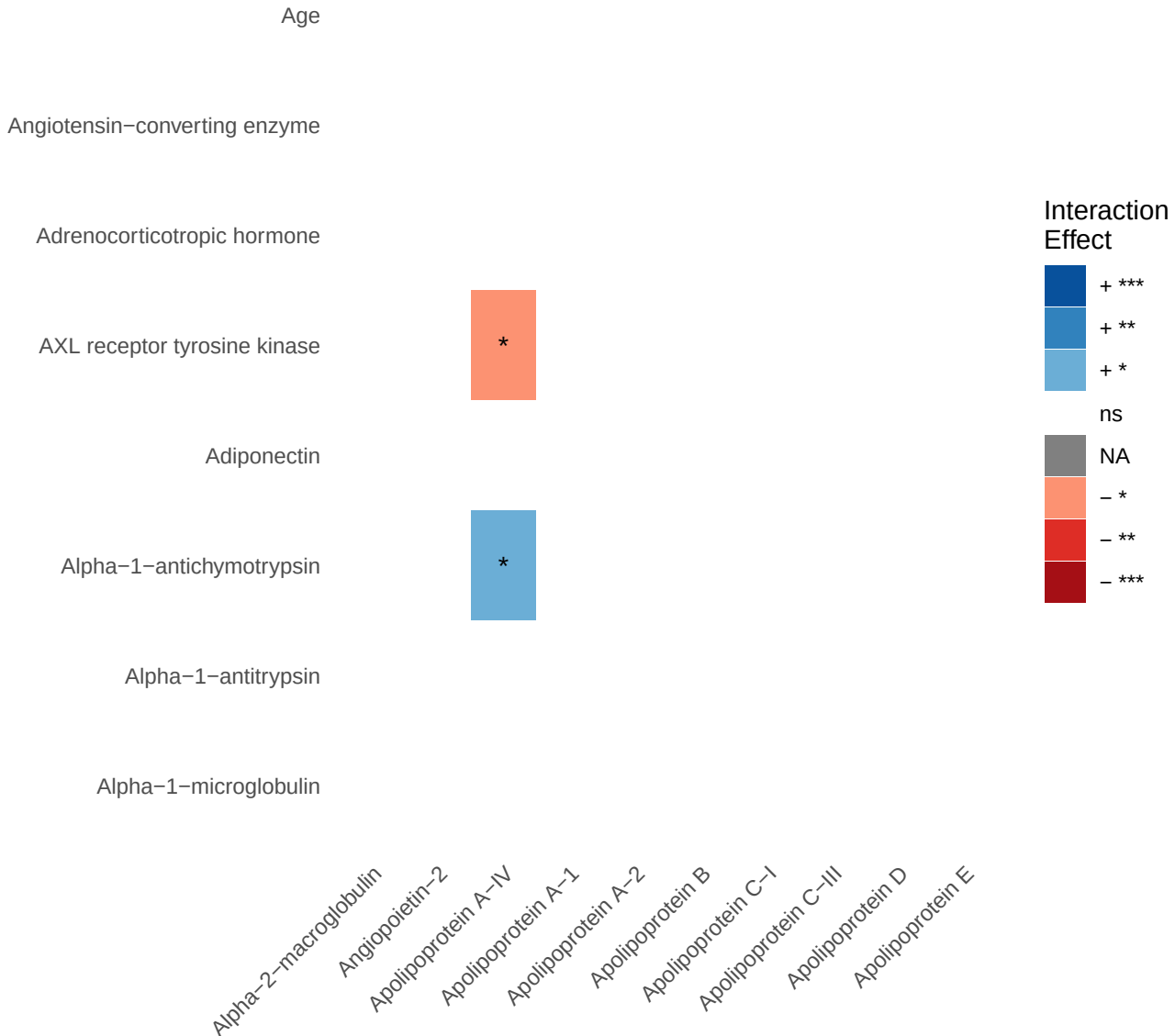
Interaction with Adiponectin

Interaction p-value: 0.7488



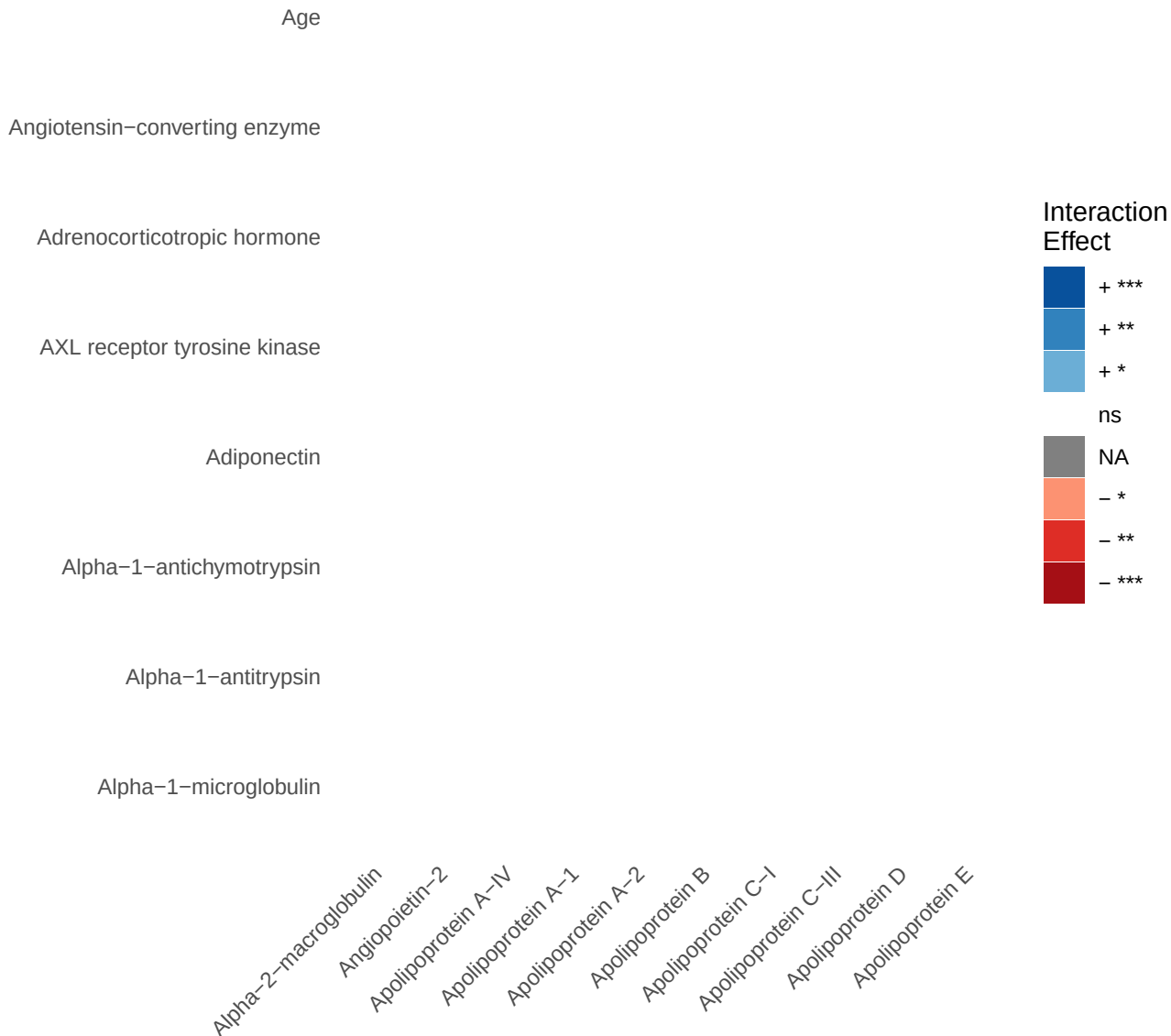
No covariates included in the model

Interaction Effects with Diagnosis



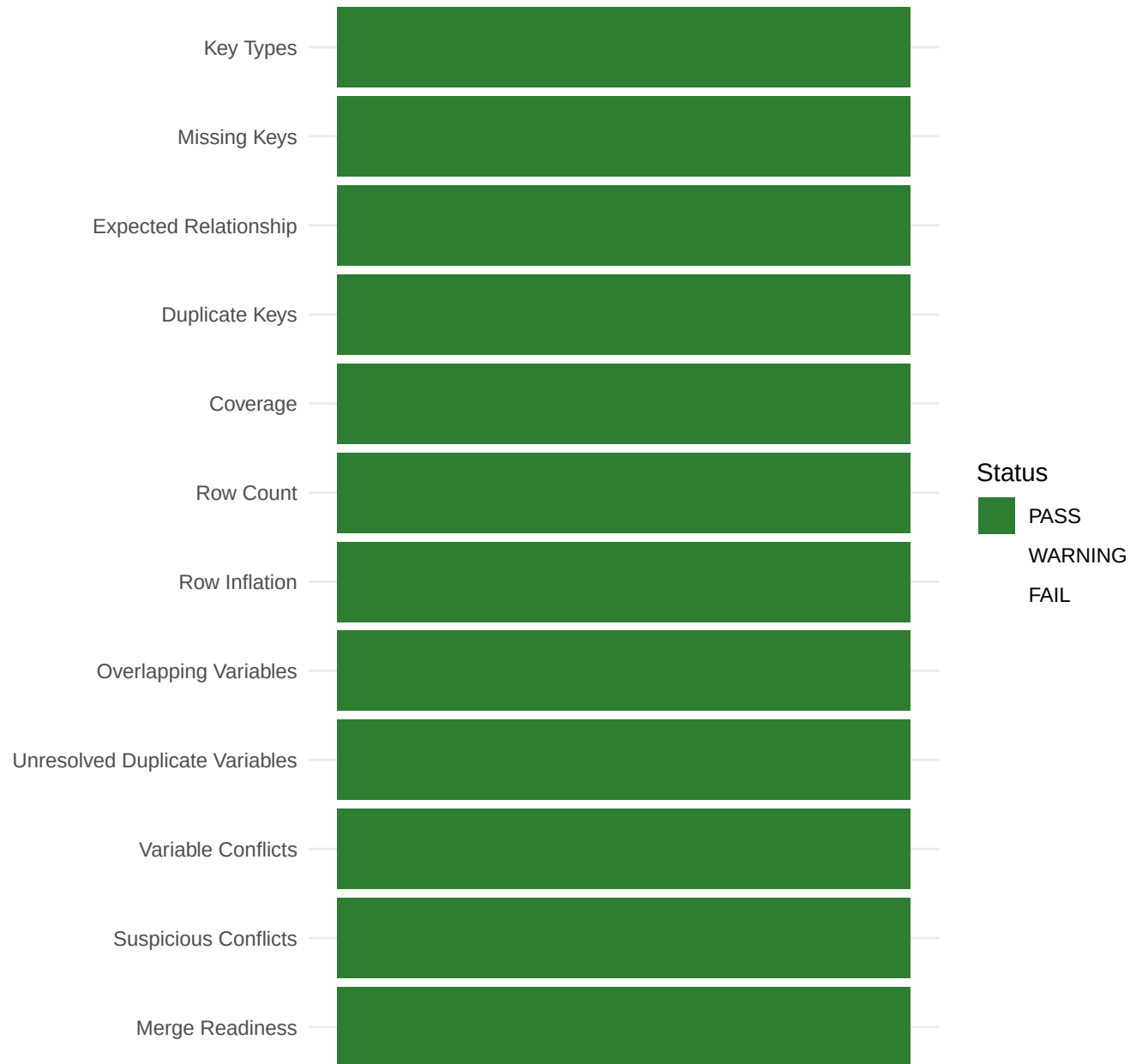
No multiple comparison correction. No covariates included.

Interaction Effects with Diagnosis



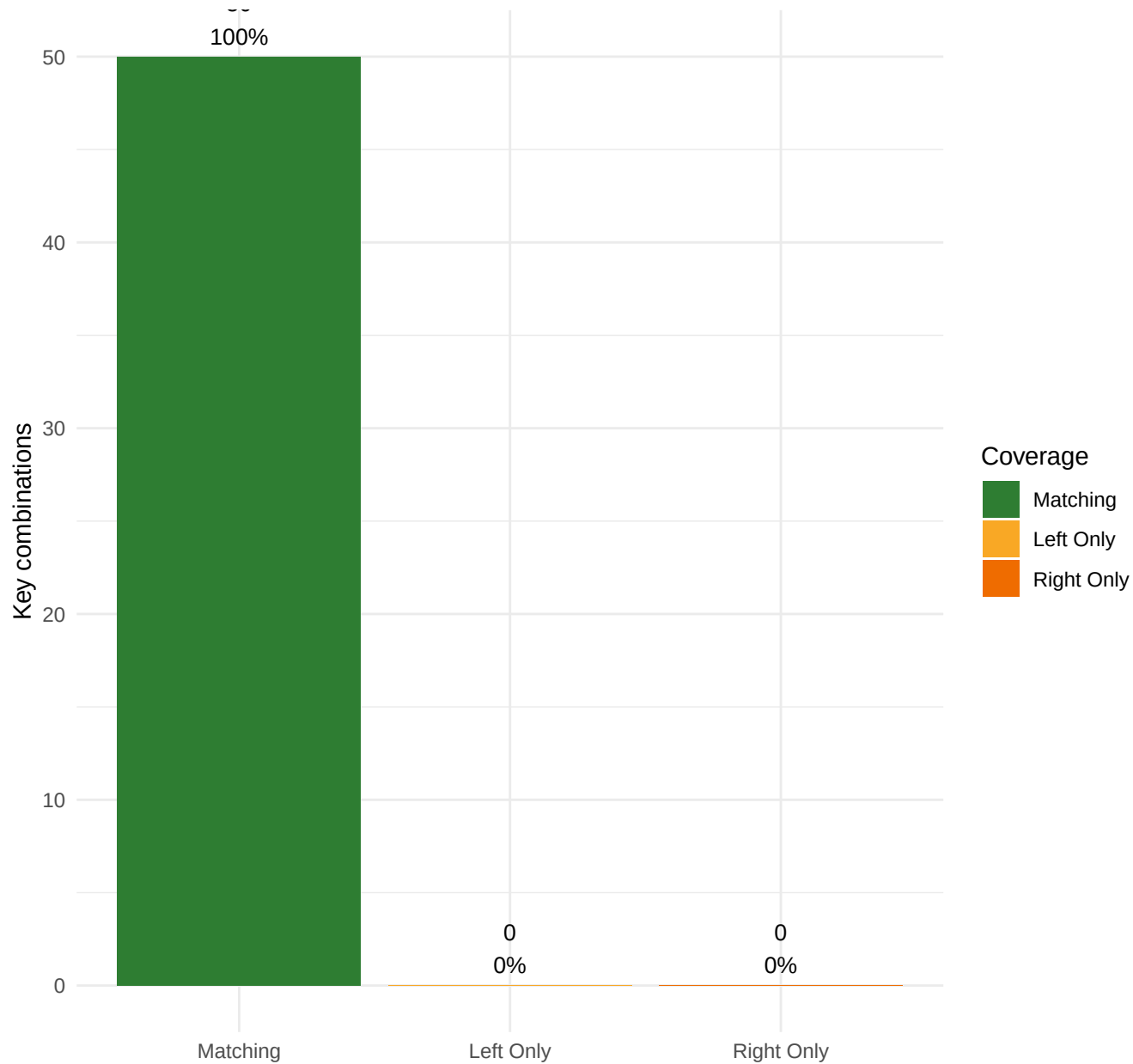
FDR corrected for multiple comparisons. No covariates included.

Merge validation checks



Merge coverage

Key combinations by source overlap



Join audit

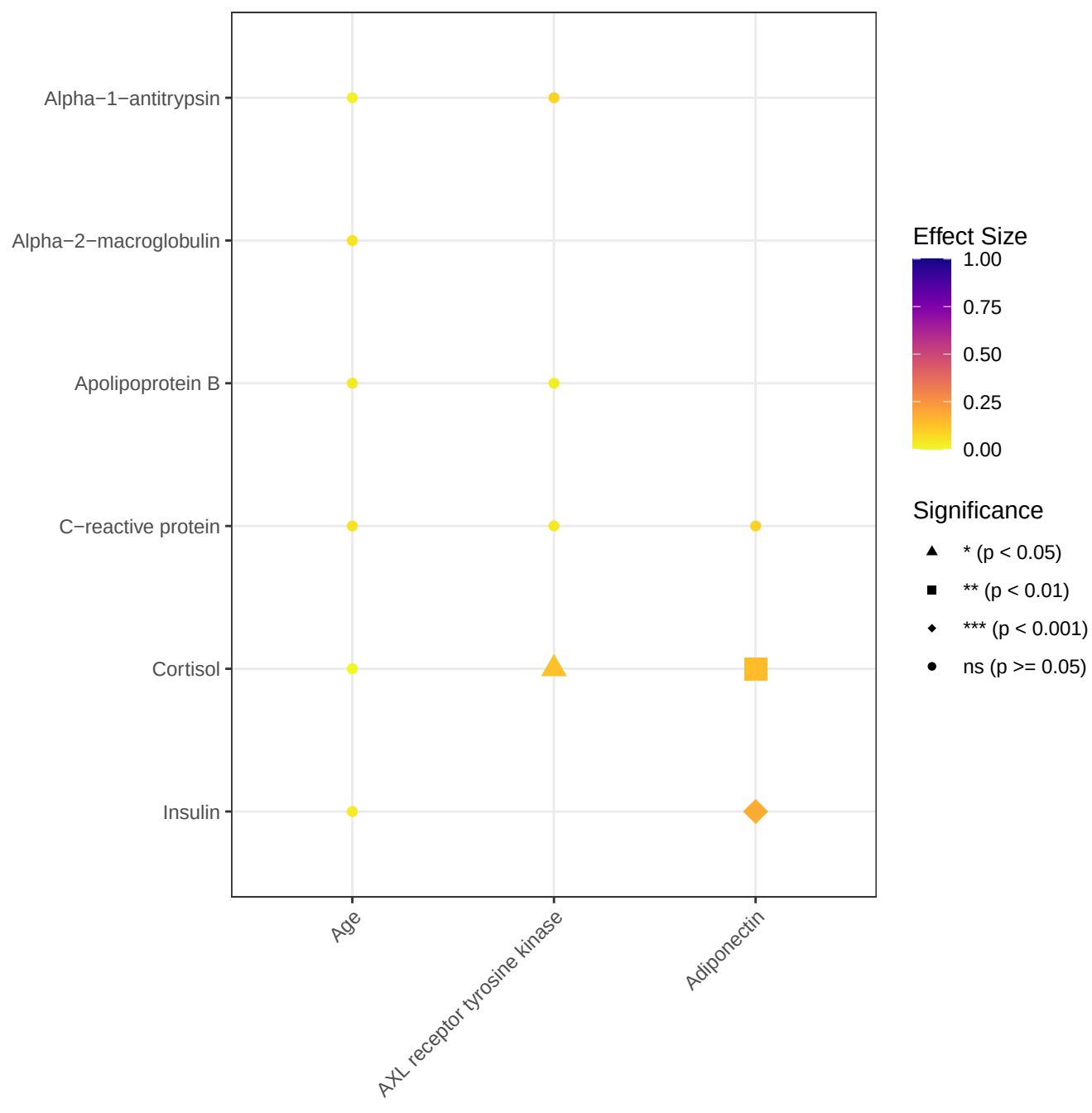


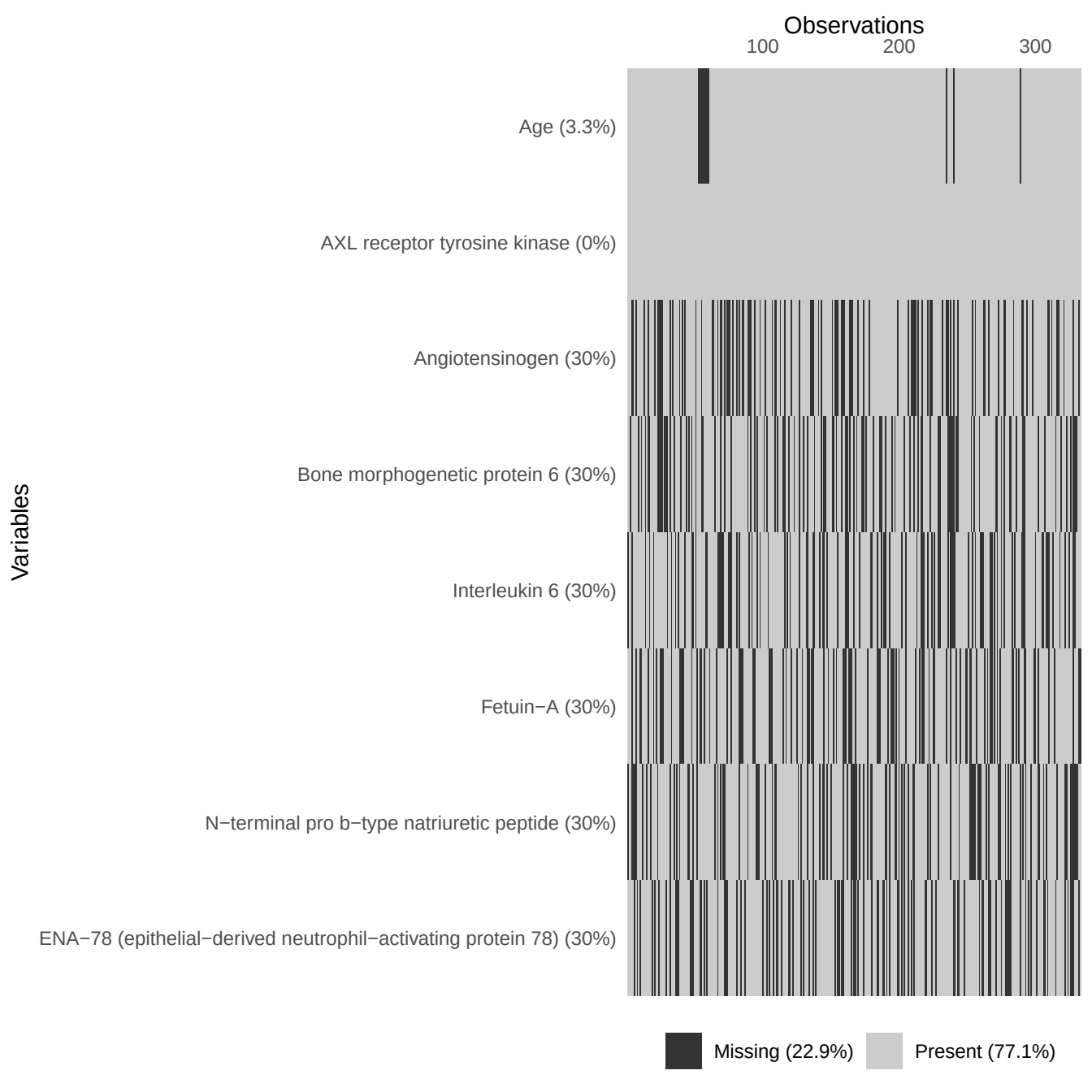
Duplicate-variable agreement

No unresolved duplicate variables detected

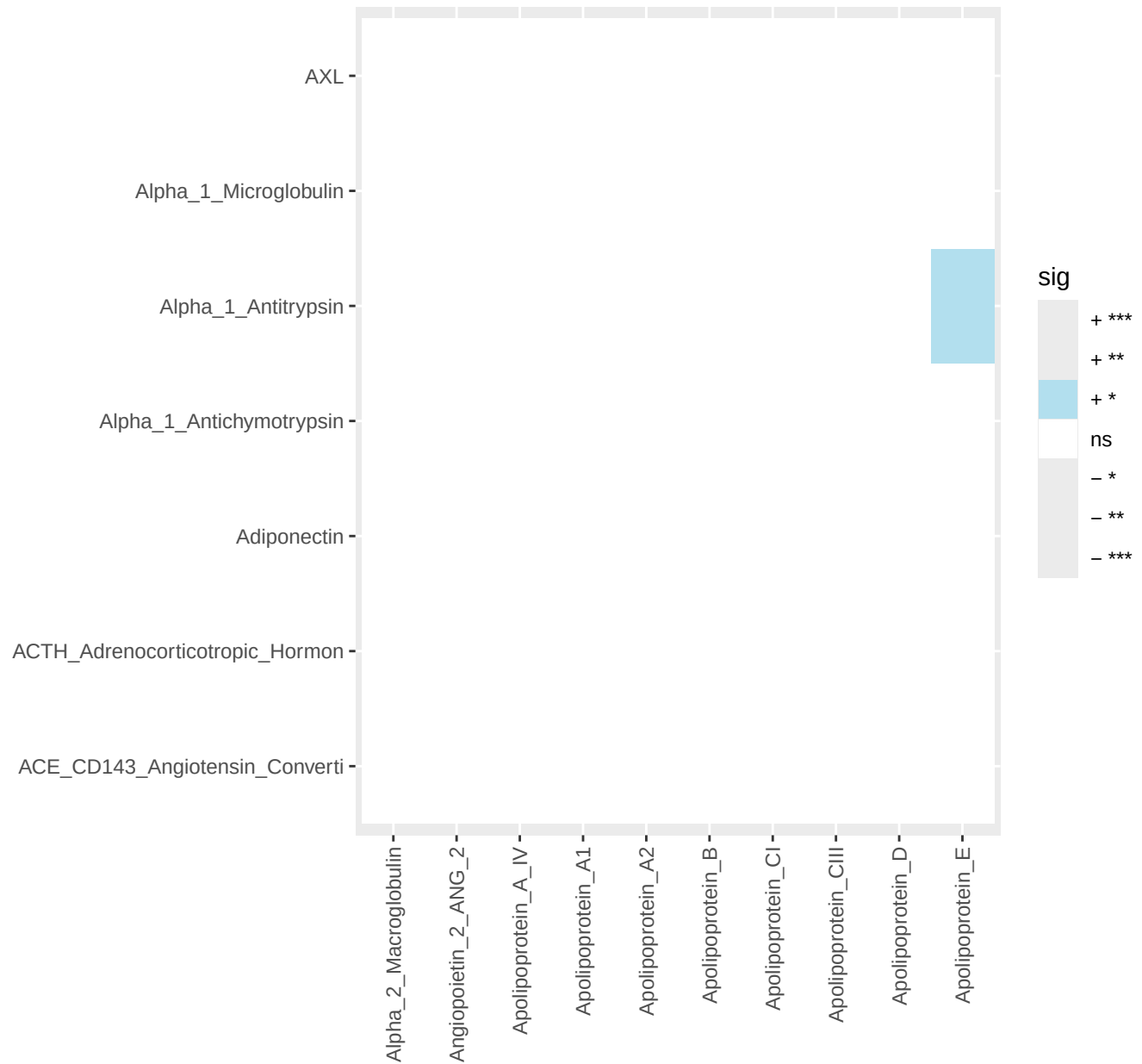
Duplicate-variable conflicts

No variable conflicts detected

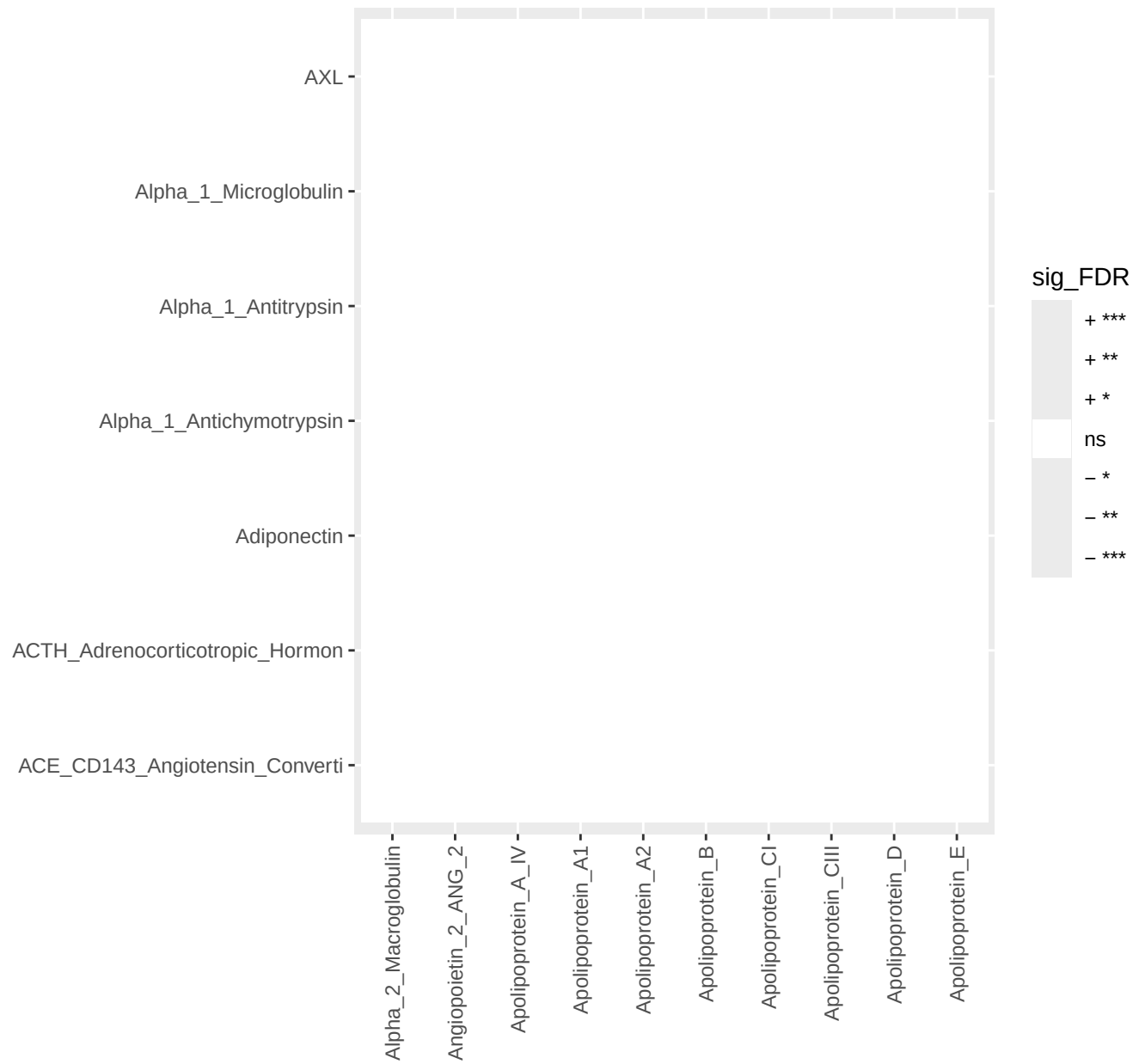




No Multiple Comparison Correction

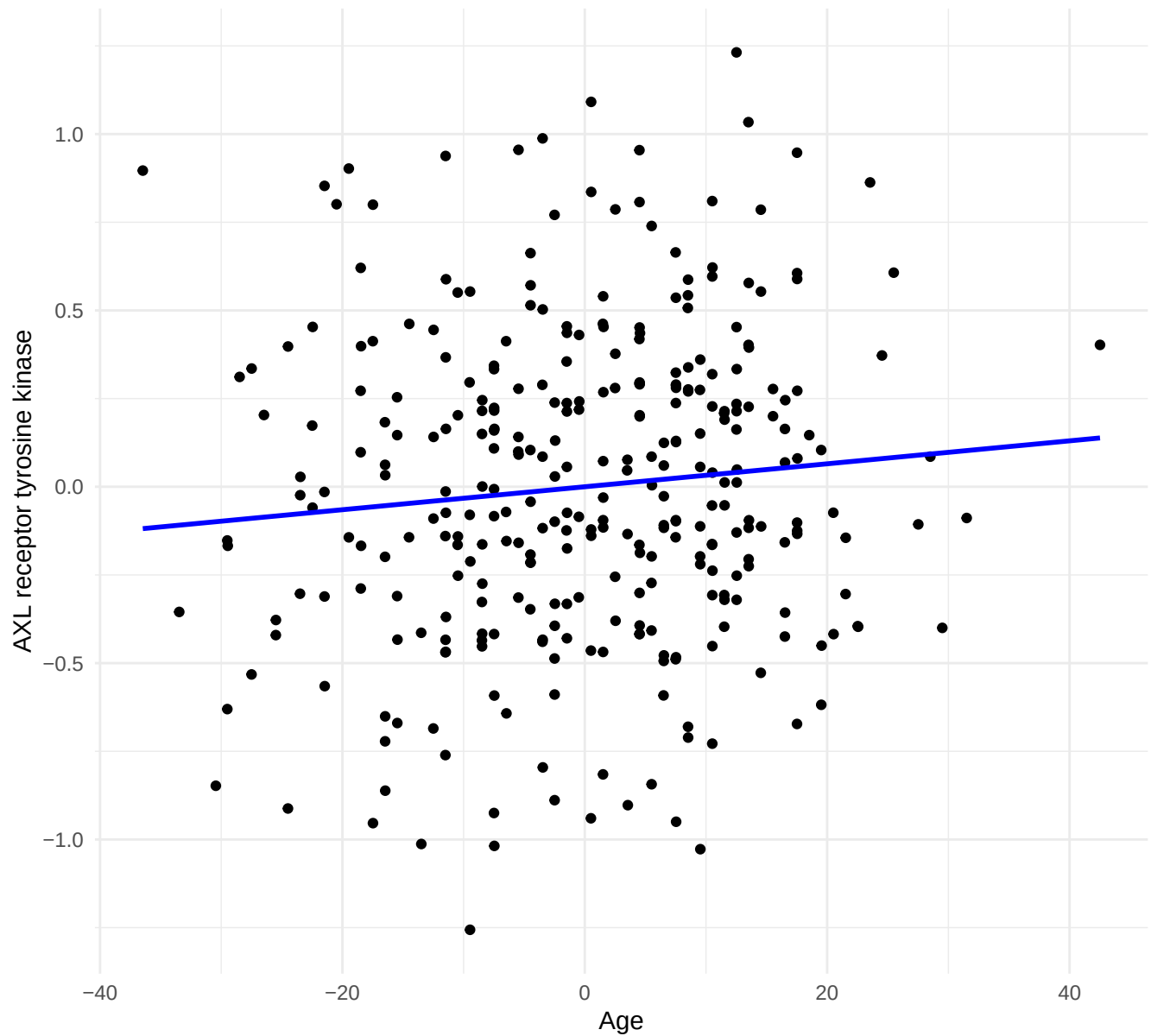


FDR Correction

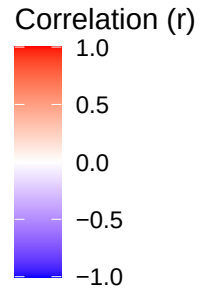
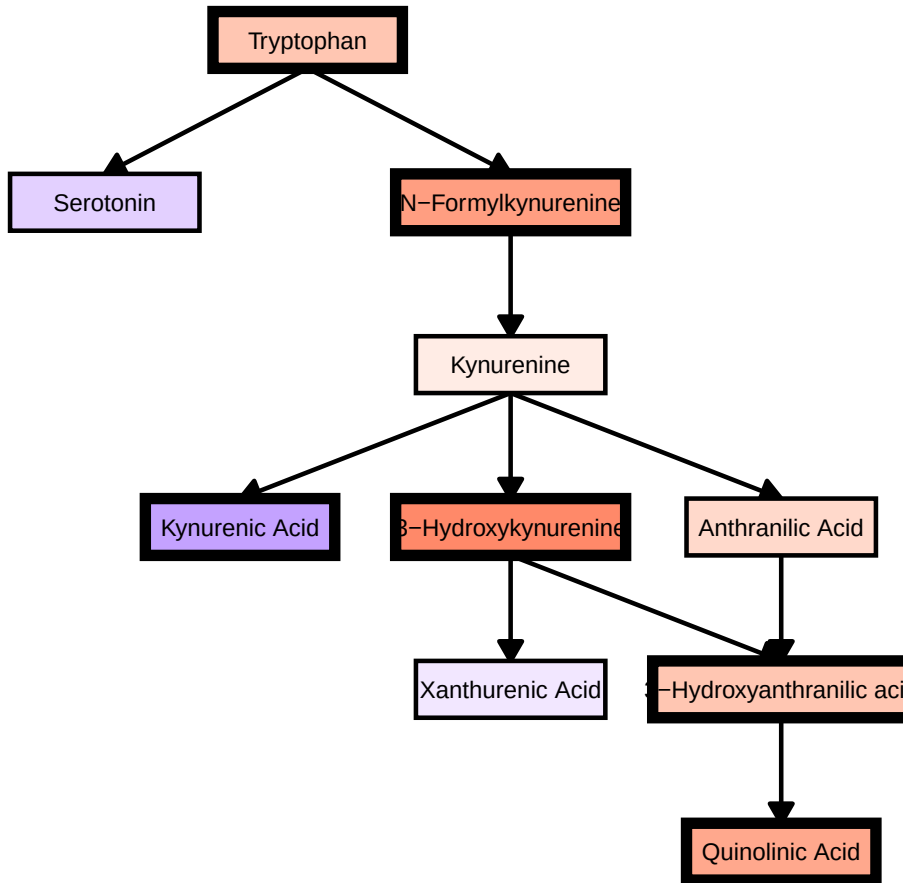


Partial Regression Plot

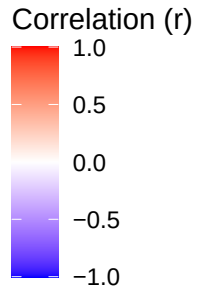
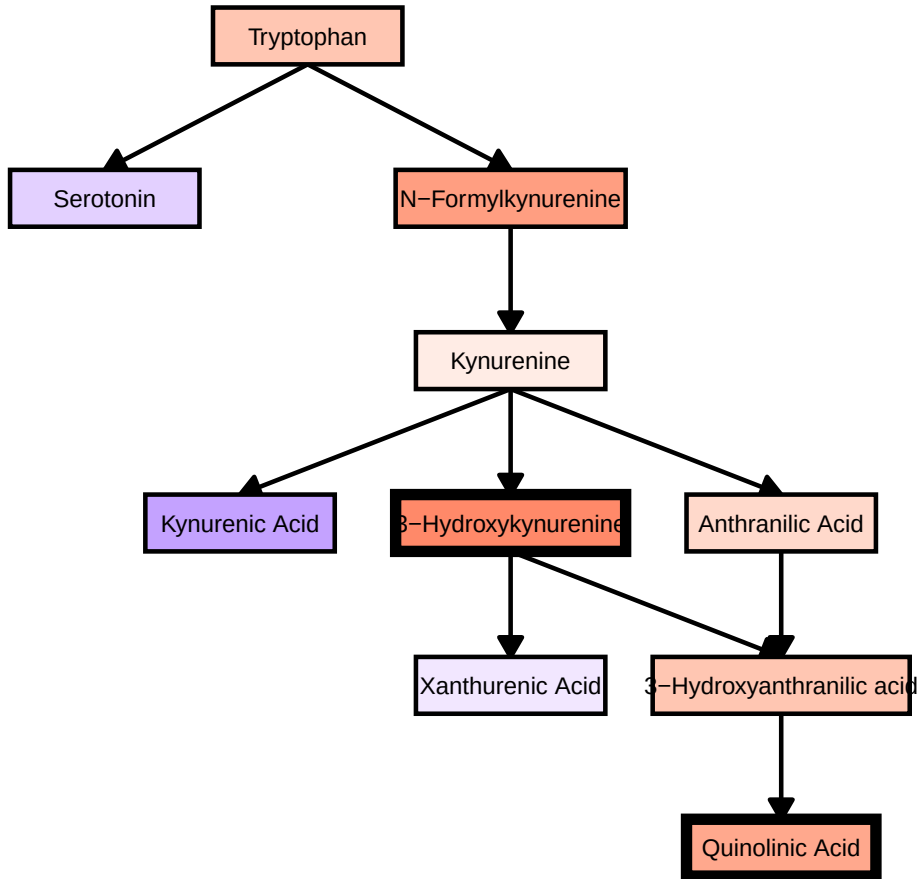
Residual AXL = $0 + 0 \cdot \text{age_resid}$ | R-squared = 0.01, n = 322, p = 0.085



Kynurenine pathway



Kynurenine pathway



Sex

*

Φ

1.0

0.5

0.0

-0.5

-1.0

Diagnosis

*

Diagnosis

Sex

Sex

*

Φ

1.0

0.5

0.0

-0.5

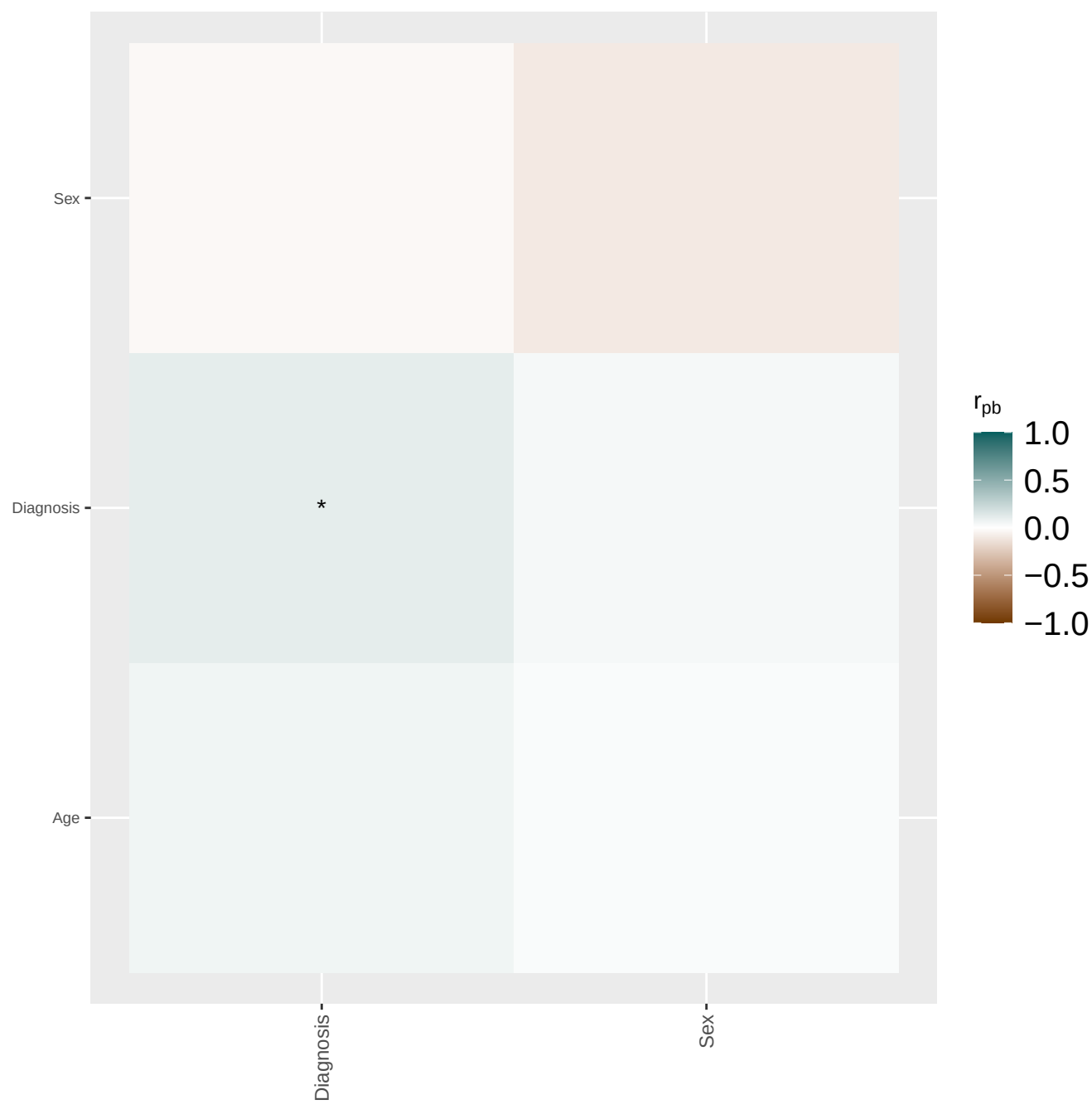
-1.0

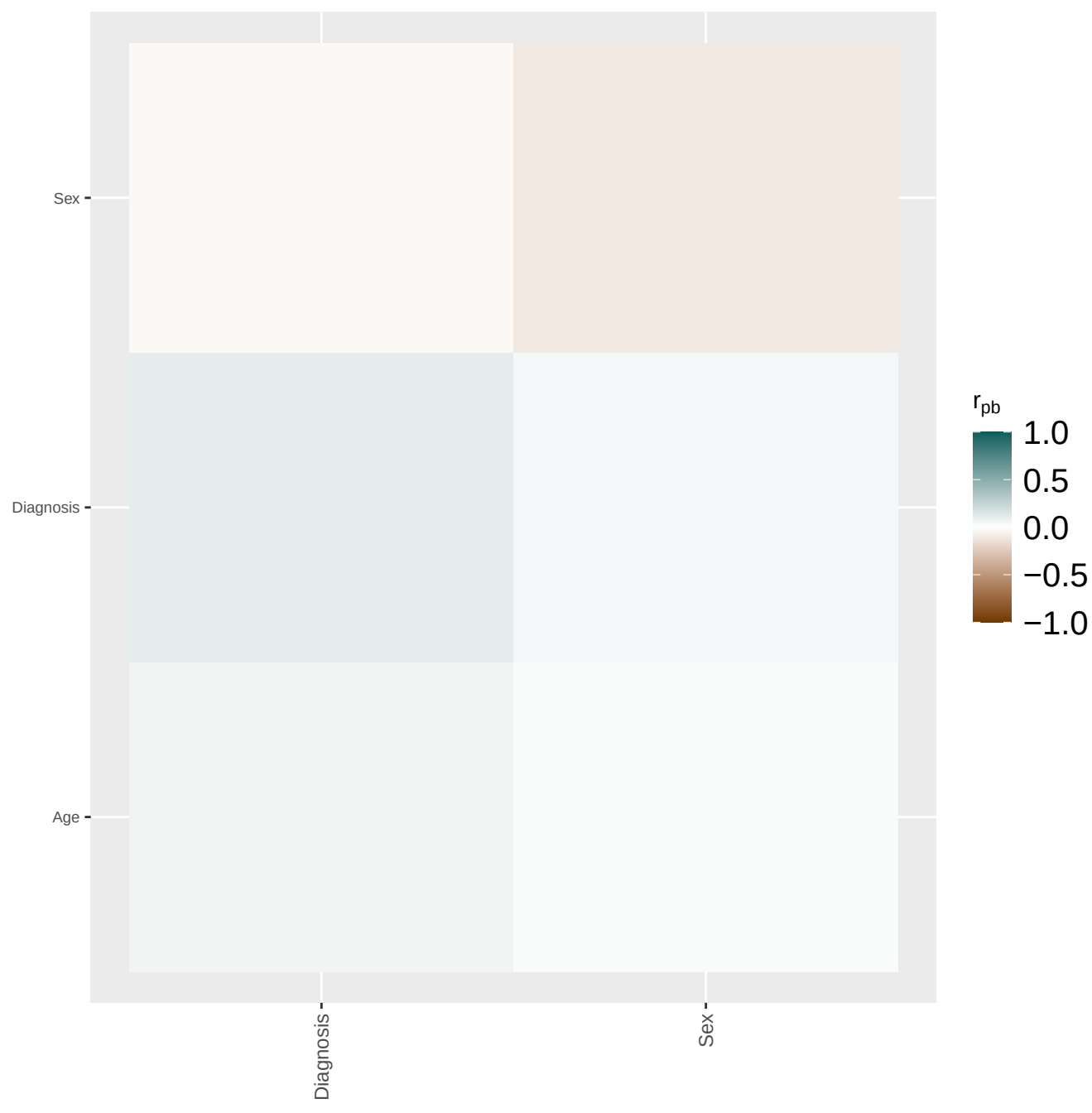
Diagnosis

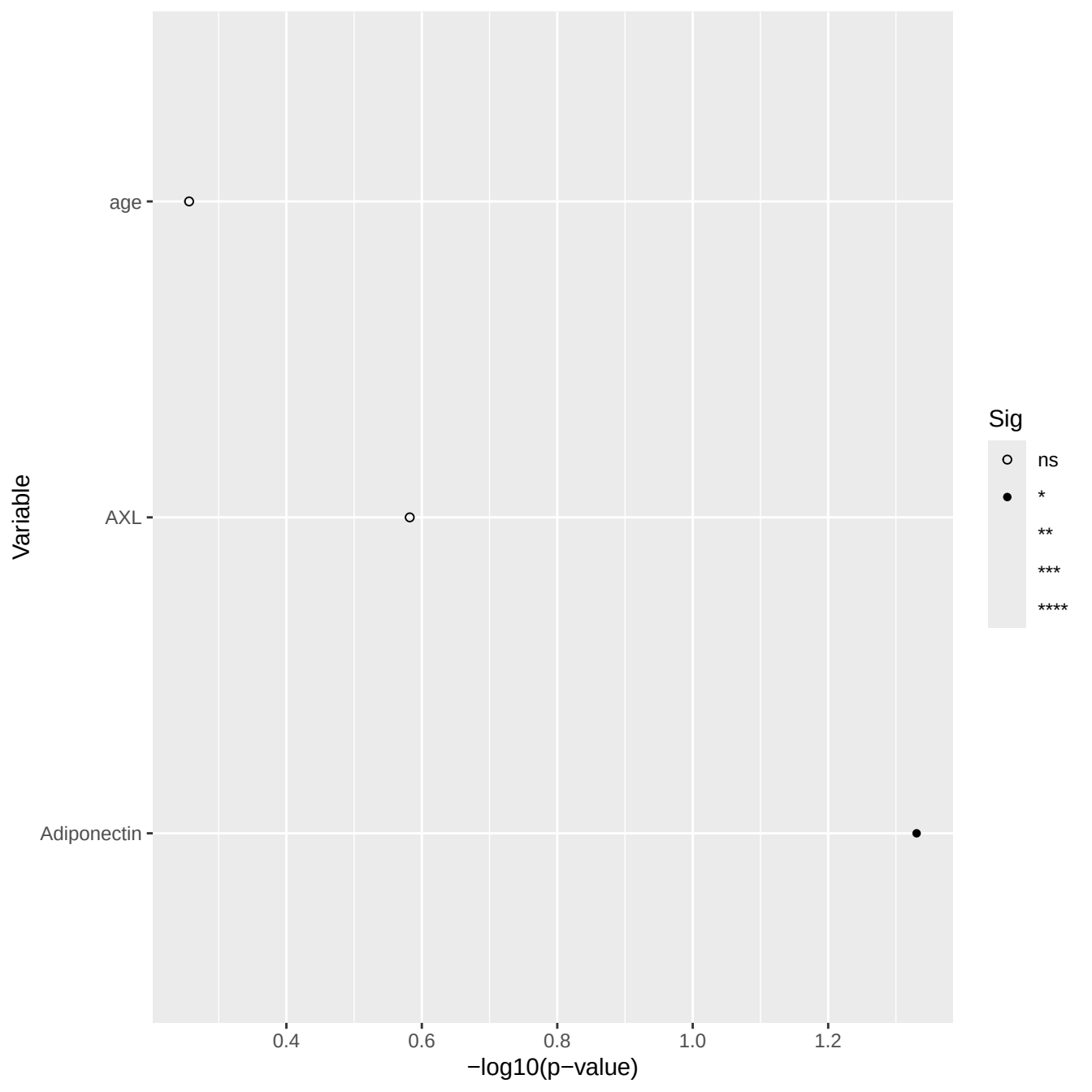
*

Diagnosis

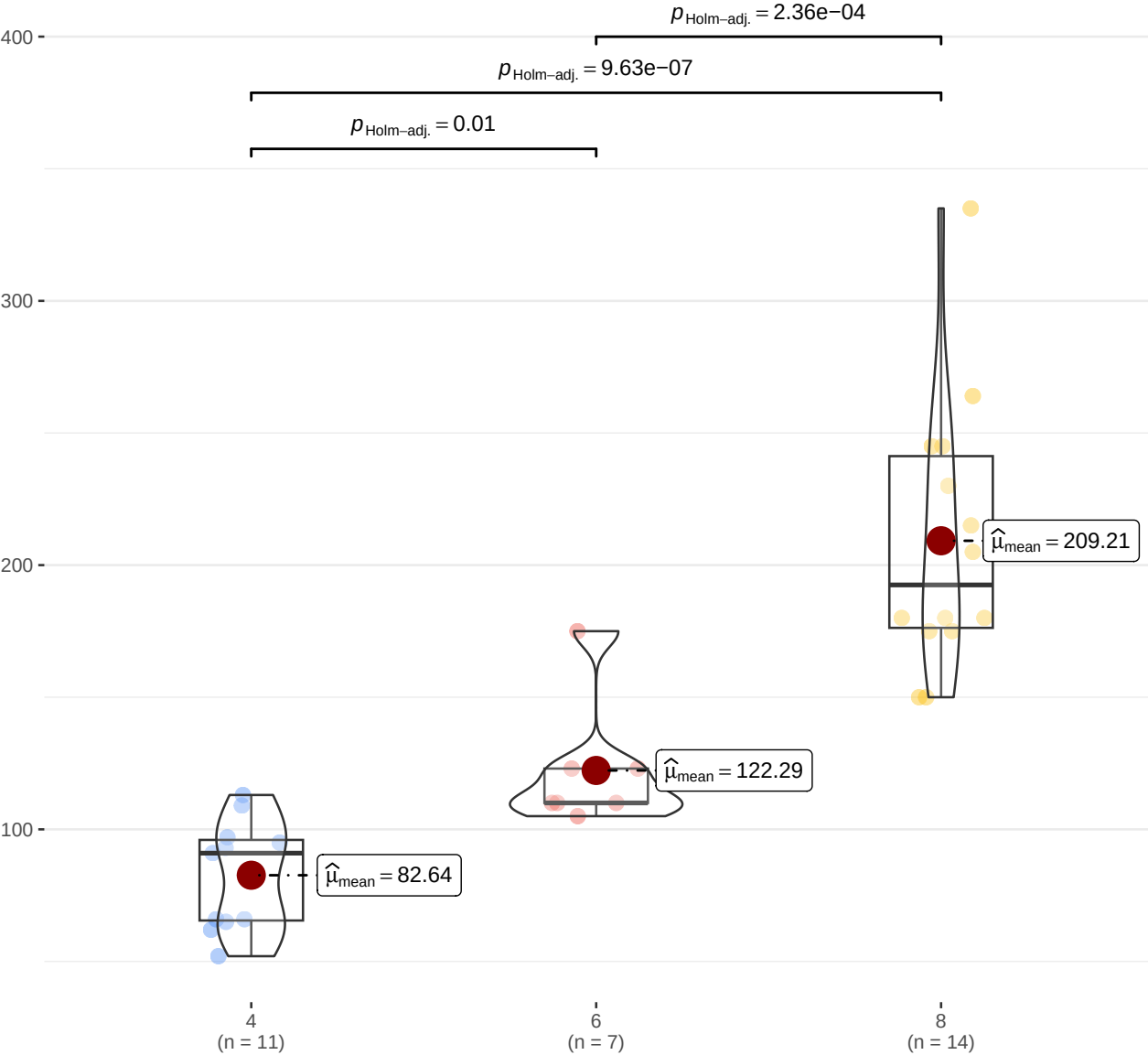
Sex



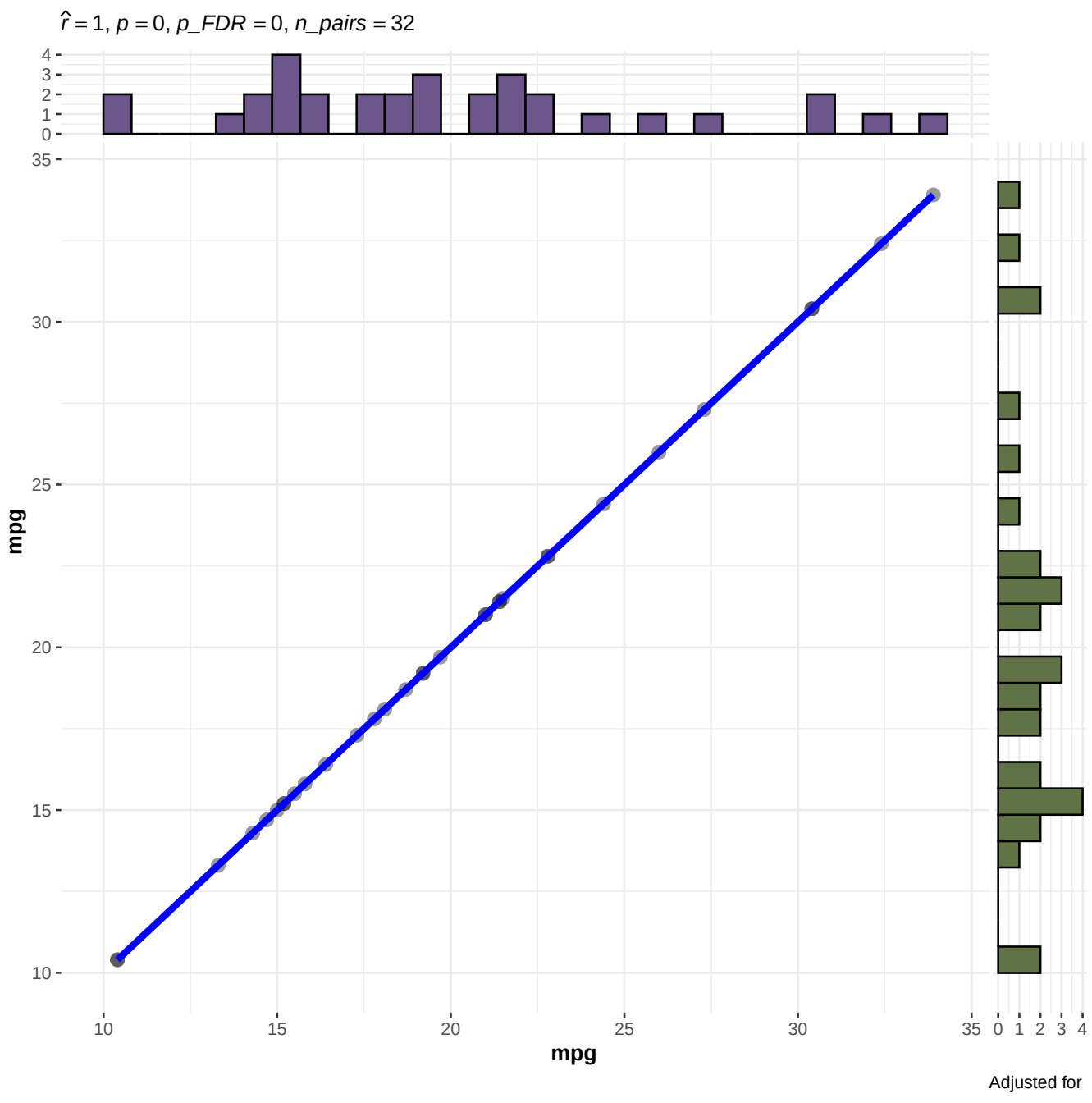


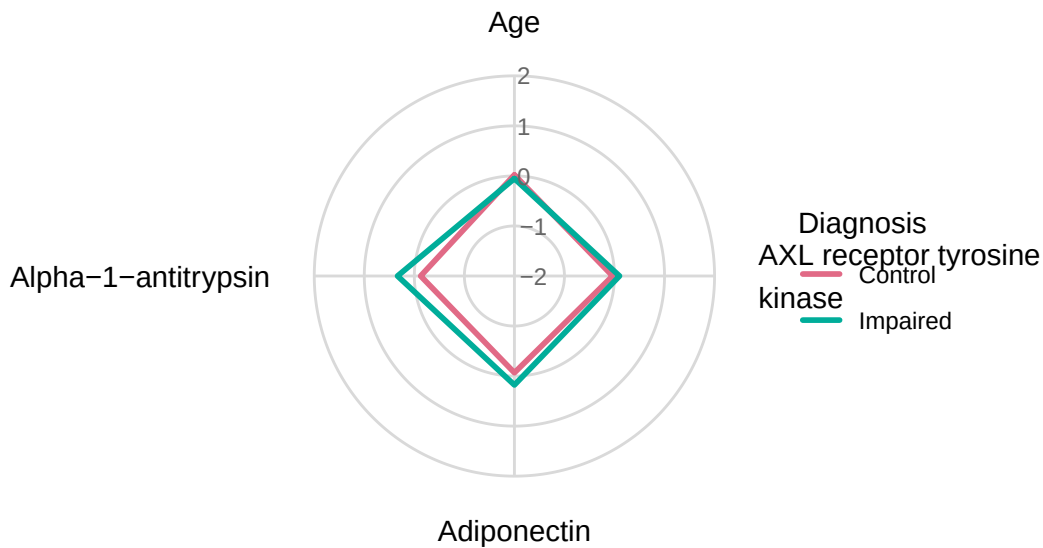


$F_{\text{Welch}}(2, 16.51) = 35.38, p = 1.07\text{e-}06, \hat{\omega}_p^2 = 0.78, \text{CI}_{95\%} [0.58, 1.00], n_{\text{obs}} = 32$



$\log_e(\text{BF}_{01}) = -14.03, \hat{R}^2_{\text{Bayesian}} = 0.69, \text{CI}_{95\%}^{\text{HDI}} [0.55, 0.77], r_{\text{Cauchy}}^{\text{JZS}} = 0.71$







Control (n=242)



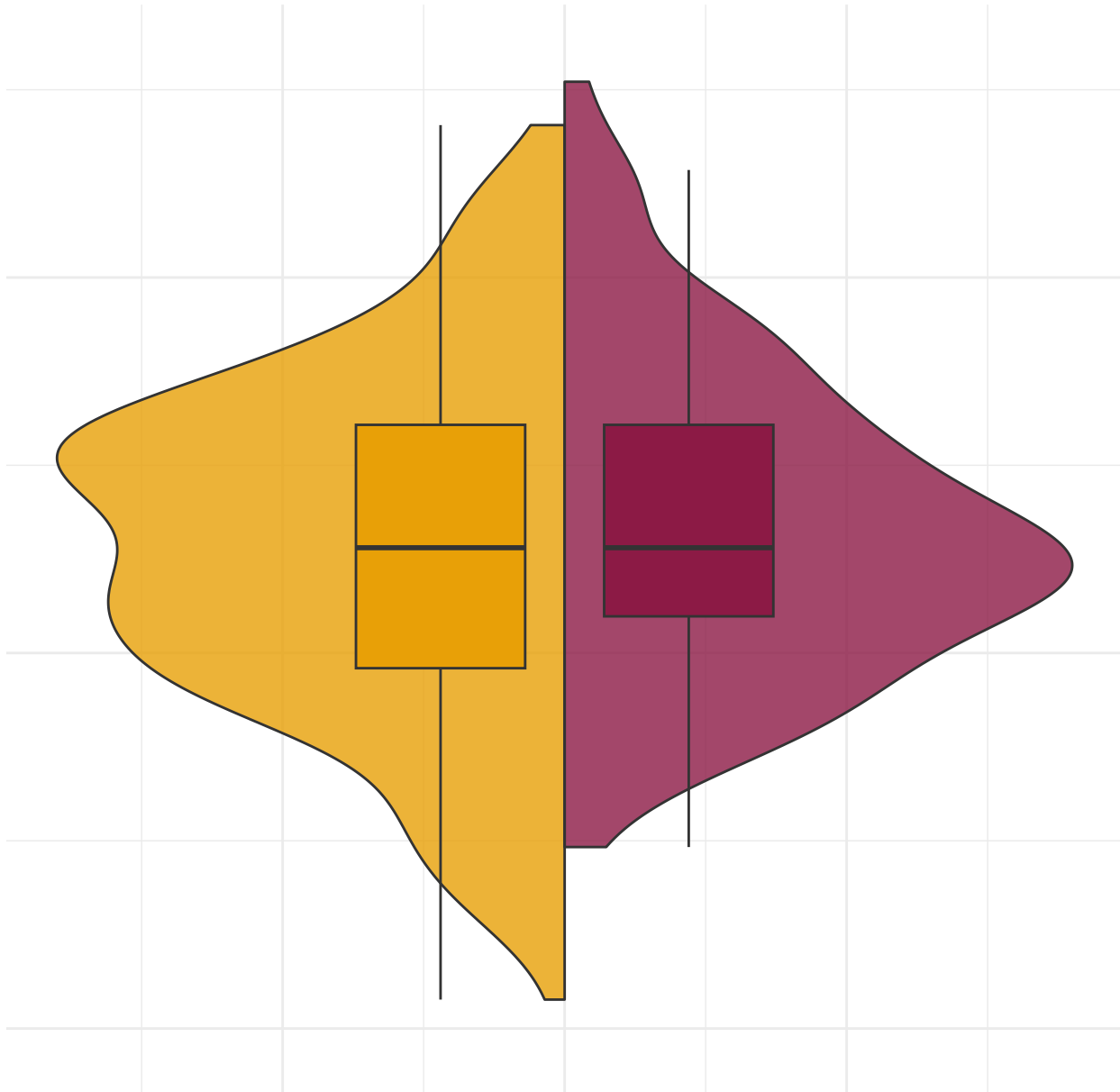
Impaired (n=91)

AXL

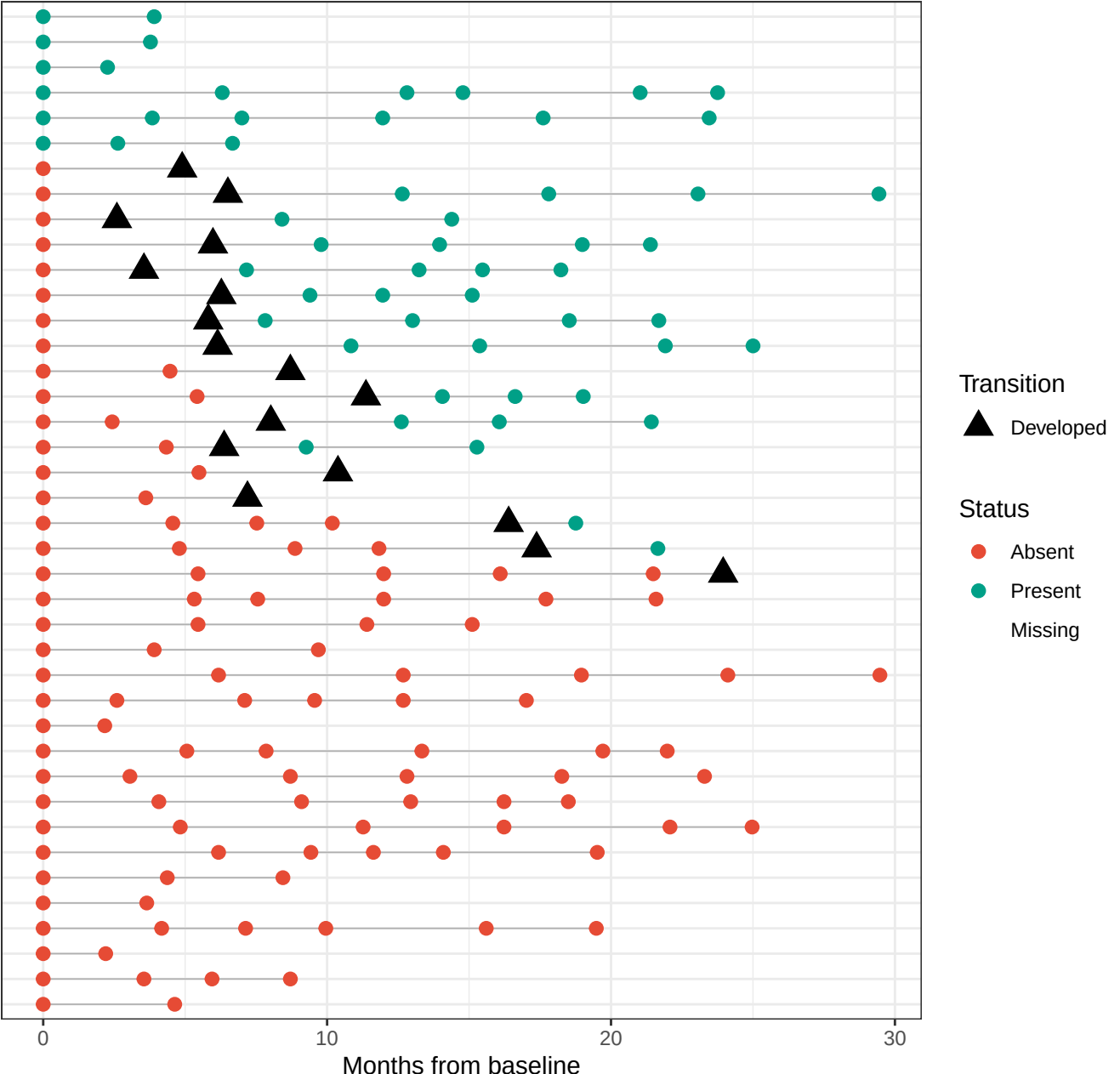
1

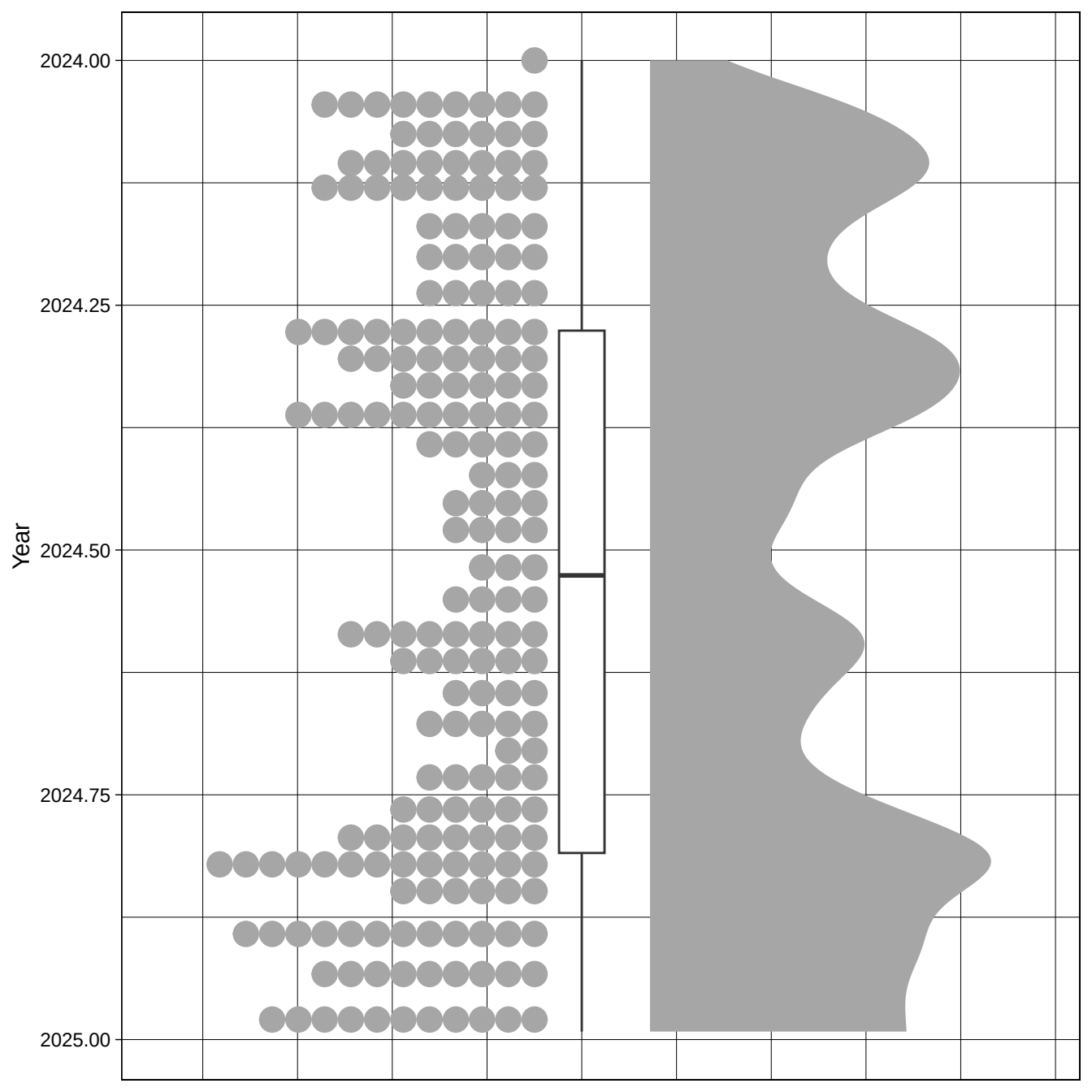
0

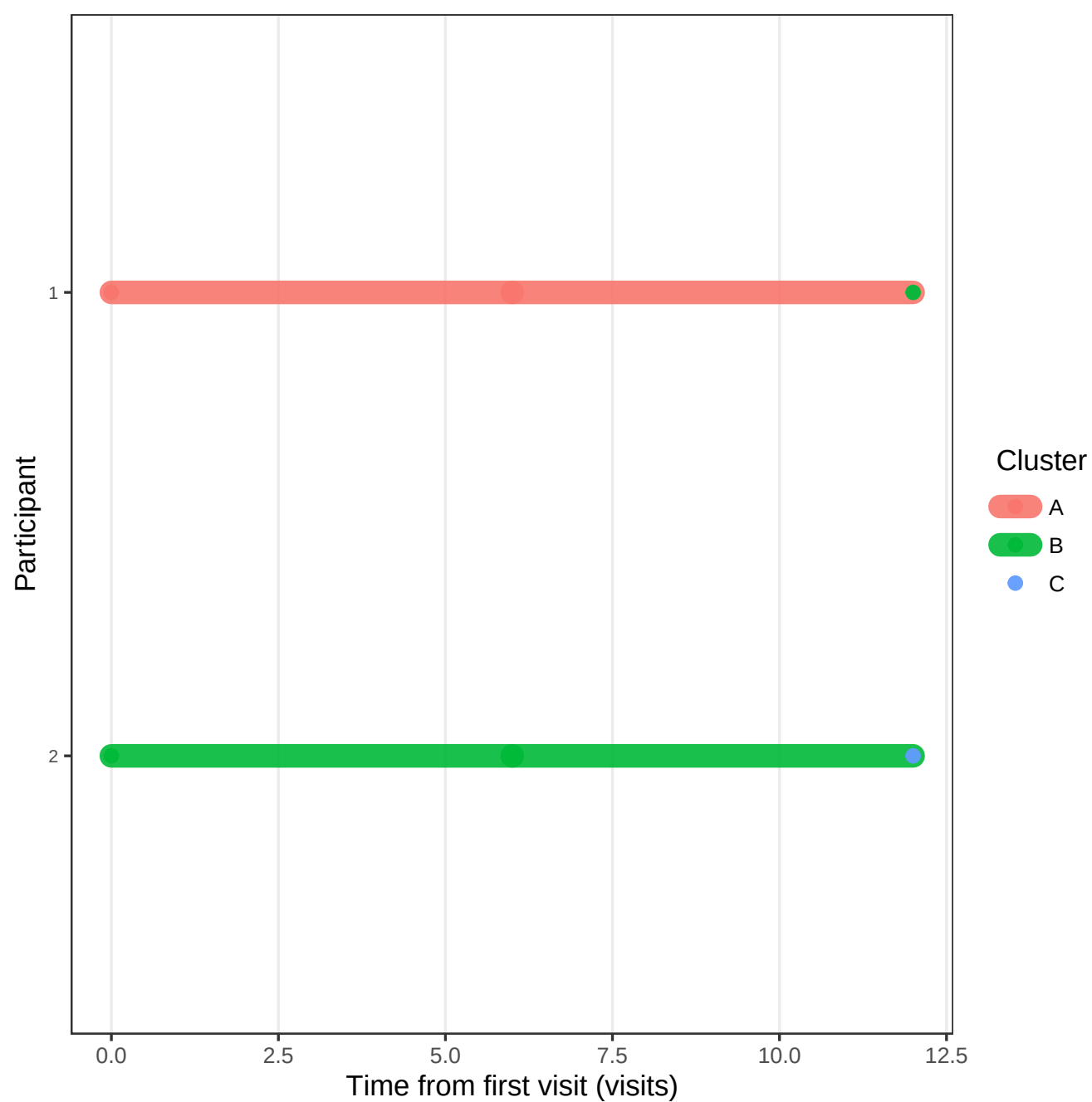
-1



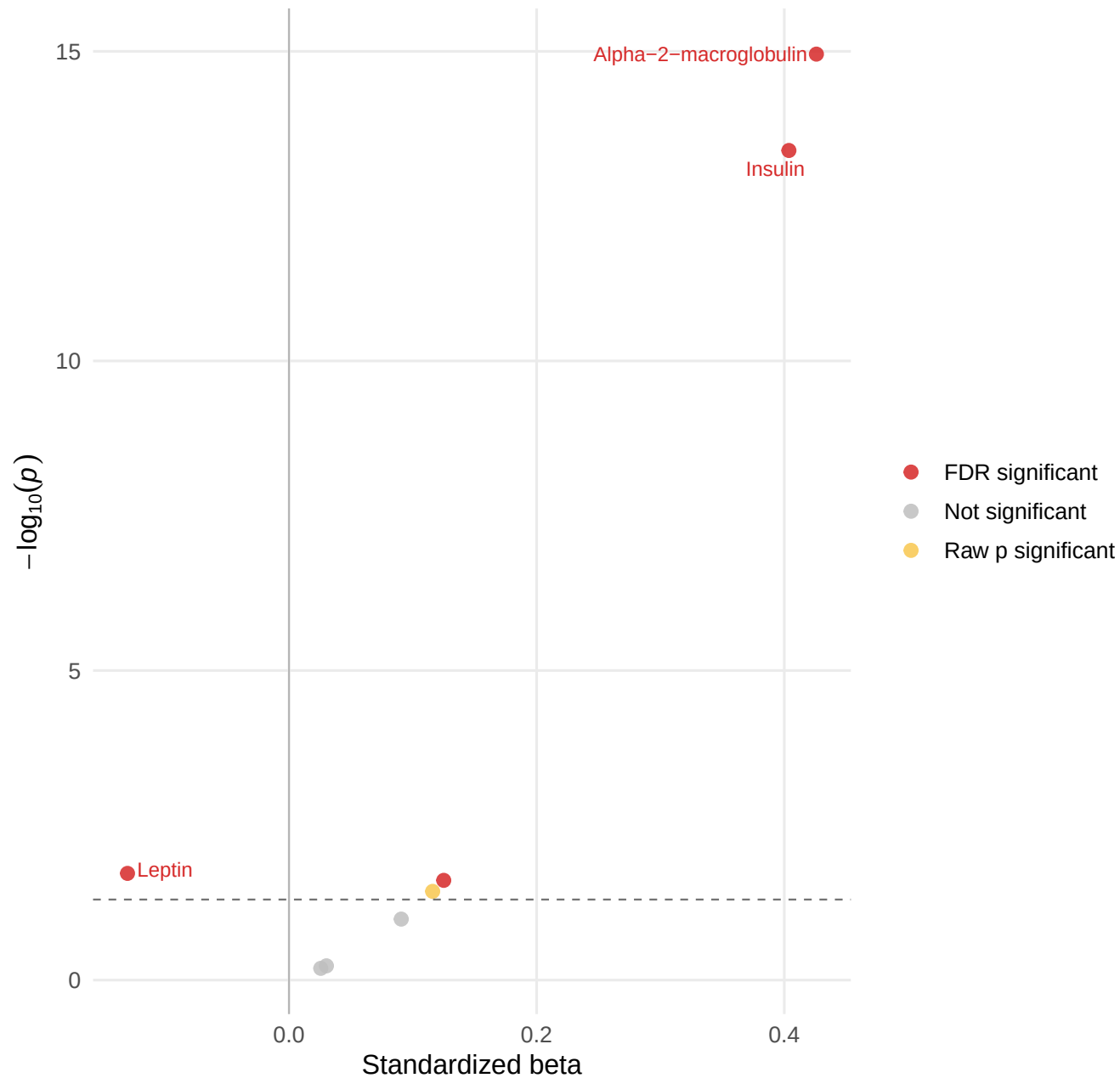
Swimmer plot of Recurrence across visits



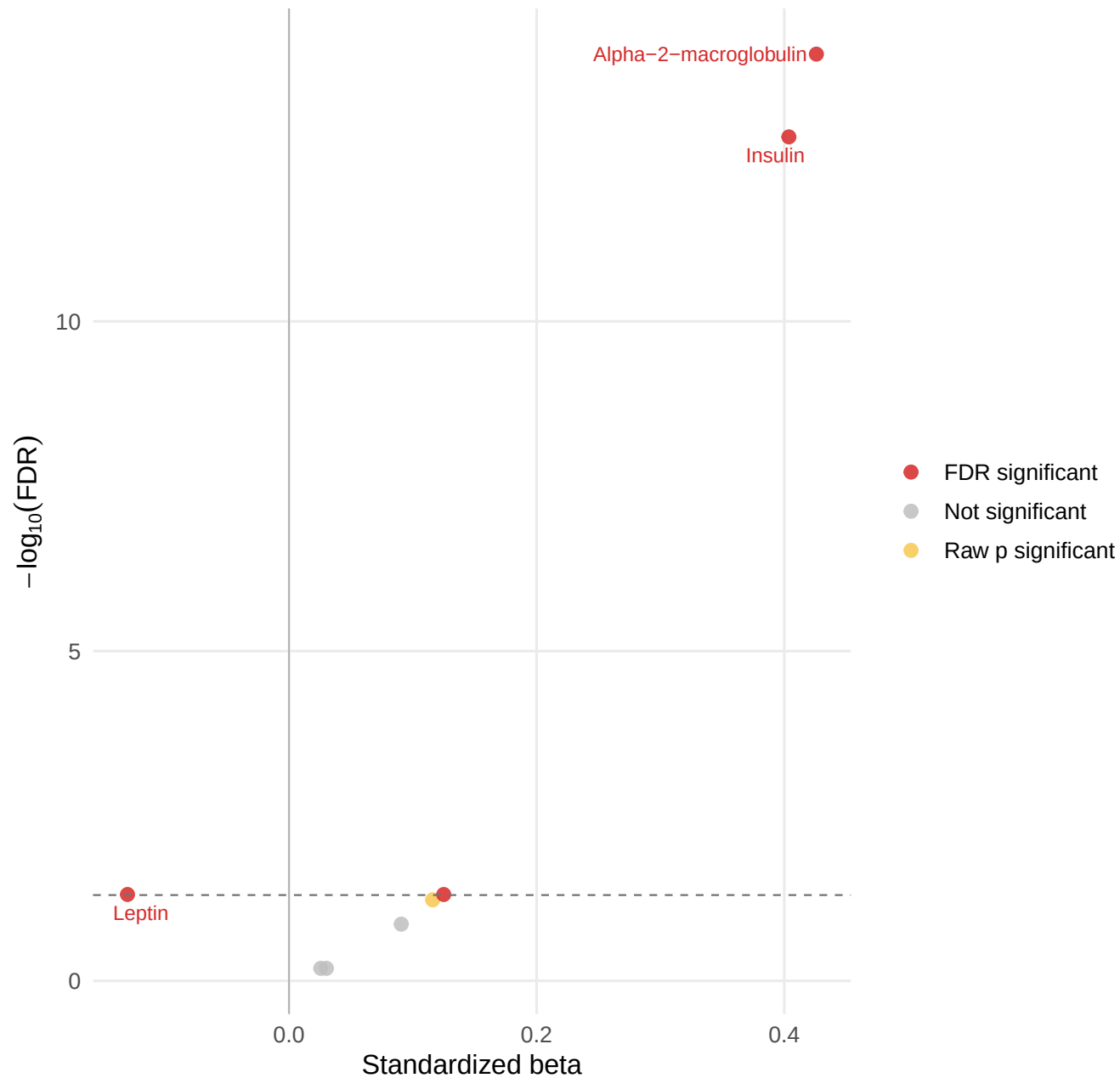




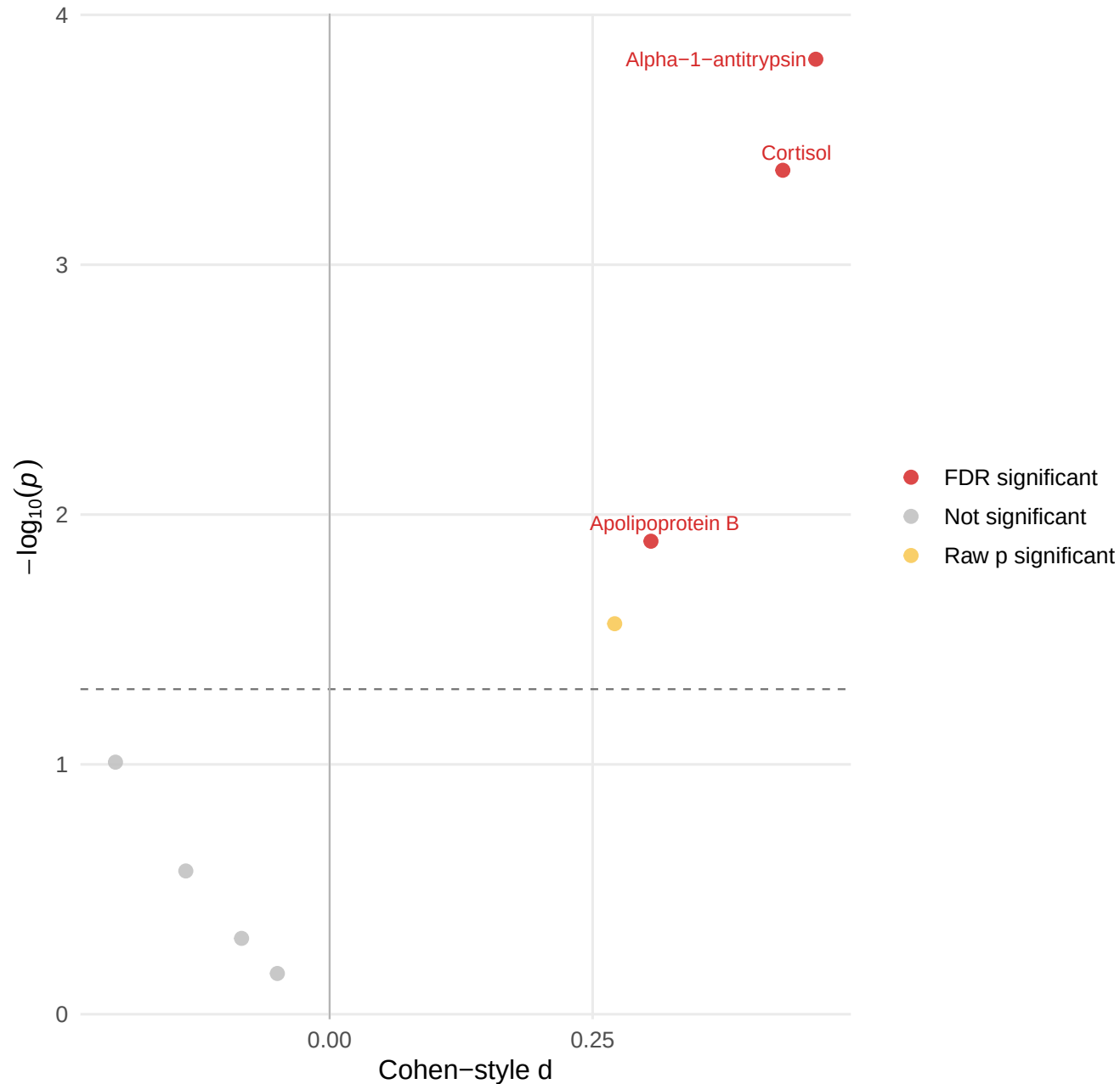
Volcano plot using raw p-values: AXL



Volcano plot using FDR-adjusted p-values: AXL



Volcano plot using raw p-values: Diagnosis



Volcano plot using FDR-adjusted p-values: Diagnosis

